Assessing Prey Fish-Seascape Relationships Through Spatially Nested Models

Supplemental Material 4: Logistic Regression Observed Versus Posterior Plots

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The density plots below depict the distribution of the posterior probabilities for categories of observed data (0 = no fish; 1 fish observed and > 20 fish observed) at the focal samples. The first set of plots are inclusive of all gear types for model 2C (figures 1 to 9) and 2G (figures 10 to 18) and correspond to Table 1 for prey fish. The second set of plots divide the model output versus the observed abundance by each individual gear type (figures 19-72) and corresponds with tables 2, 3, and 4. For predator fish, the first set of plots are inclusive of all gear types for model 2I (figures X to X) and correspond to Table 5. The second set of plots divide the model output versus the observed abundance by each individual gear type (figures 73-XX) and corresponds with tables 6, 7, and 8. Our expectation is that the distribution of the posteriors will align with the observed data. In this case, samples with no

fish should have lower predicted probabilities of occurrence (i.e., peak over zero) while samples with > 20 fish should have higher posterior probabilities of occurrence (i.e., peak over one). We did not evaluate observations of only 1 prey fish individual in our interpretation of evidence as these prey fish are unlikely to be alone but rather in larger groups. Also note that predator fish are unlikely to be in large groups which may influence some of the interpretations of the >20 evaluations. notice the change in the axis among the plots with each category.

Table 1. The table below summarizes our interpretation of the evidence when all gear types are in consideration:

Model 2C			
Species	Fish observed =0	Fish observed >20	Overall
Grouped Prey Fish	Inconclusive	Strong	Credible
Herring	Strong	Inconclusive	Credible
Anchovy	Strong	Weak	Credible
Mullet	Strong	Strong	Strong
Mojarra	Inconclusive	Strong	Credible
Silver Perch	Strong	Strong	Strong
Pinfish	Strong	Strong	Strong
Pigfish	Strong	Strong	Strong
Silversides	Weak	Weak	Weak
Model 2G			
Species	Fish observed =0	Fish observed >20	Overall
Grouped Prey Fish	Weak	Strong	Convincing
Herring	Strong	Inconclusive	Convincing
Anchovy	Strong	Inconclusive	Convincing
Mullet	Strong	Strong	Strong
Mojarra	Weak	Strong	Strong
Silver Perch	Strong	Strong	Strong
Pinfish	Strong	Strong	Strong
Pigfish	Strong	Strong	Strong
Silversides	Strong	Strong	Weak

Table 2. The table below summarizes our interpretation of the evidence when only gear type 1 (small seines) is in consideration:

Model 2C			
Species	Fish observed =0	Fish observed >20	Overall
Grouped Prey Fish	Weak	Strong	Credible

Herring	Strong	Weak	Credible
Anchovy	Strong	Dubious	Credible
Mullet	Strong	Weak	Credible
Mojarra	Weak	Strong	Convincing
Silver Perch	Strong	Weak	Convincing
Pinfish	Strong	Strong	Strong
Pigfish	Strong	Inconclusive	Convincing
Silversides	Strong	Weak	Weak
Model 2G			
Species	Fish observed =0	Fish observed >20	Overall
Grouped Prey Fish	Weak	Strong	Convincing
Herring	Strong	Inconclusive	Convincing
Anchovy	Strong	Inconclusive	Convincing
Mullet	Strong	Strong	Strong
Mojarra	Weak	Strong	Strong
Silver Perch	Strong	Strong	Strong
Pinfish	Strong	Strong	Strong
Pigfish	Strong	Strong	Strong
Silversides	Strong	Weak	Credible

Table 3. The table below summarizes our interpretation of the evidence when only gear type 2 (large seines) is in consideration:

Model 2C			
Species	Fish observed =0	Fish observed >20	Overall
Grouped Prey Fish	Inconclusive	Strong	Credible
Herring	Strong	Weak	Credible
Anchovy	Strong	Weak	Credible
Mullet	Inconclusive	Dubious	Dubious
Mojarra	Dubious	Strong	Convincing
Silver Perch	Strong	Strong	Strong
Pinfish	Inconclusive	Strong	Convincing
Pigfish	Inconclusive	Strong	Convincing
Silversides	Strong	NA	Strong
Model 2G			
Species	Fish observed =0	Fish observed >20	Overall
Grouped Prey Fish	Weak	Strong	Convincing
Herring	Strong	Inconclusive	Convincing

Anchovy	Strong	Inconclusive	Convincing
Mullet	Strong	Strong	Strong
Mojarra	Weak	Strong	Strong
Silver Perch	Strong	Strong	Strong
Pinfish	Strong	Strong	Strong
Pigfish	Inconclusive	Strong	Convincing
Silversides	Strong	NA	Strong

Table 4. The table below summarizes our interpretation of the evidence when only gear type 3 (otter trawls) is in consideration:

Model 2C			
Species	Fish observed =0	Fish observed >20	Overall
Grouped Prey Fish	Weak	Strong	Credible
Herring	Strong	NA	Strong
Anchovy	Strong	Dubious	Credible
Mullet	Strong	NA	Strong
Mojarra	Weak	Strong	Convincing
Silver Perch	Strong	Weak	Convincing
Pinfish	Inconclusive	Strong	Convincing
Pigfish	Strong	Strong	Strong
Silversides	Strong	NA	Strong
Model 2G			
Species	Fish observed =0	Fish observed >20	Overall
Grouped Prey Fish	Weak	Strong	Convincing
Herring	Strong	Inconclusive	Convincing
Anchovy	Strong	Inconclusive	Convincing
Mullet	Strong	Strong	Strong
Mojarra	Weak	Strong	Strong
Silver Perch	Strong	Strong	Strong
Pinfish	Strong	Strong	Strong
Pigfish	Strong	Strong	Strong
Silversides	Strong	Strong	Weak

Table 5. The table below summarizes our interpretation of the evidence when all gear types are in consideration for predator fish:

Model 2C			
Species	Fish observed =0	Fish observed >20	Overall
Grouped Predator Fish	Strong	Strong	Strong
Common Snook	Credible	Credible	Credible
Trout	Strong	Weak	Convincing
Red Drum	Strong	Weak	Convincing
Snapper	Strong	Strong	Strong
Crevalle Jack	Strong	Convincing	Credible

Table 6. The table below summarizes our interpretation of the evidence when only gear type 1 (small seines) is in consideration:

Model 2C			
Species	Fish observed =0	Fish observed >20	Overall
Grouped Predator Fish	Strong	Convincing	Credible
Common Snook	Strong	Credible	Credible
Trout	Strong	Convincing	Credible
Red Drum	Strong	Convincing	Credible
Snapper	Convincing	Convincing	Convincing
Crevalle Jack	Strong	Weak	Convincing

Table 7. The table below summarizes our interpretation of the evidence when only gear type 2 (large seine) is in consideration:

Model 2C			
Species	Fish observed =0	Fish observed >20	Overall
Grouped Predator Fish	Convincing	Credible	Convincing
Common Snook	Strong	Weak	Convincing
Trout	Strong	Weak	Convincing
Red Drum	Strong	Weak	Convincing
Snapper	Strong	Inconclusive	Inconclusive
Crevalle Jack	Strong	Convincing	Credible

Table 8. The table below summarizes our interpretation of the evidence when only gear type 3 (otter trawls) is in consideration:

Model 2C			
Species	Fish observed =0	Fish observed >20	Overall
Grouped Predator Fish	Credible	Convincing	Convincing
Common Snook	Convincing	Inconclusive	Convincing
Trout	Strong	Convincing	Credible
Red Drum	Strong	Inconclusive	Inconclusive
Snapper	Convincing	Inconclusive	Inconclusive
Crevalle Jack	Strong	Inconclusive	Inconclusive

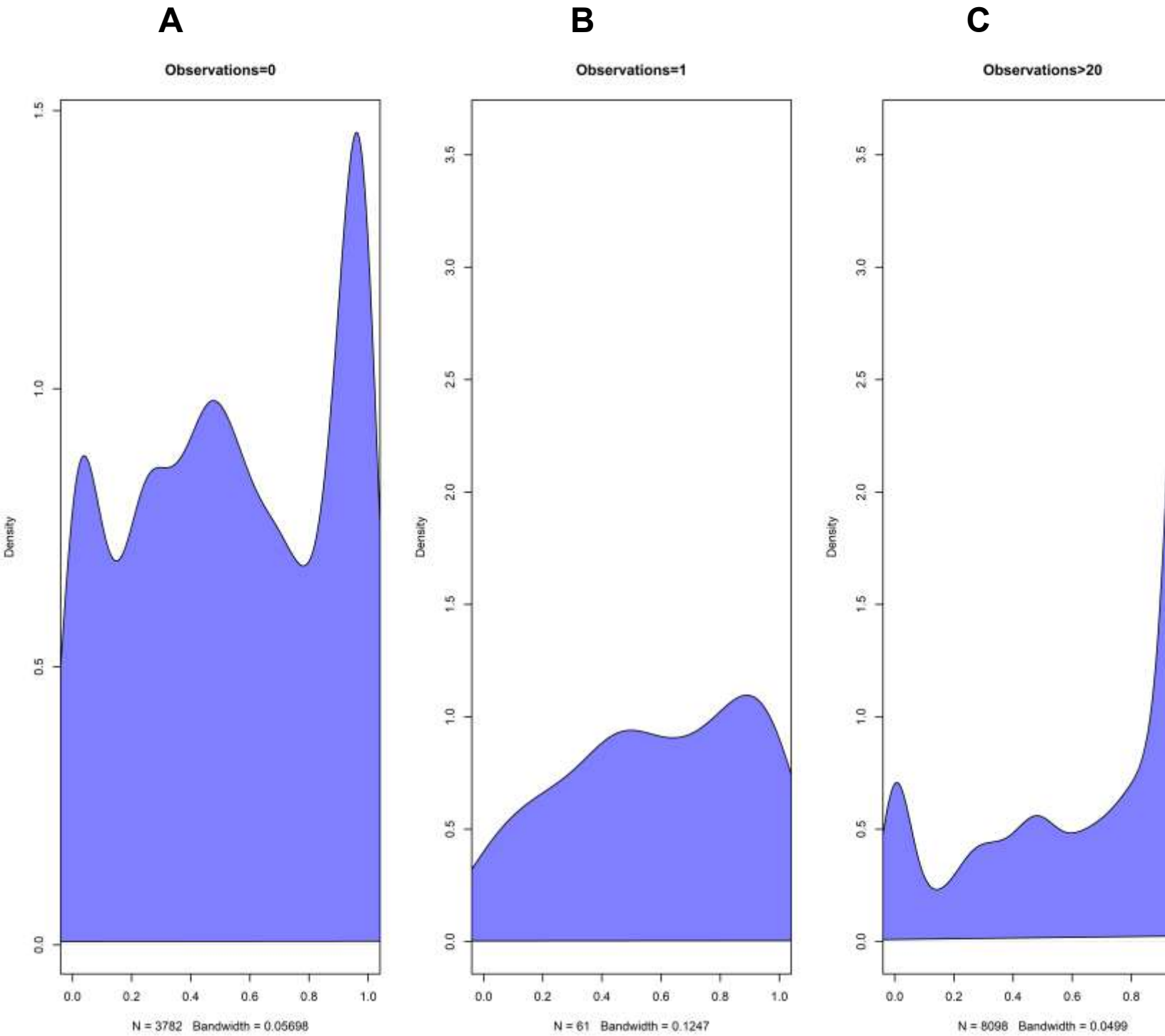
A
Observations=0
Density
N = 3782 Bandwidth = 0.05698
B
Observations=1
Density
N = 61 Bandwidth = 0.1247
C
Observations>20
Density
N = 8098 Bandwidth = 0.0499

Figure 1. Logistic regression observed versus posterior plots for Grouped Prey Fish Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

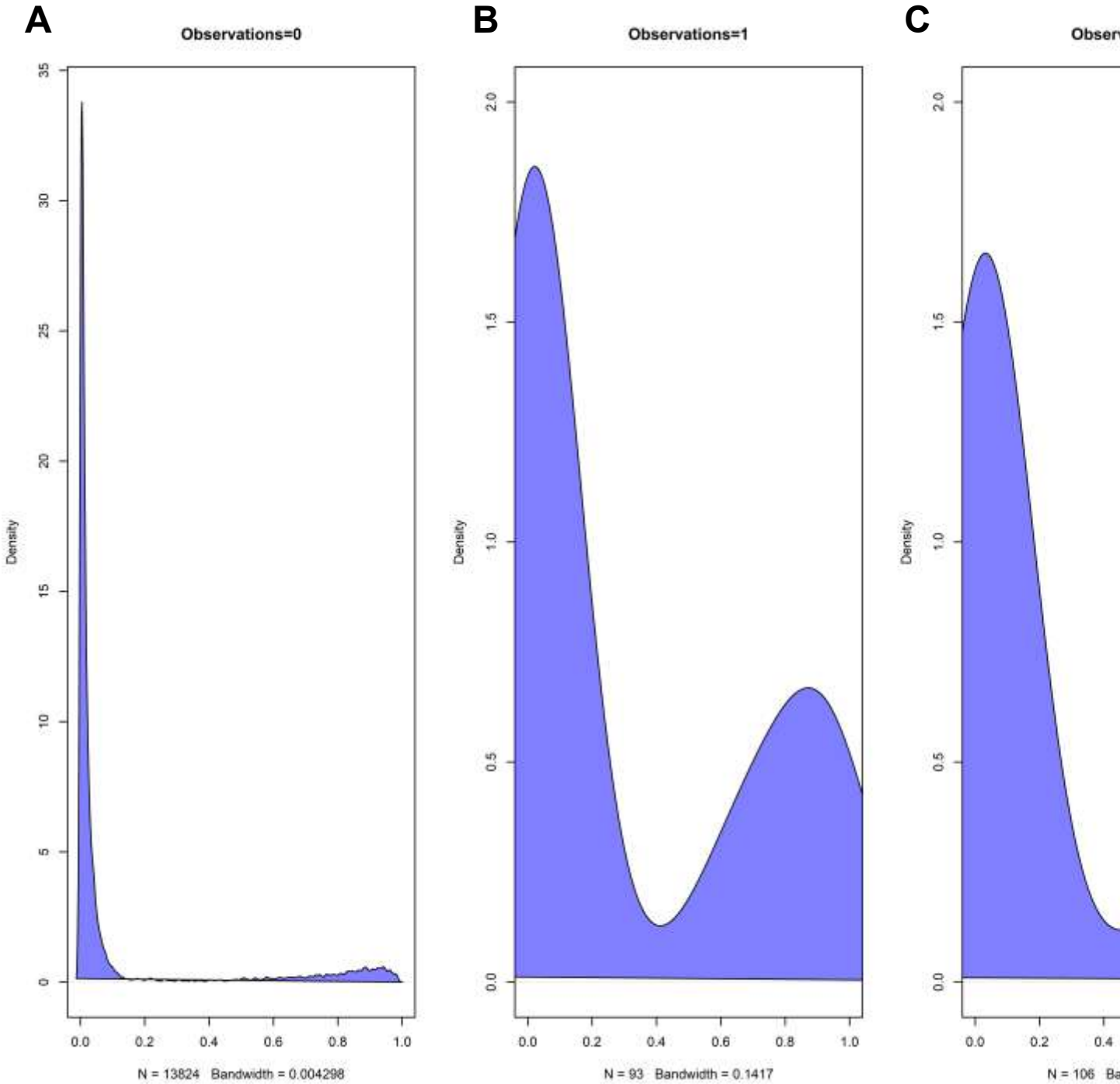

Figure 2. Logistic regression observed versus posterior plots for Herring Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

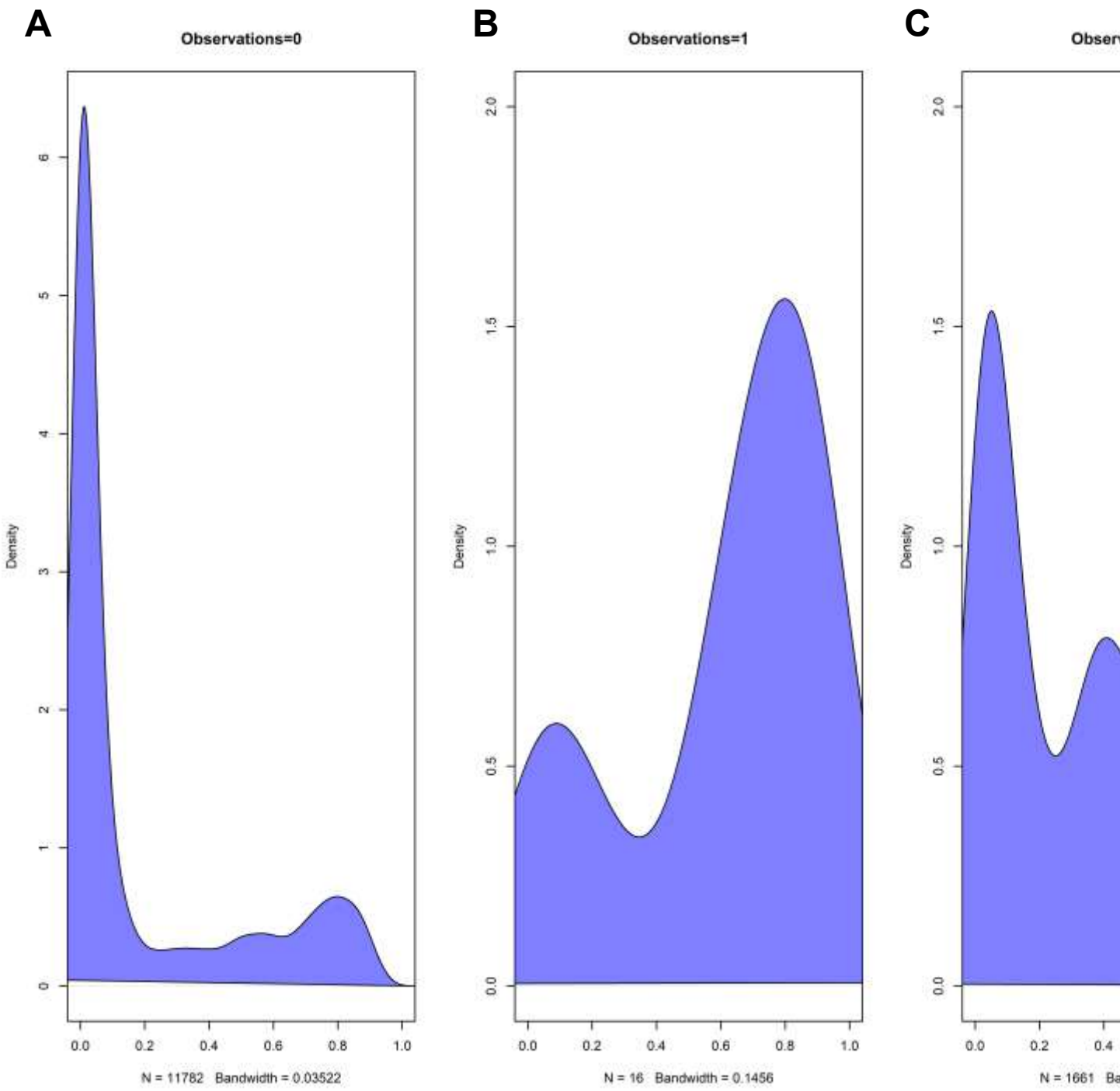

Figure 3. Logistic regression observed versus posterior plots for Anchovy Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

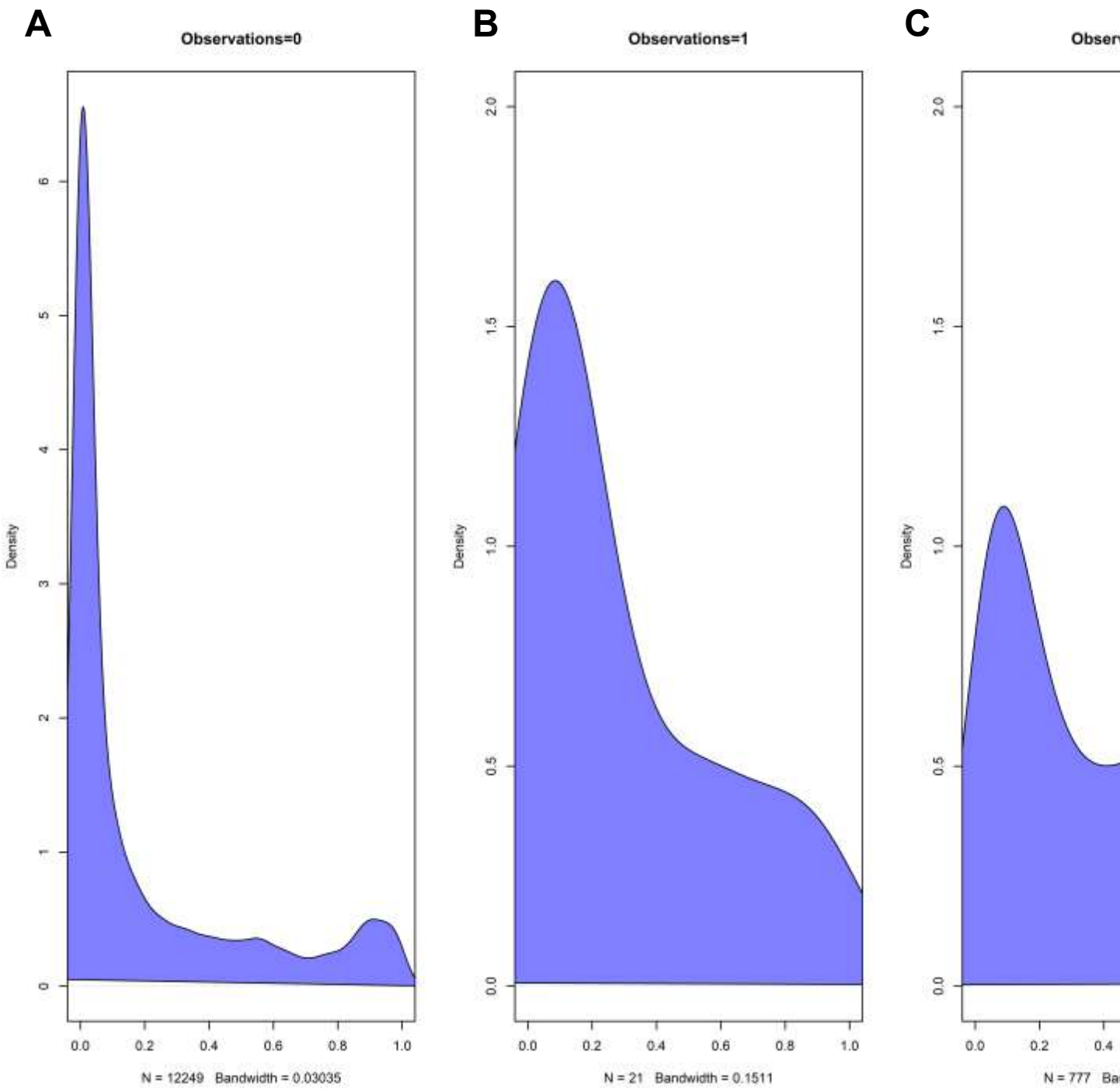

Figure 4. Logistic regression observed versus posterior plots for Mullet Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

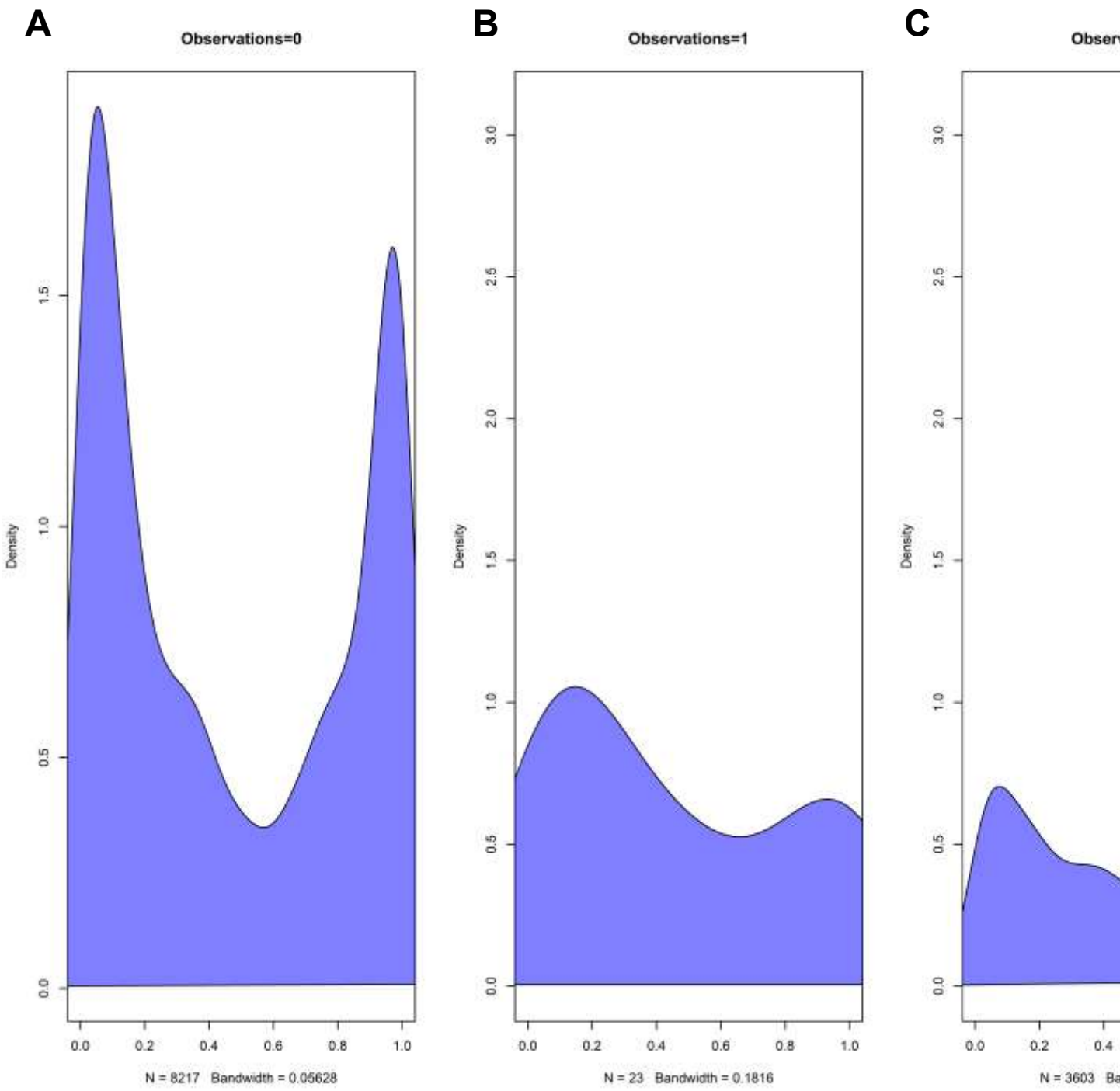

Figure 5. Logistic regression observed versus posterior plots for Mojarra Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

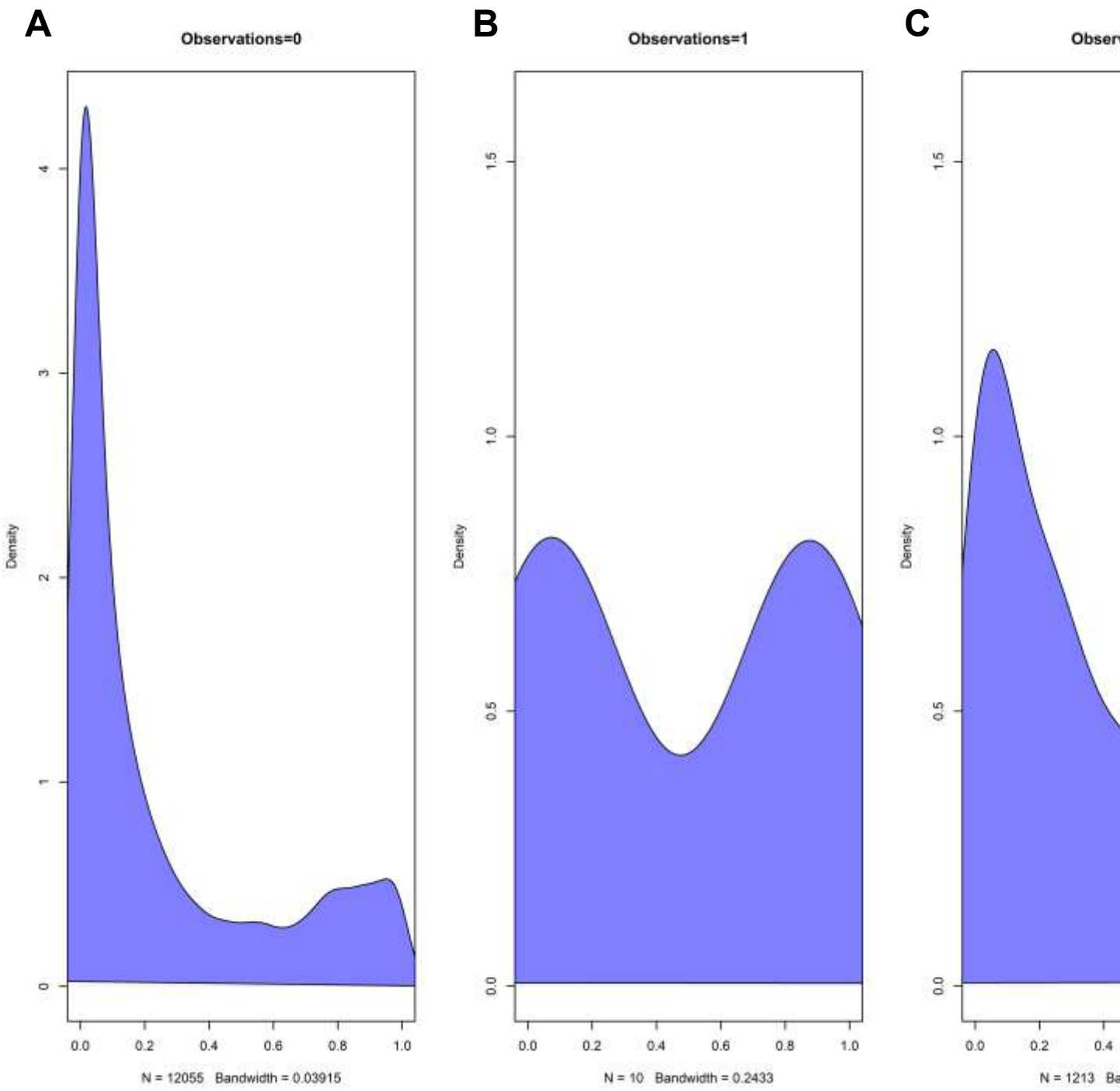

Figure 6. Logistic regression observed versus posterior plots for Silver Perch Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

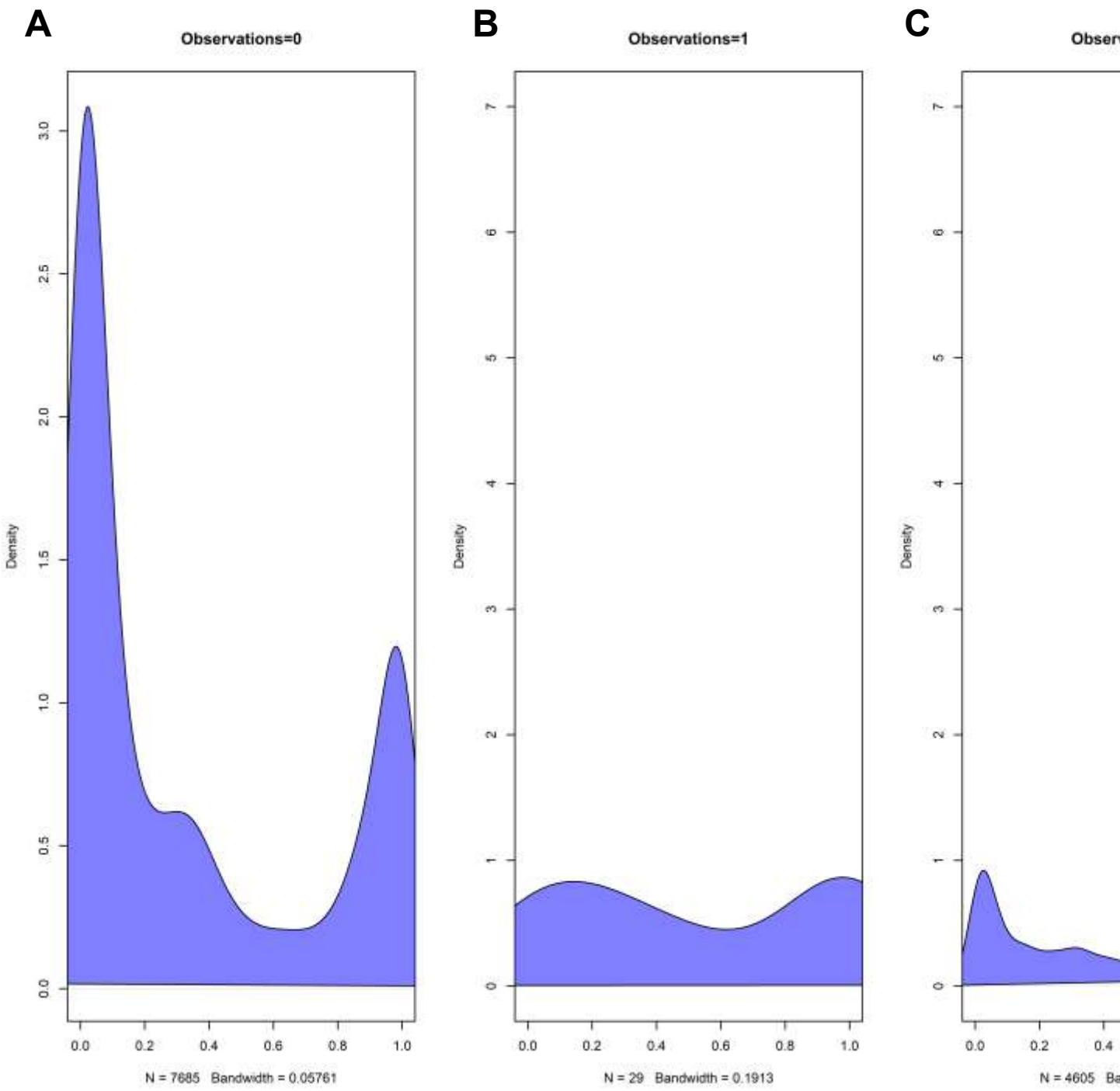

Figure 7. Logistic regression observed versus posterior plots for Pinfish Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

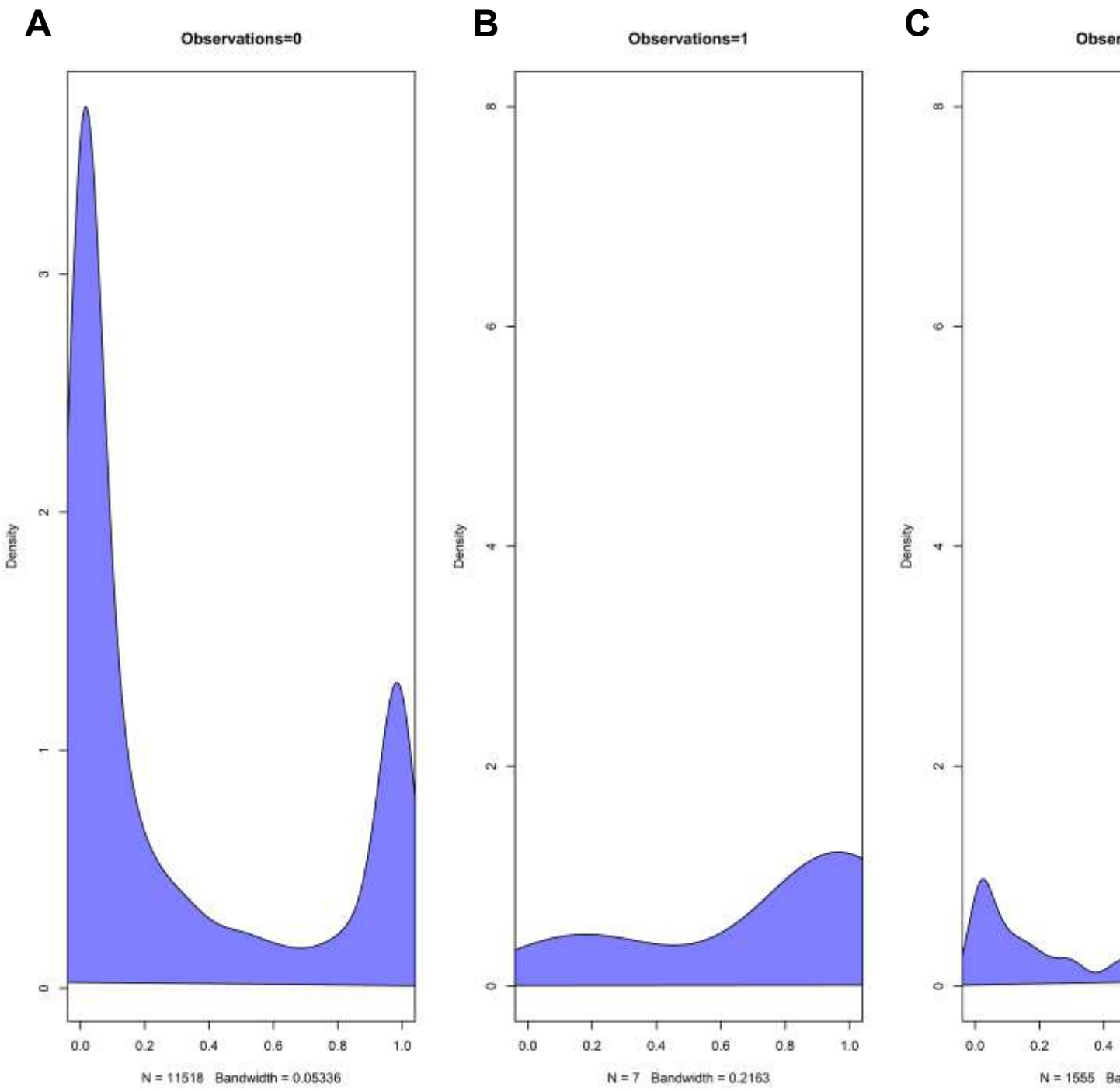

Figure 8. Logistic regression observed versus posterior plots for Pigfish Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

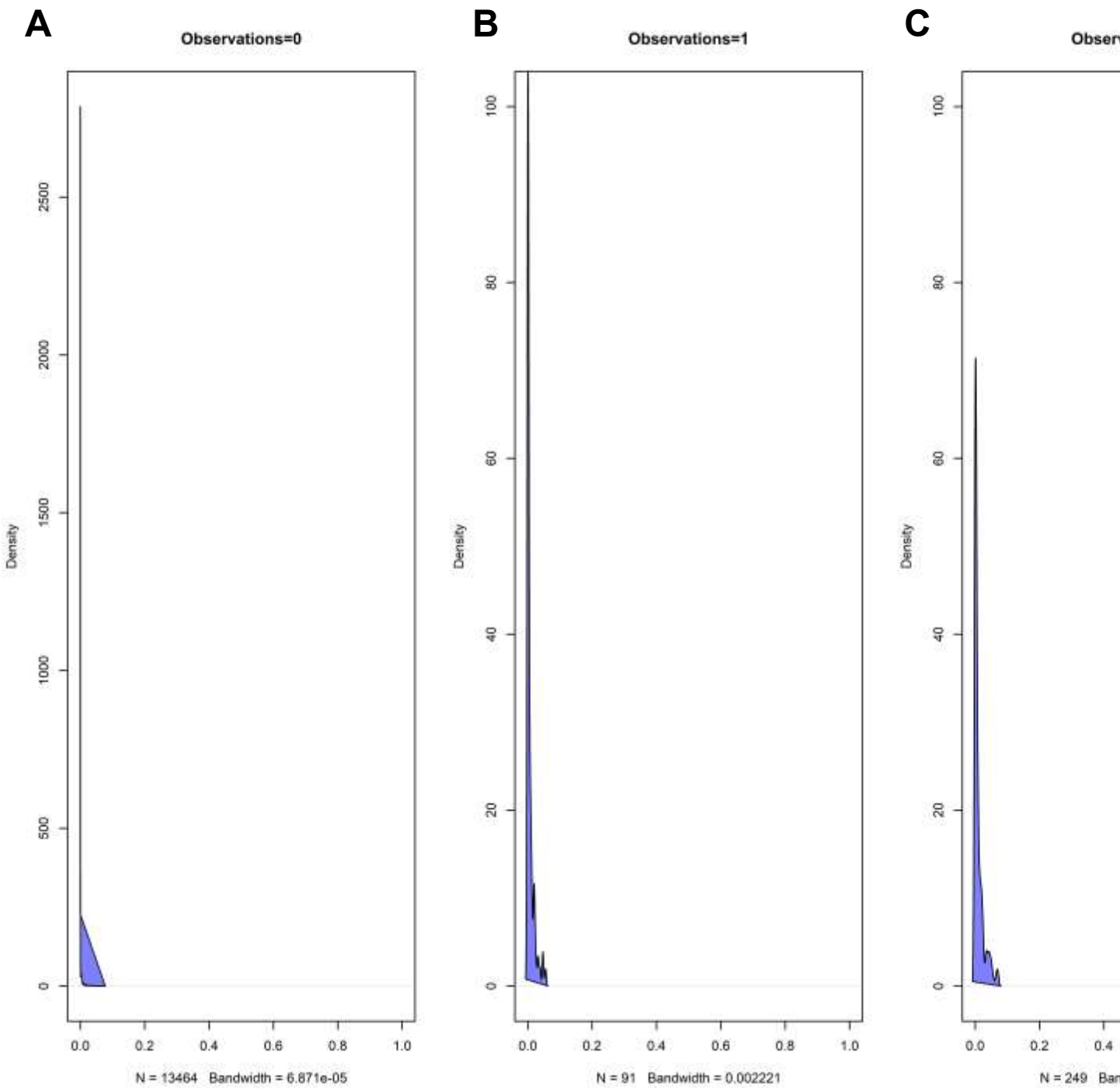

Figure 9. Logistic regression observed versus posterior plots for Silversides Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

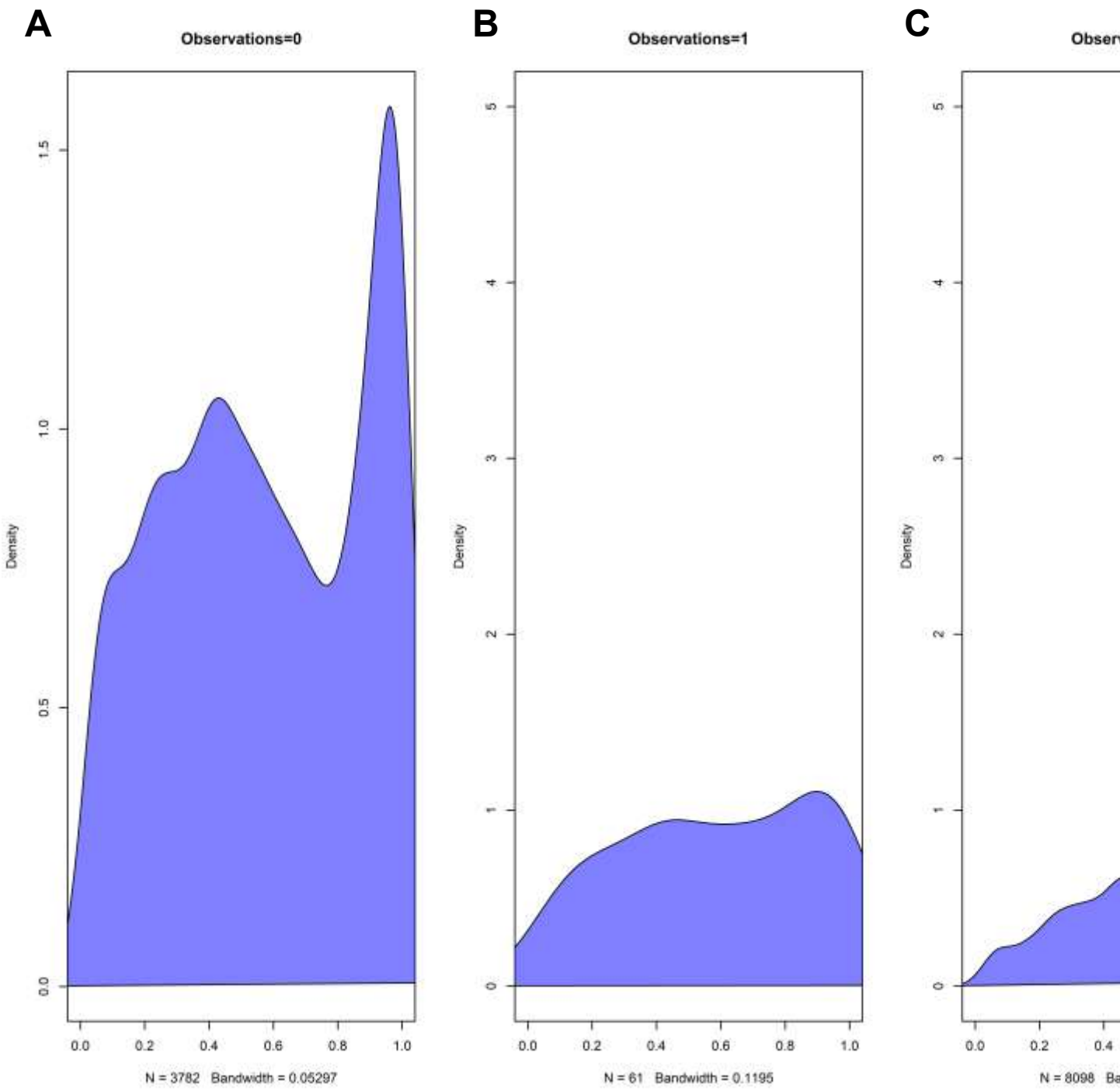

Figure 10. Logistic regression observed versus posterior plots for Grouped Prey Fish Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

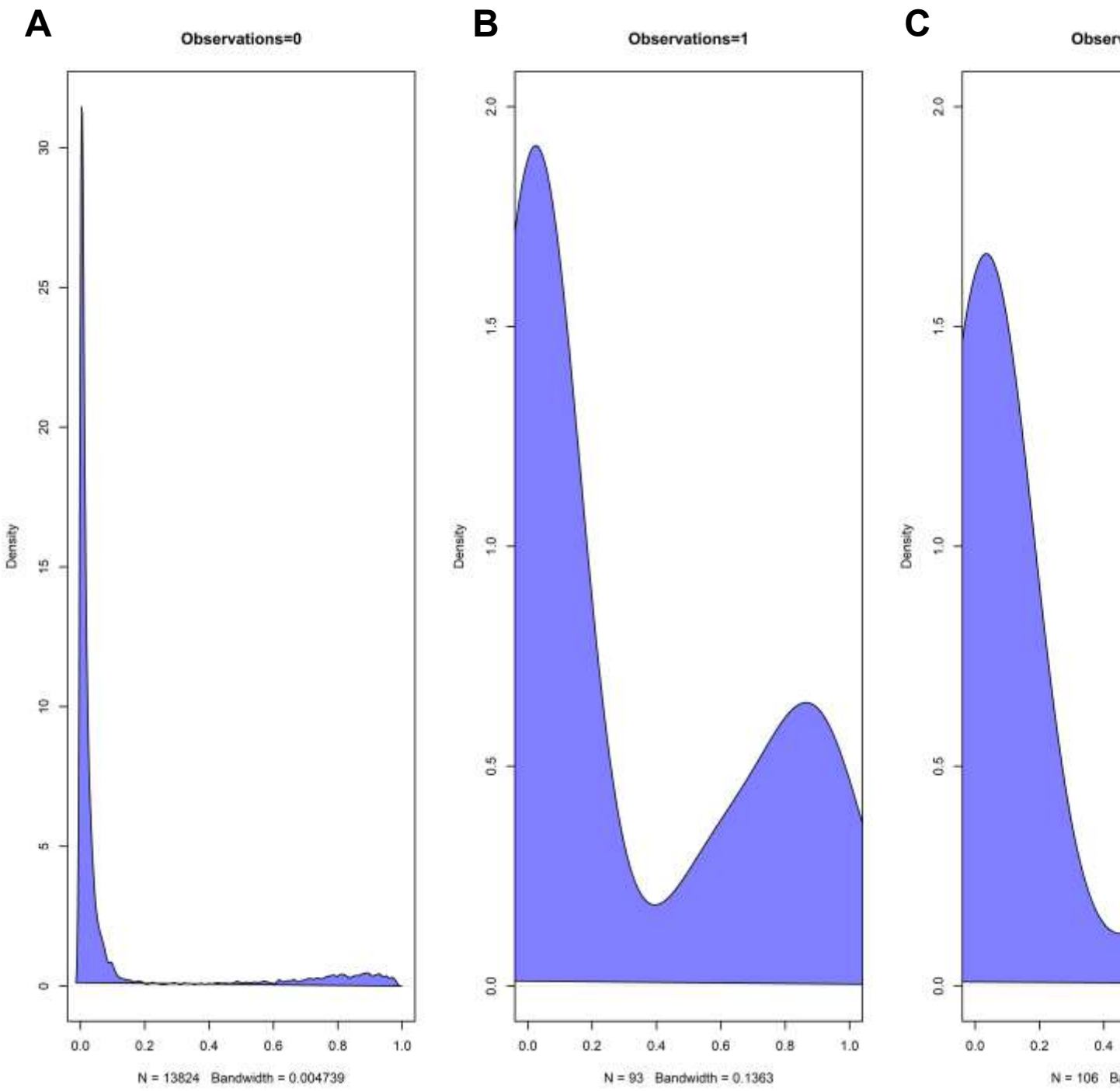

Figure 11. Logistic regression observed versus posterior plots for Herring Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

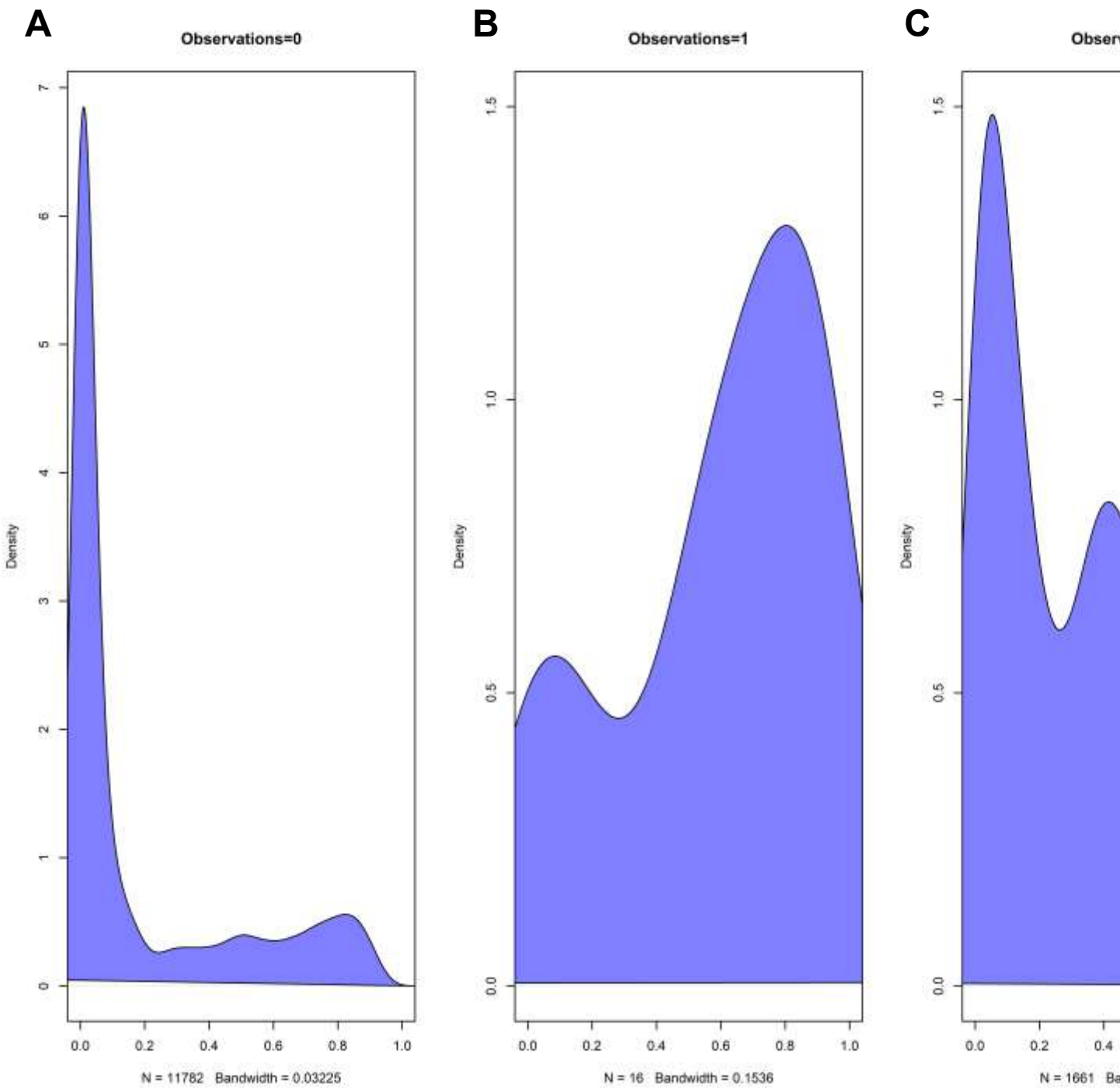

Figure 12. Logistic regression observed versus posterior plots for Anchovy Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

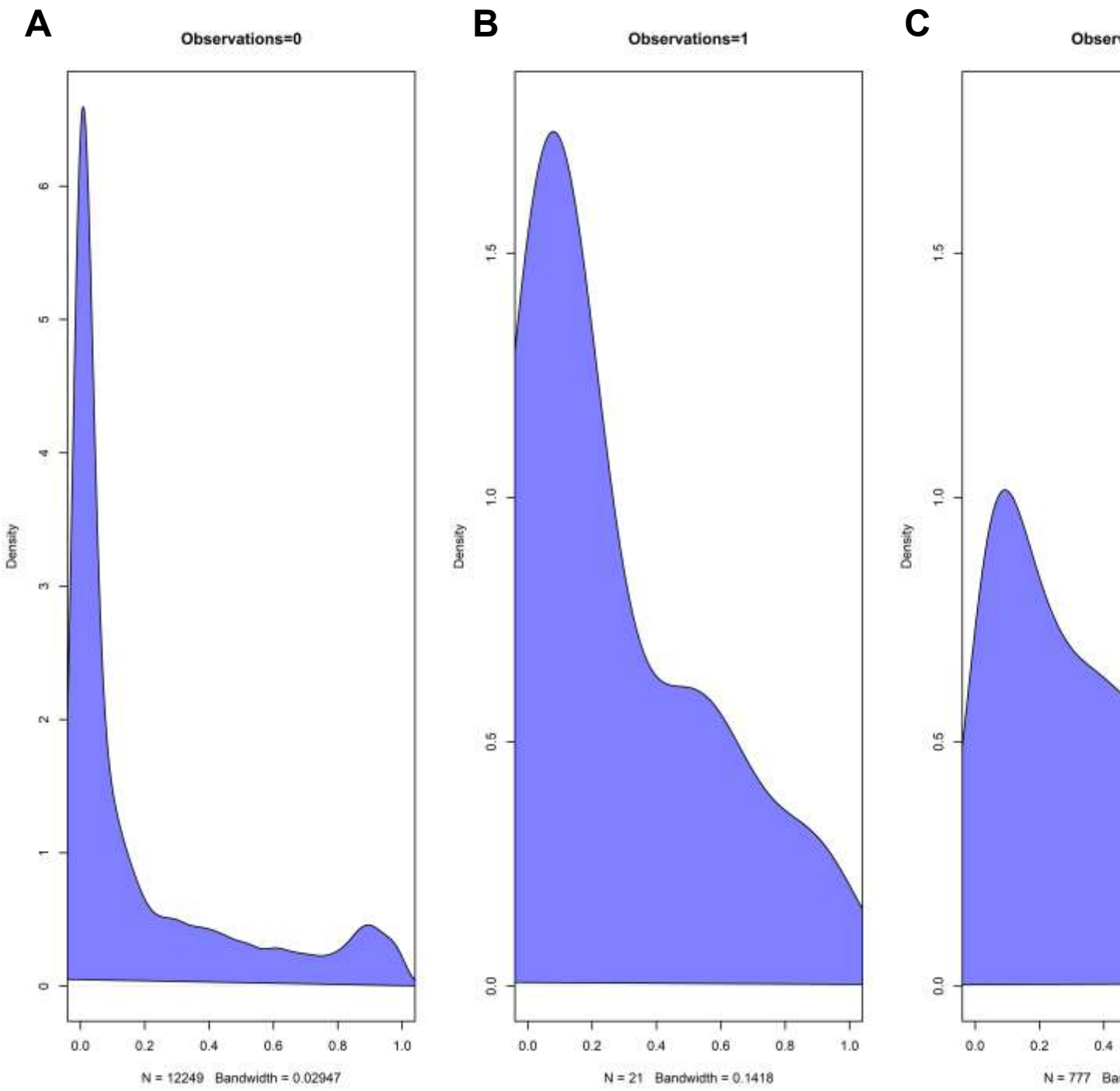

Figure 13. Logistic regression observed versus posterior plots for Mullet Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

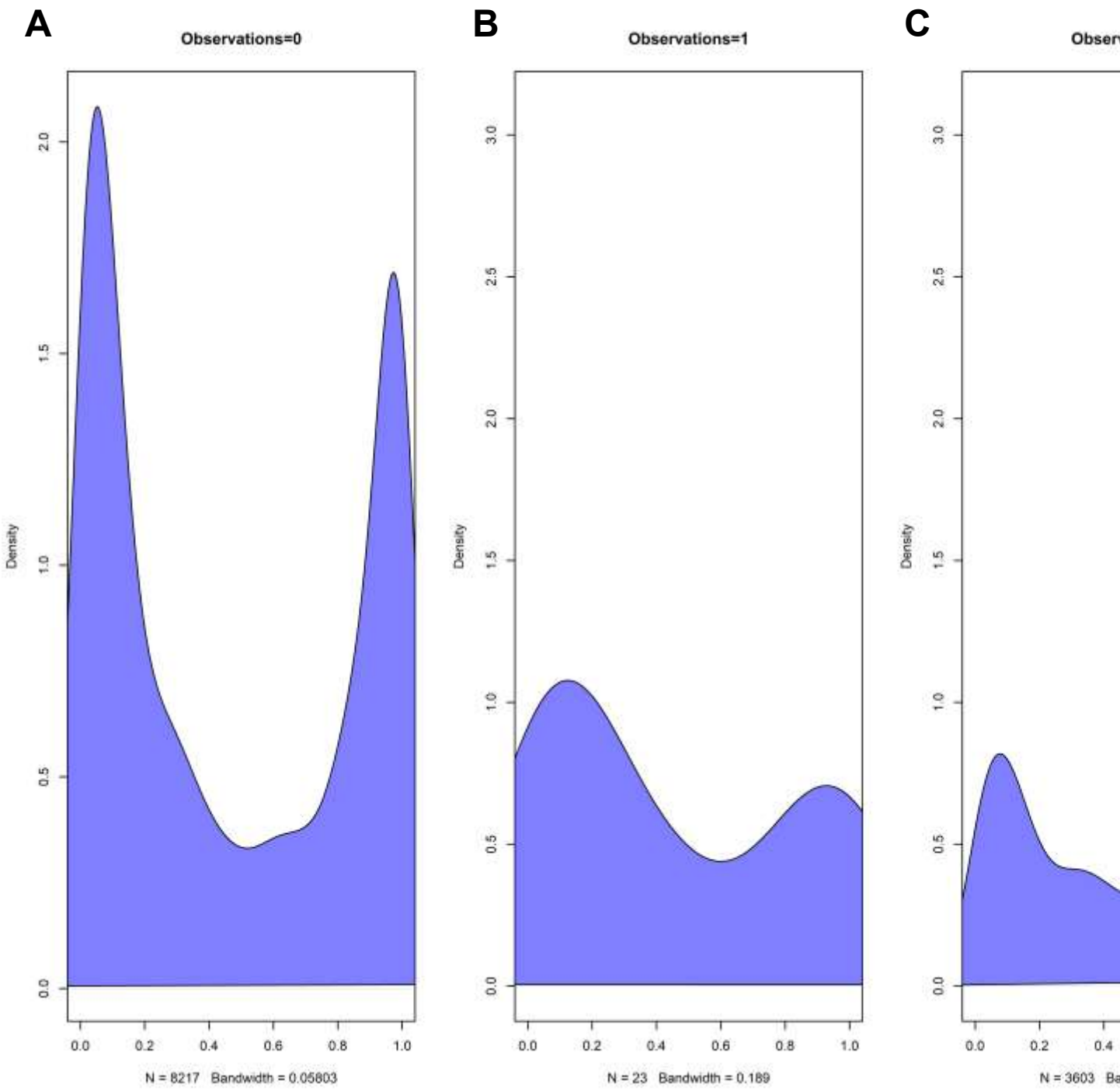

Figure 14. Logistic regression observed versus posterior plots for Mojarra Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

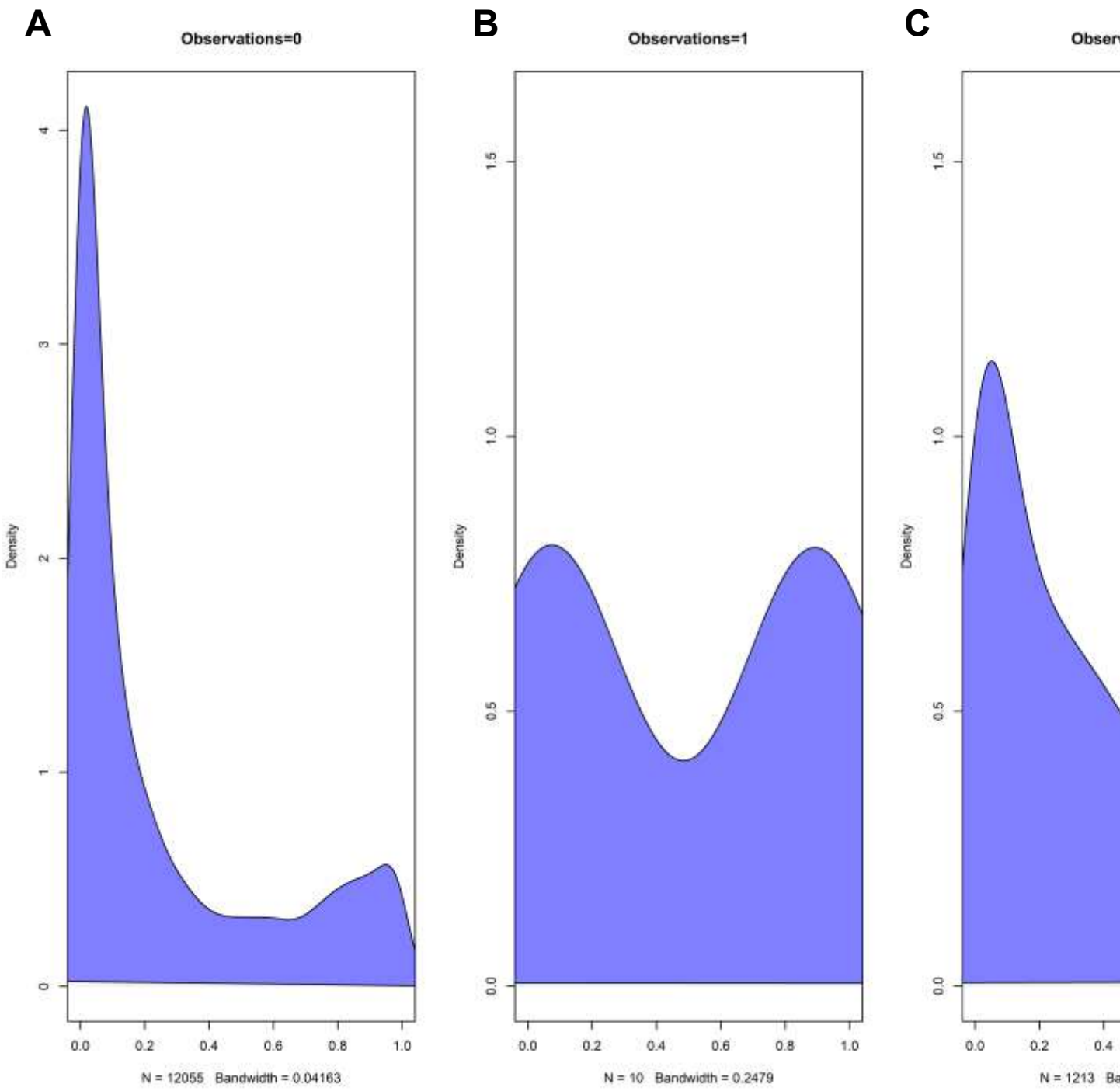

Figure 15. Logistic regression observed versus posterior plots for Silver Perch Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

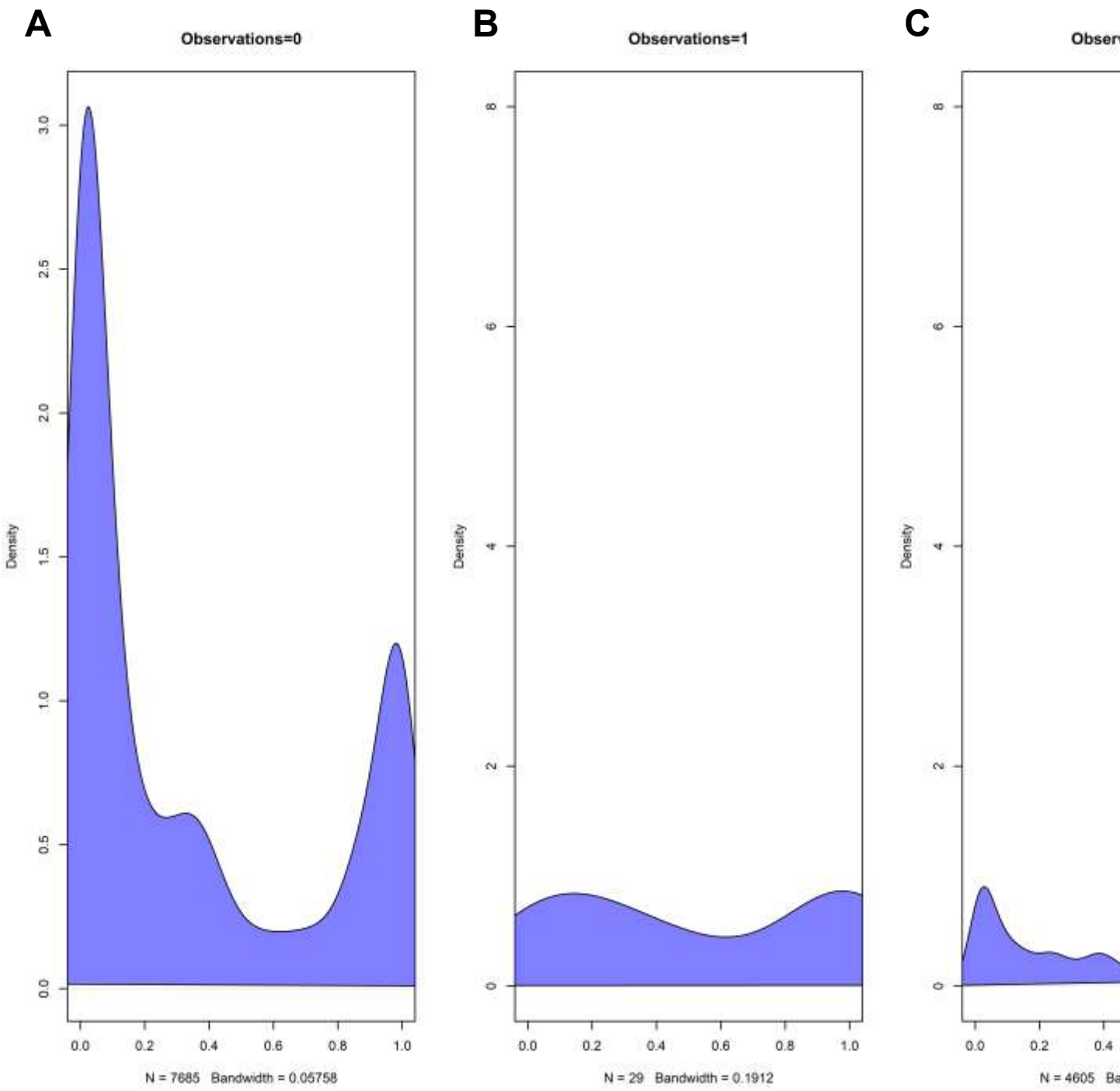

Figure 16. Logistic regression observed versus posterior plots for Pinfish Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

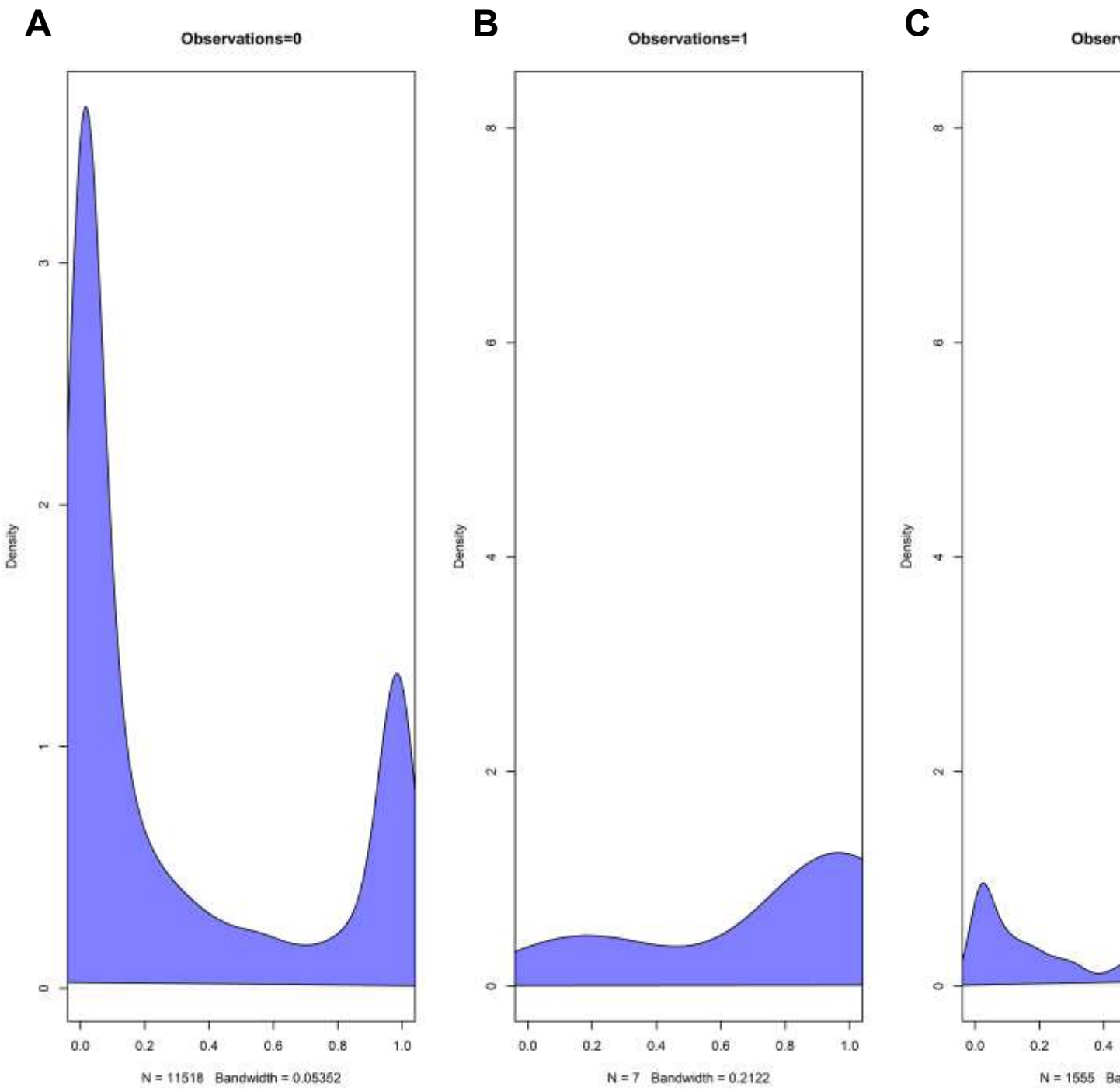

Figure 17. Logistic regression observed versus posterior plots for Pigfish Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

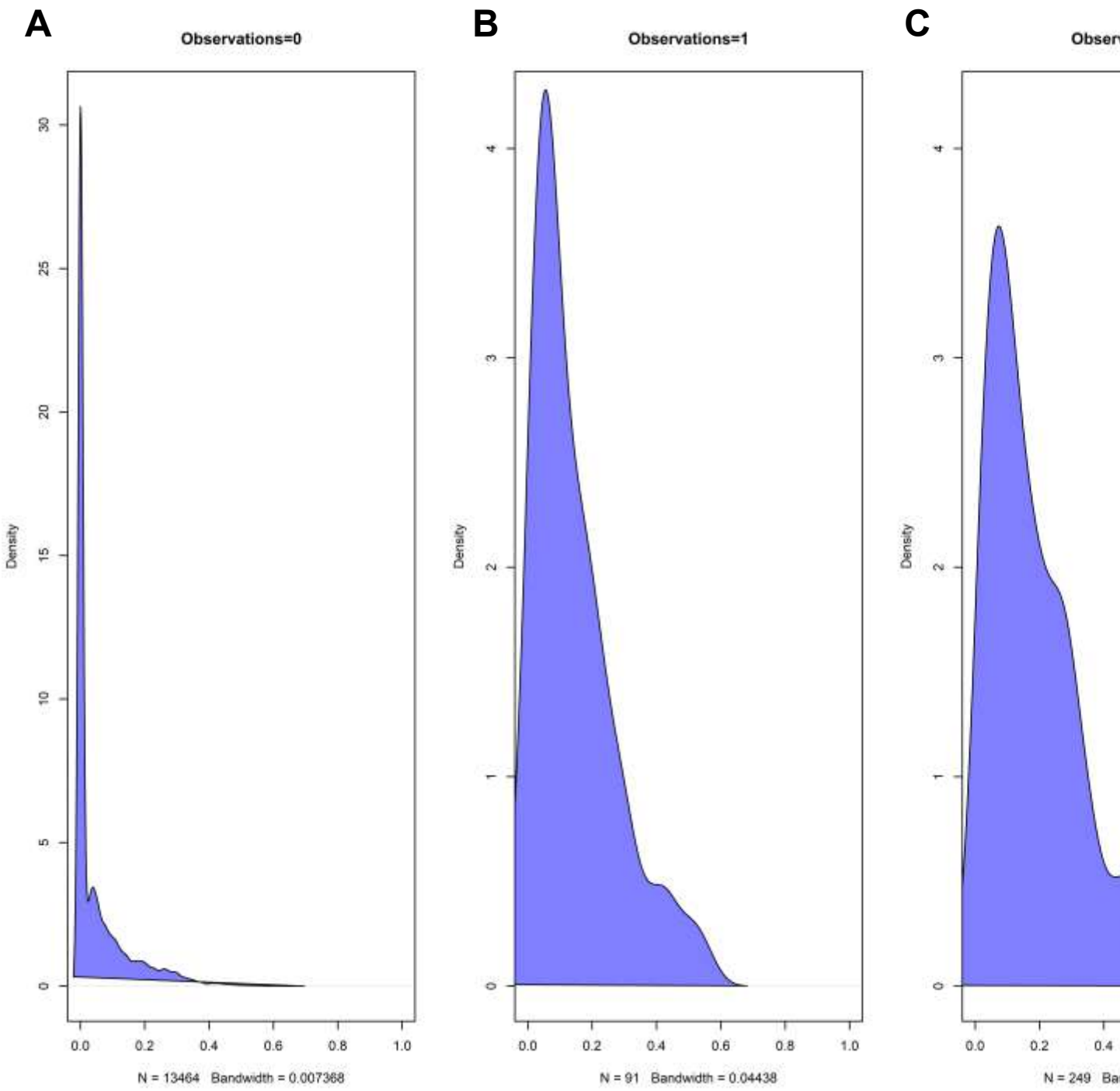

Figure 18. Logistic regression observed versus posterior plots for Silversides Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

A B C

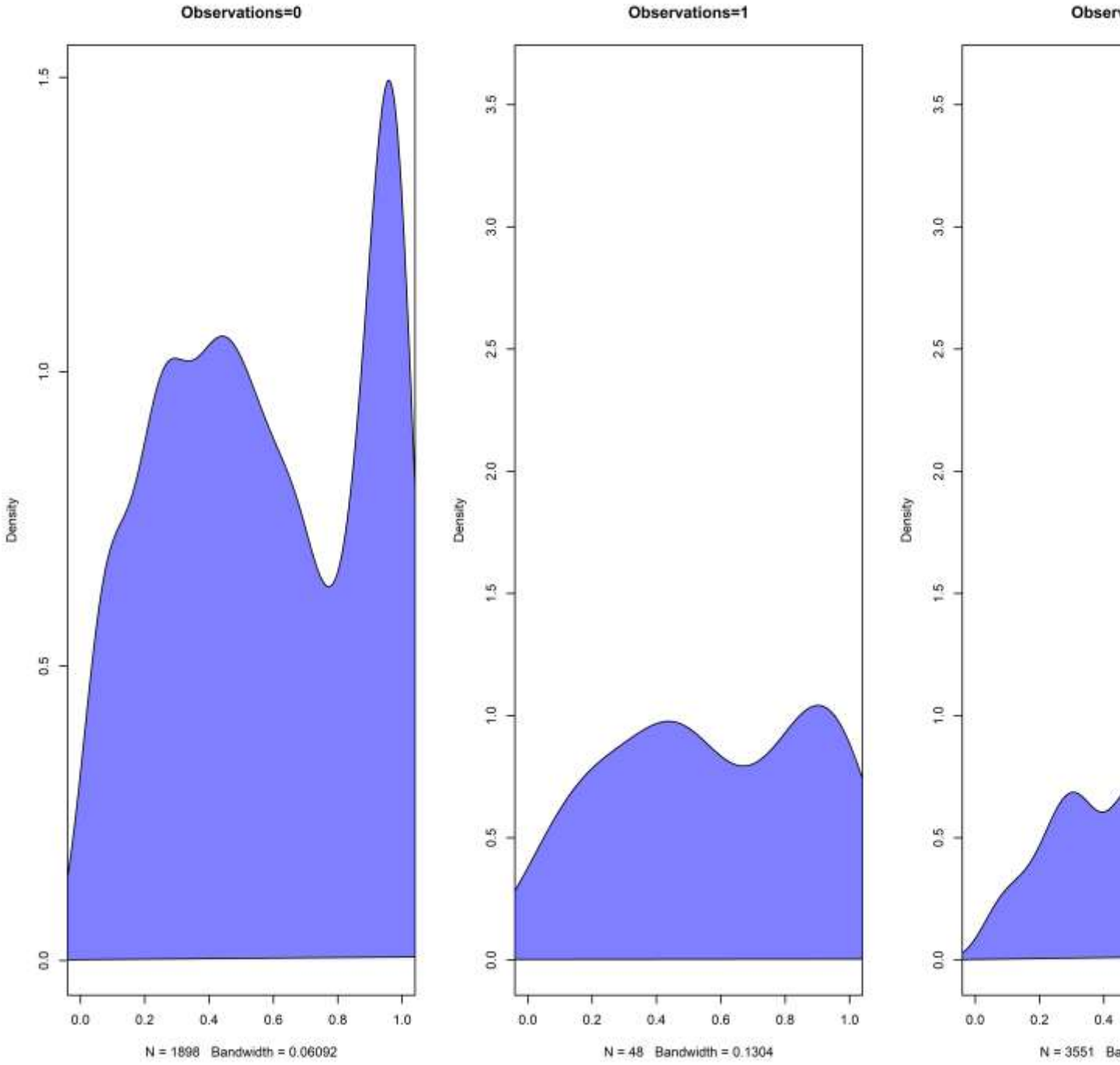

Figure 19. Logistic regression observed versus posterior plots for Grouped Prey Fish Gear 1 Only Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

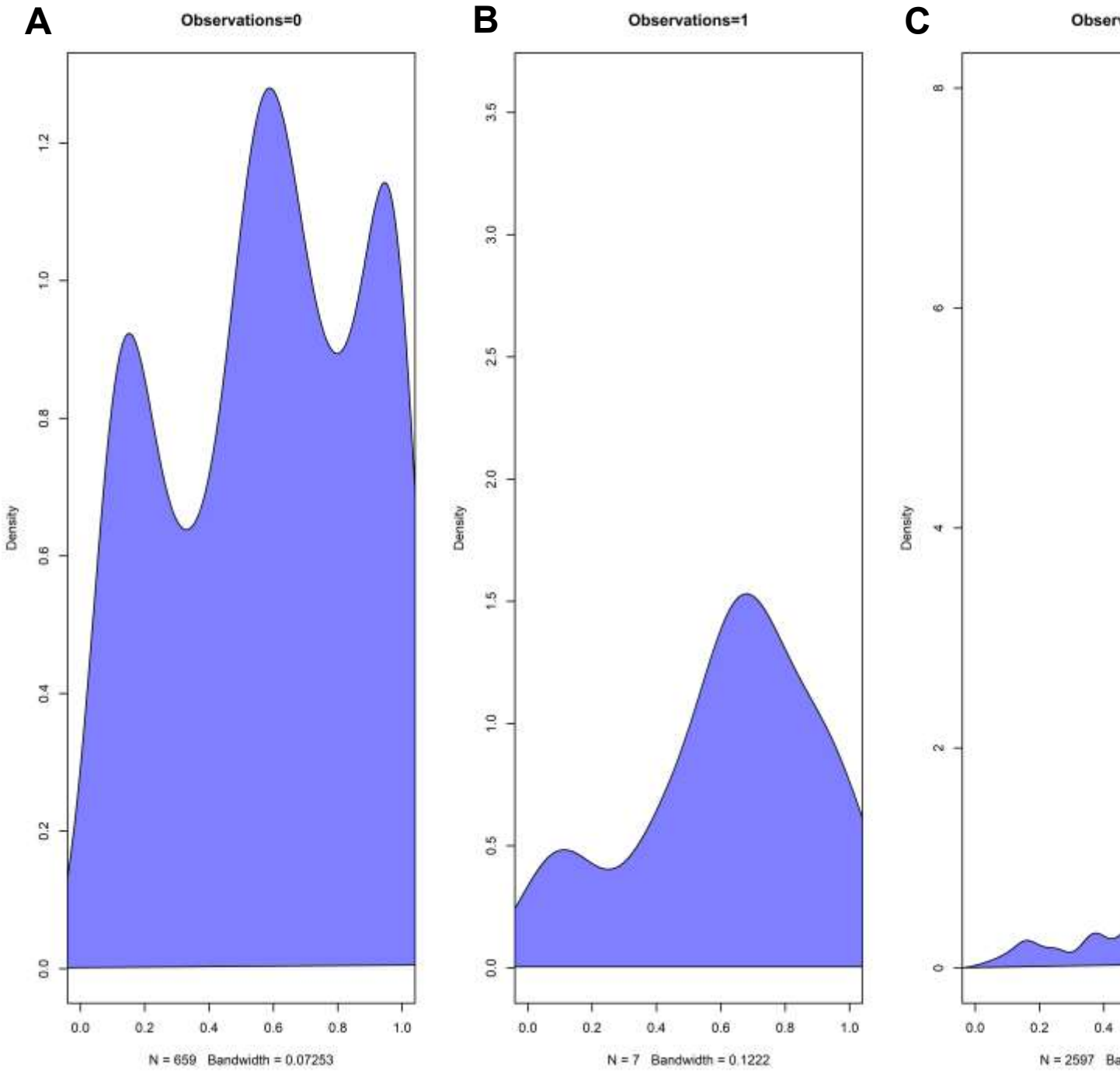

Figure 20. Logistic regression observed versus posterior plots for Grouped Prey Fish Gear 2 Only Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

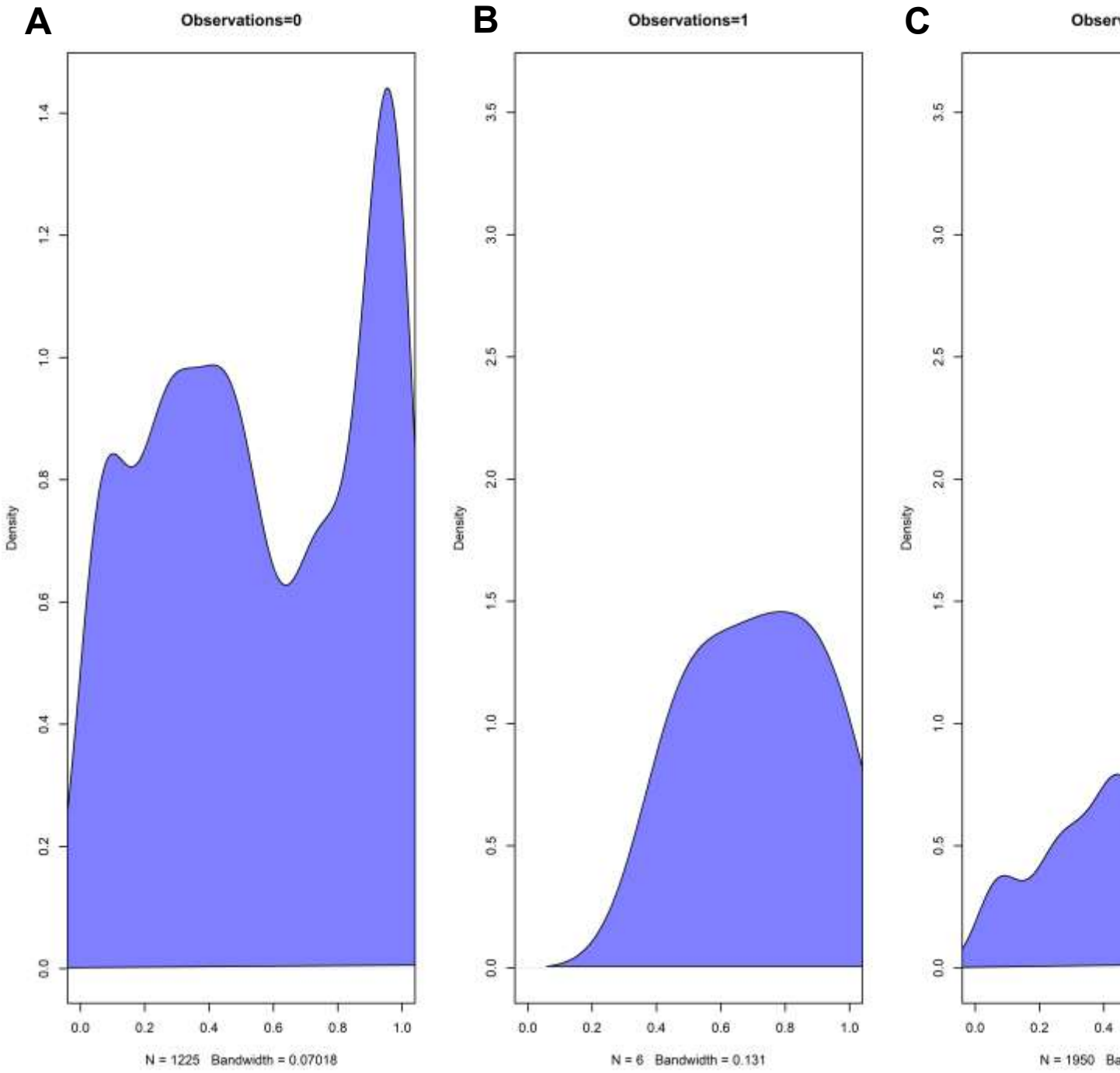

Figure 21. Logistic regression observed versus posterior plots for Grouped Prey Fish Gear 3 Only Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

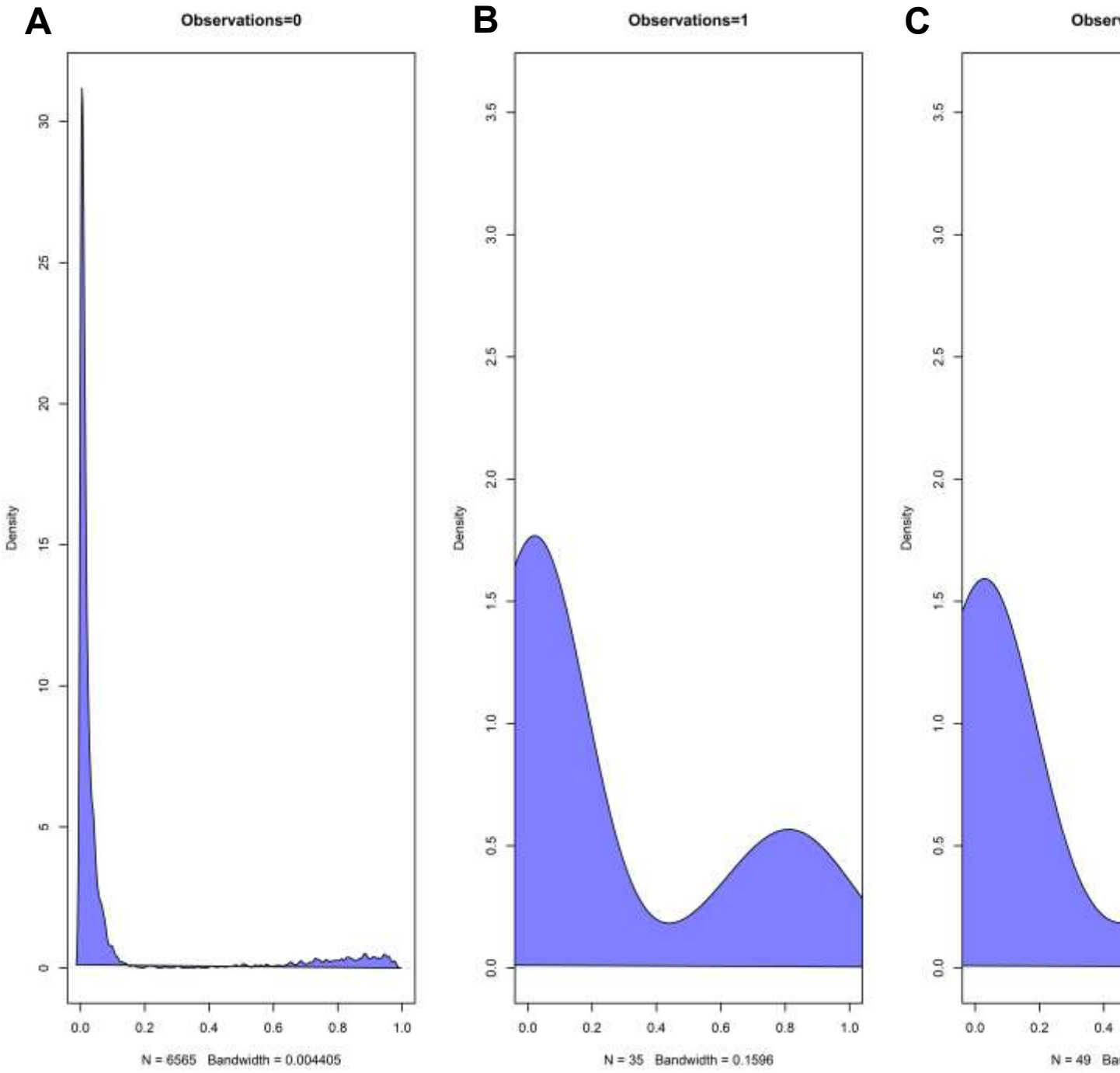

Figure 22. Logistic regression observed versus posterior plots for Herring Gear 1 Only Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

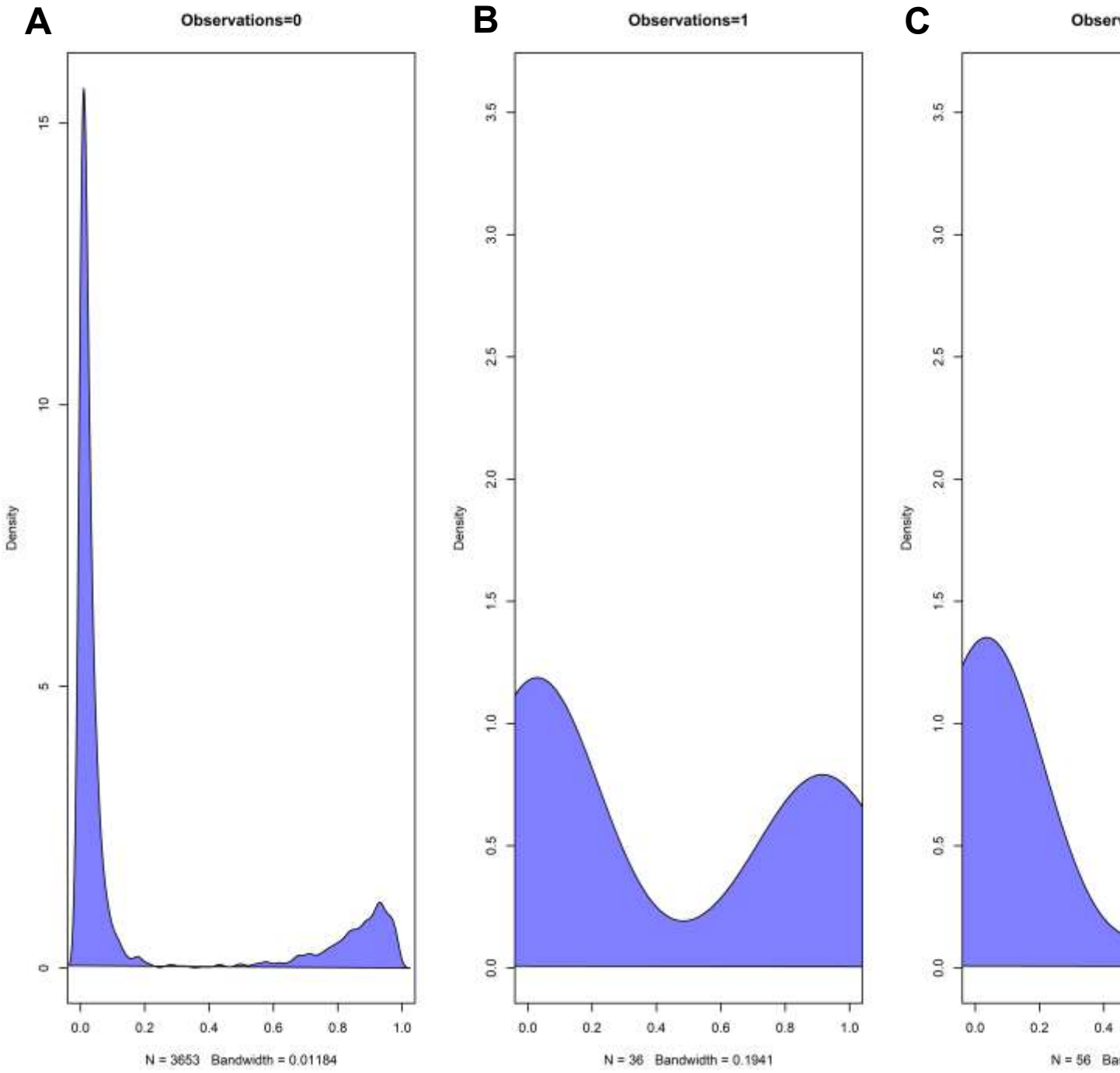

Figure 23. Logistic regression observed versus posterior plots for Herring Gear 2 Only Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

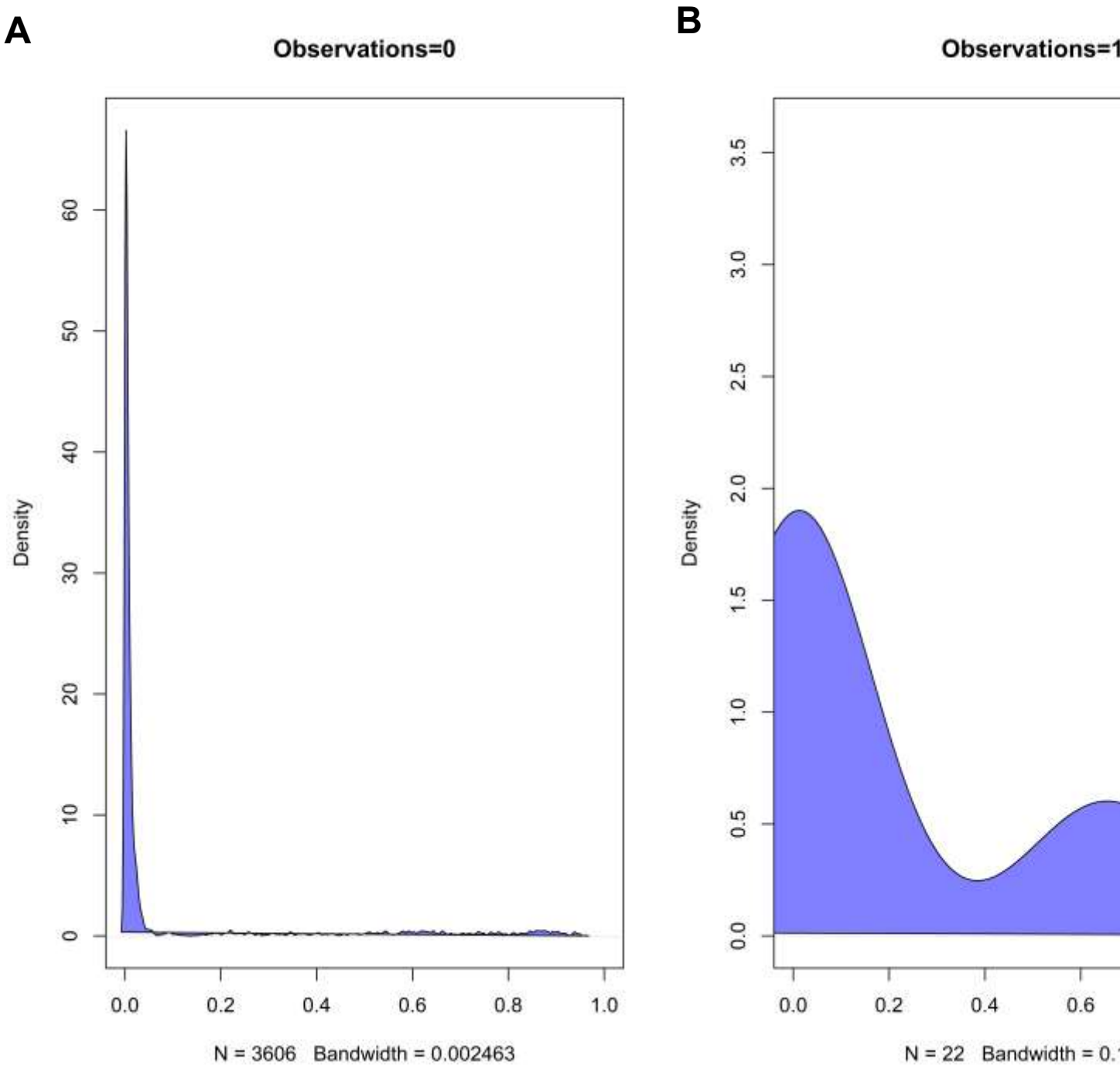

Figure 24. Logistic regression observed versus posterior plots for Herring Gear 3 Only Model 2C for (A) 0 observations and (B) 1 observation. There were no multiple observations of more than 20 individuals for Herring collected by Gear 3 (otter trawl).

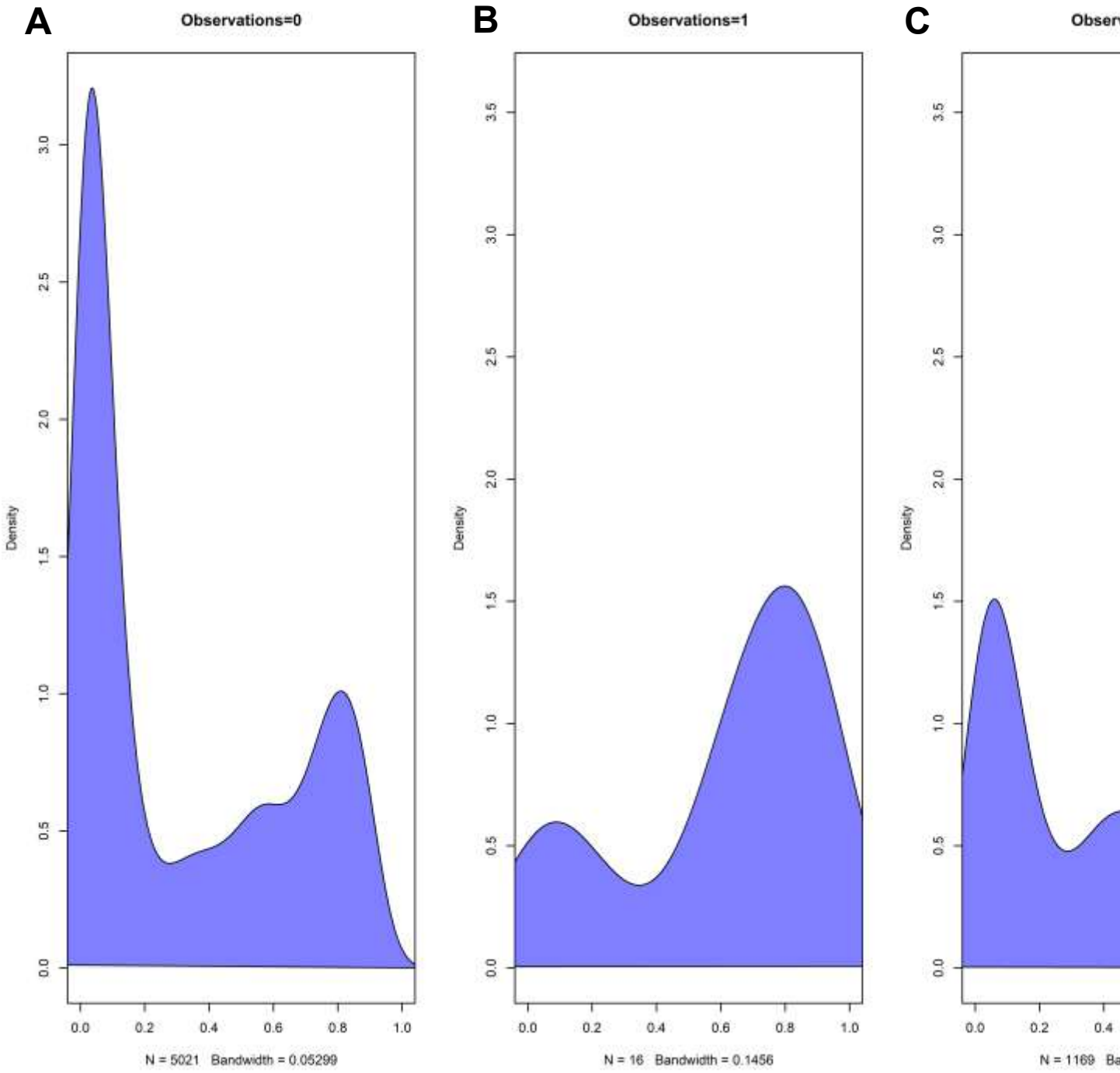

Figure 25. Logistic regression observed versus posterior plots for Anchovy Gear 1 Only Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

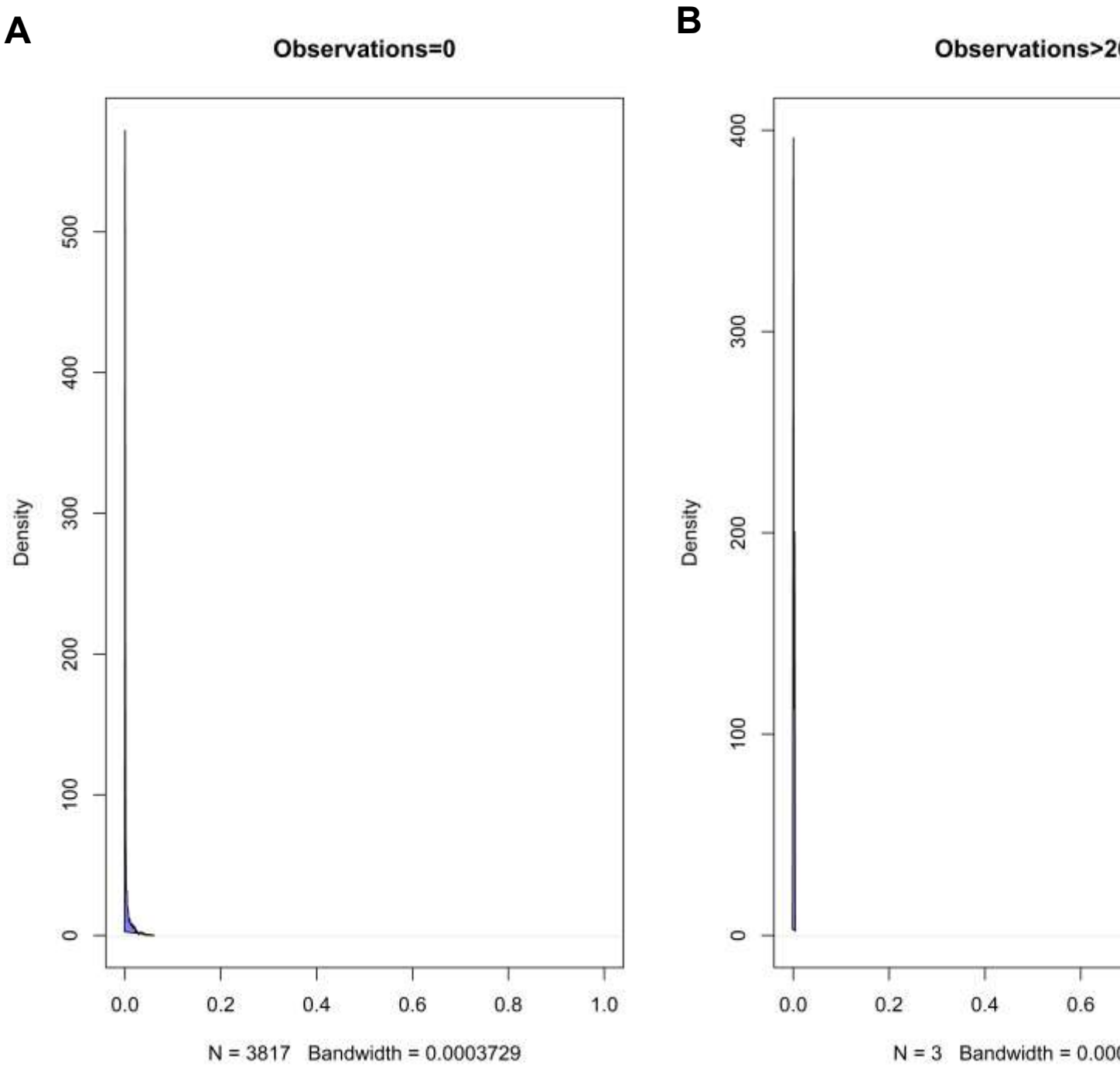

Figure 26. Logistic regression observed versus posterior plots for Anchovy Gear 2 Only Model 2C for (A) 0 observations and (B) more than 20 observations. There were no multiple observations of only 1 individual for Anchovy collected by Gear 2 (large seine).

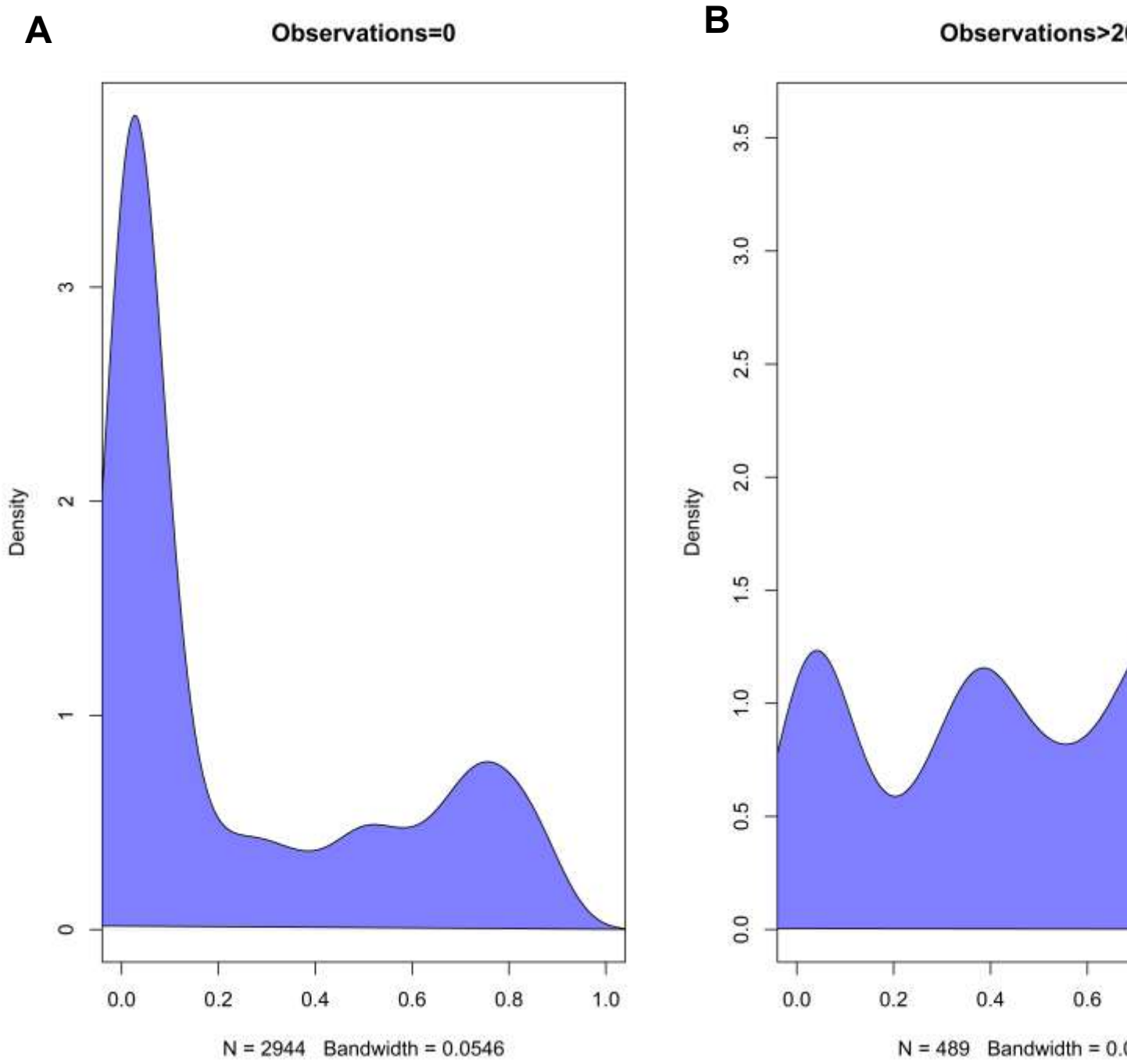

Figure 27. Logistic regression observed versus posterior plots for Anchovy Gear 3 Only Model 2C for (A) 0 observations and (B) more than 20 observations. There were no multiple observations of only 1 individual Anchovy collected by Gear 3 (otter trawl).

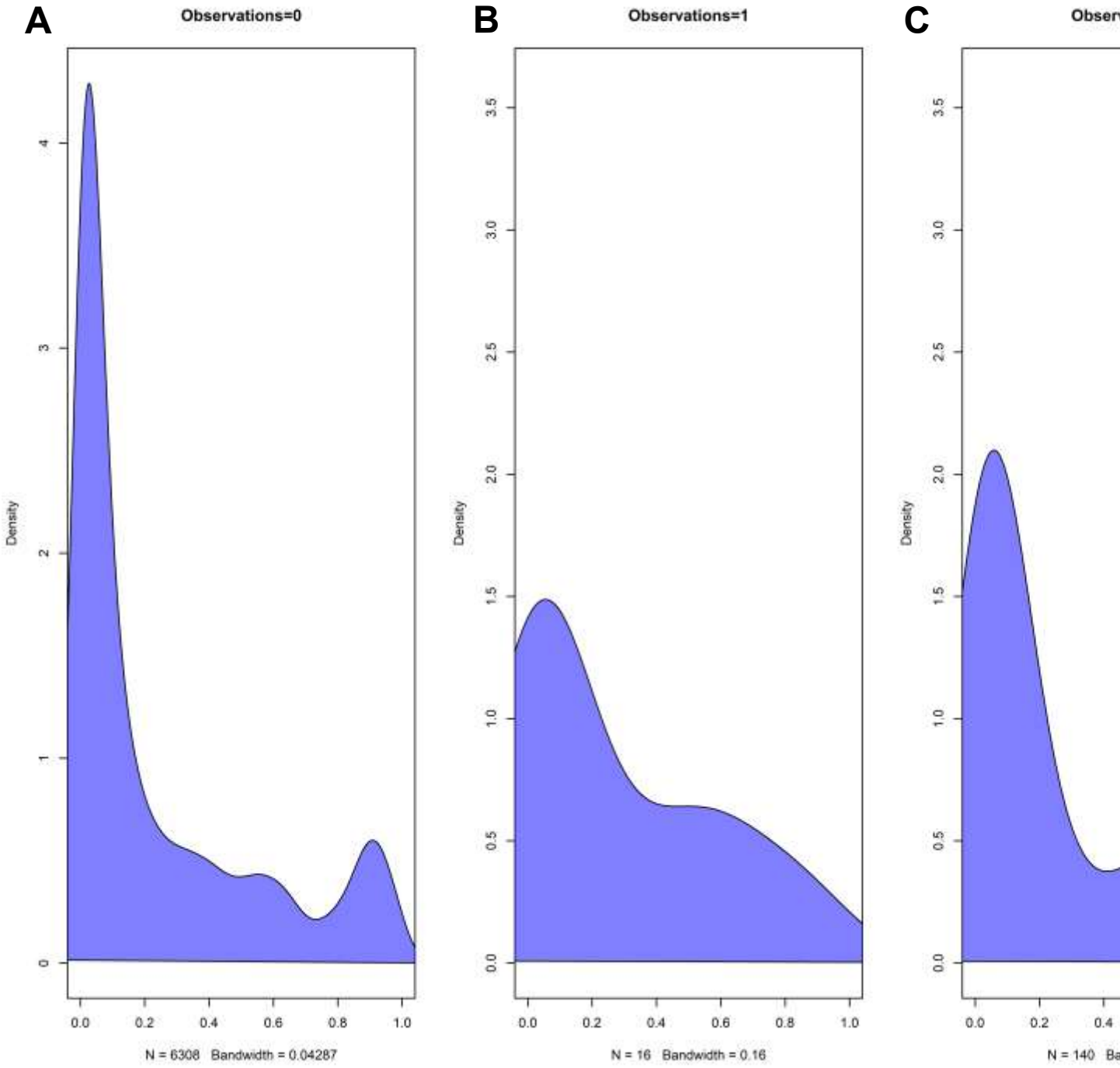

Figure 28. Logistic regression observed versus posterior plots for Mullet Gear 1 Only Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

A B C

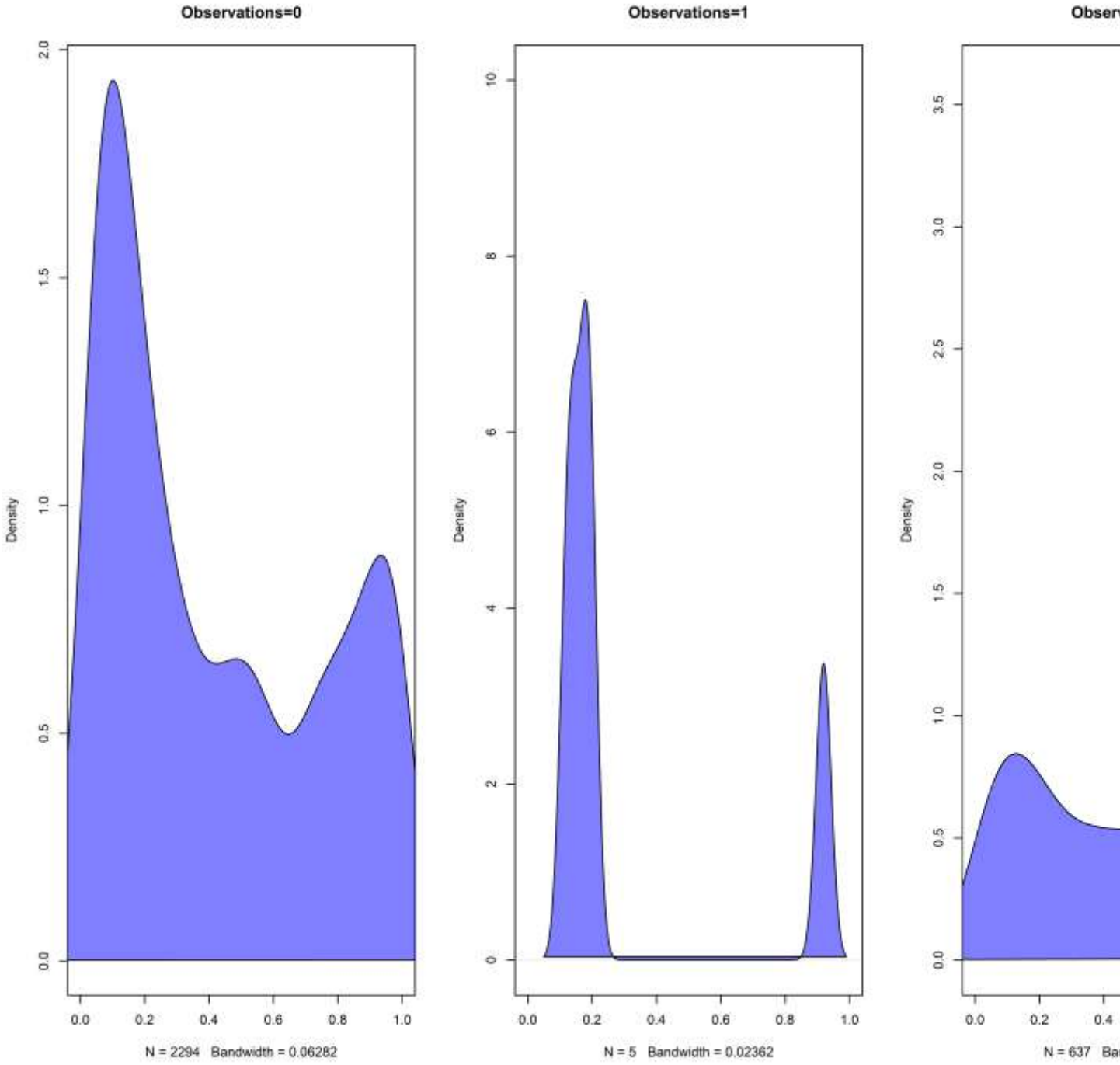

Figure 29. Logistic regression observed versus posterior plots for Mullet Gear 2 Only Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

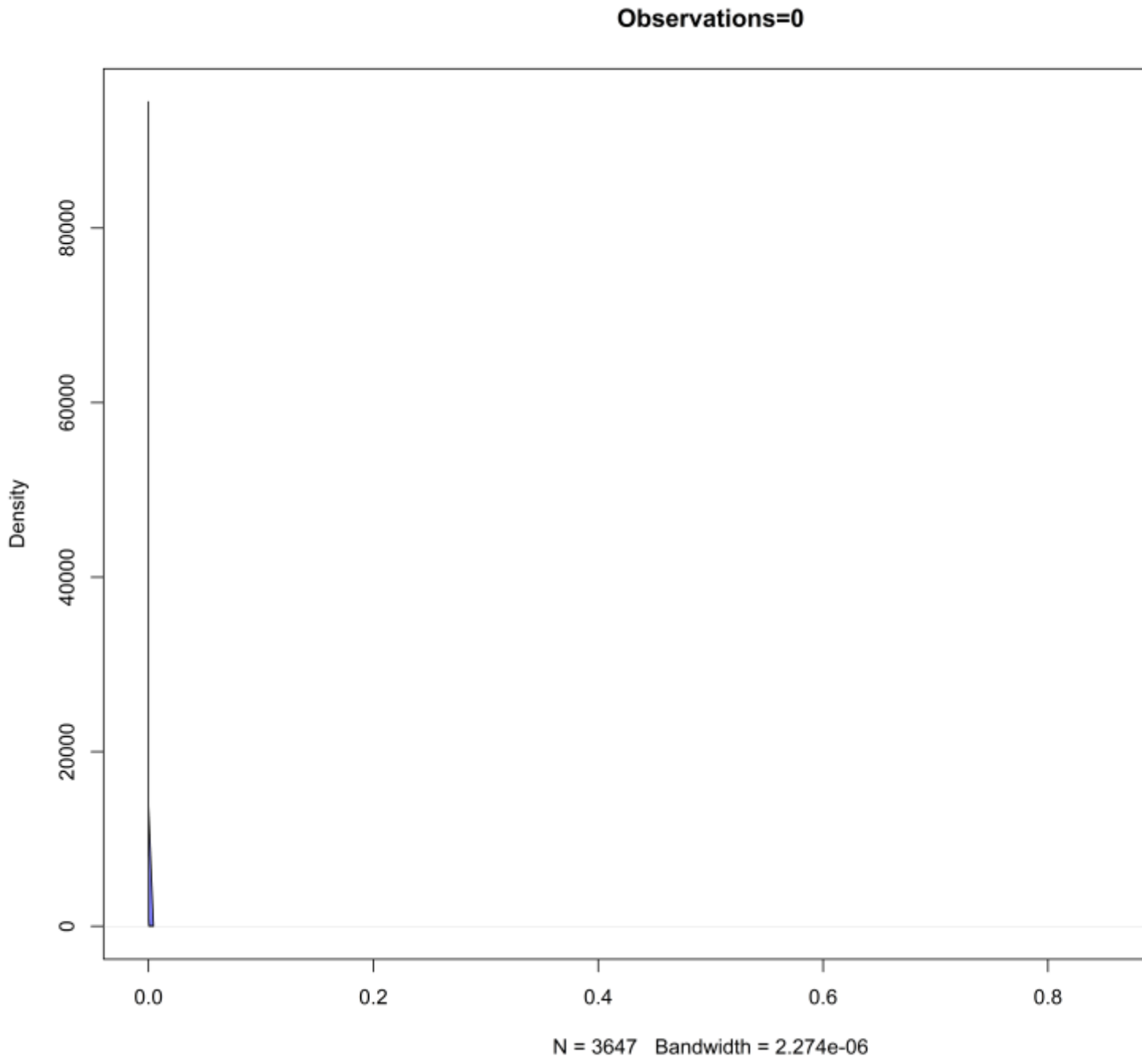

Figure 30. Logistic regression observed versus posterior plots for Mullet Gear 3 Only Model 2C for 0 observations. There were no multiple observations of only 1 individual or more than 20 individuals for Mullet collected by Gear 3 (otter trawl).

A B C

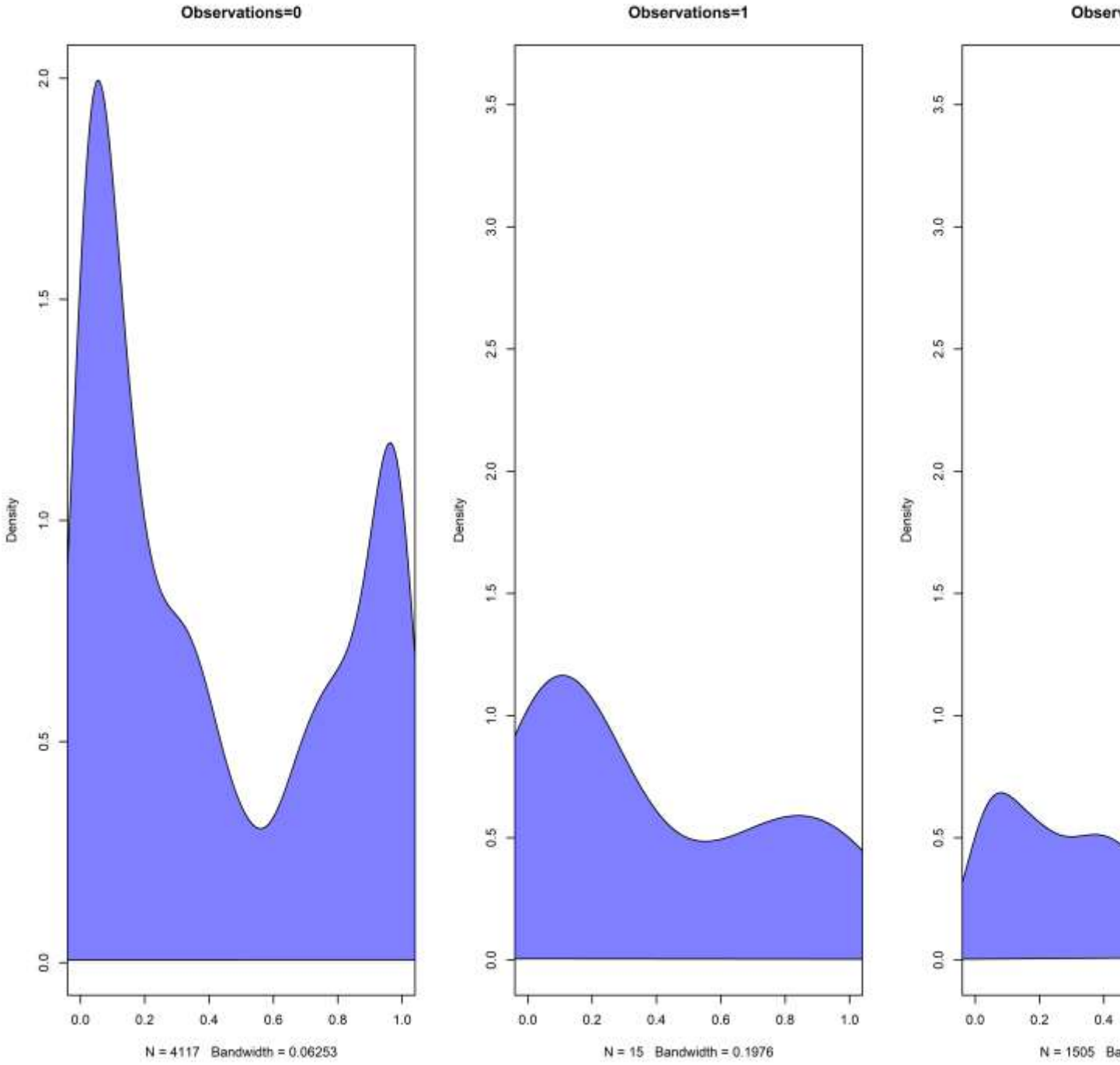

Figure 31. Logistic regression observed versus posterior plots for Mojarra Gear 1 Only Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

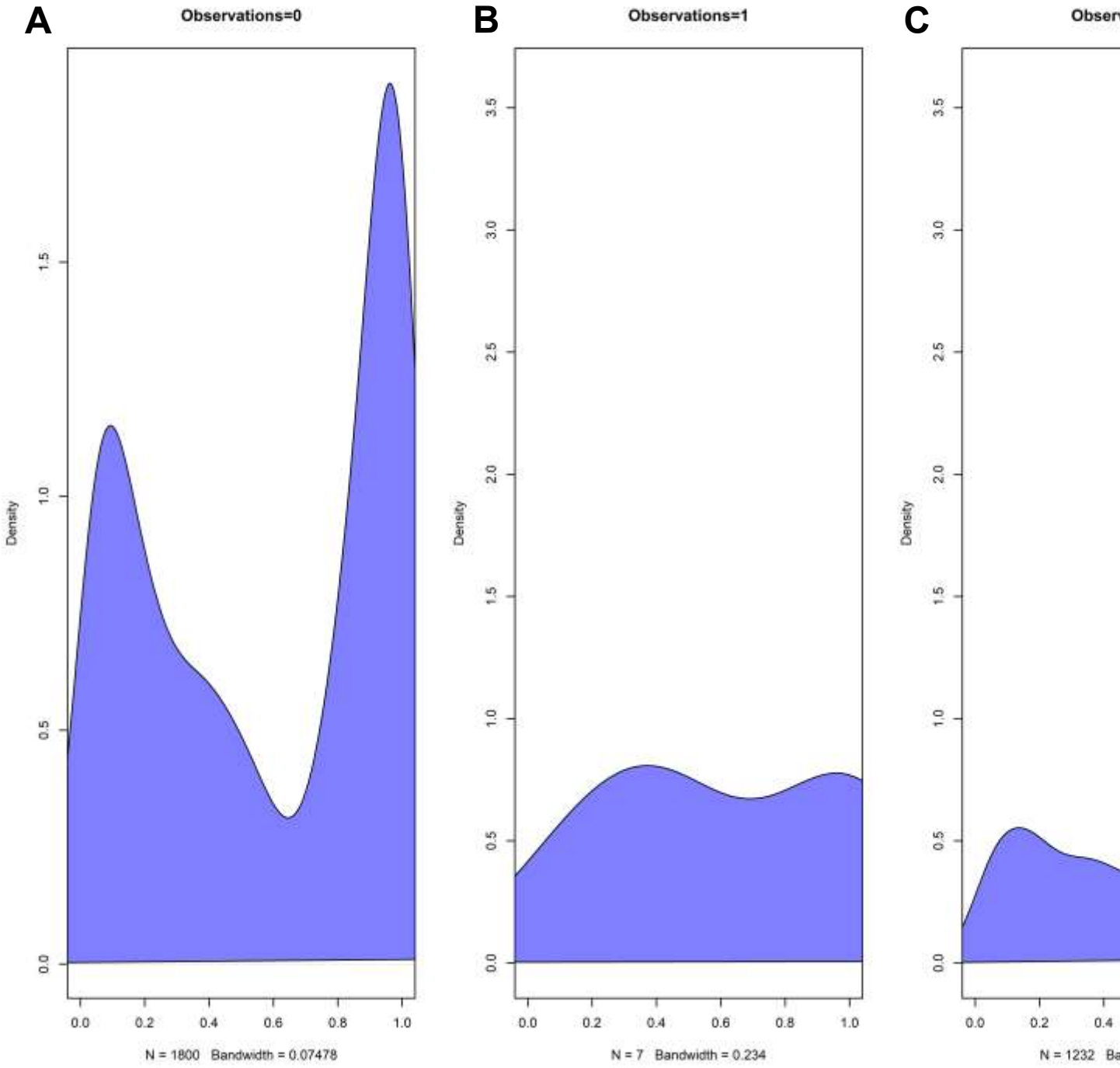

Figure 32. Logistic regression observed versus posterior plots for Mojarra Gear 2 Only Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

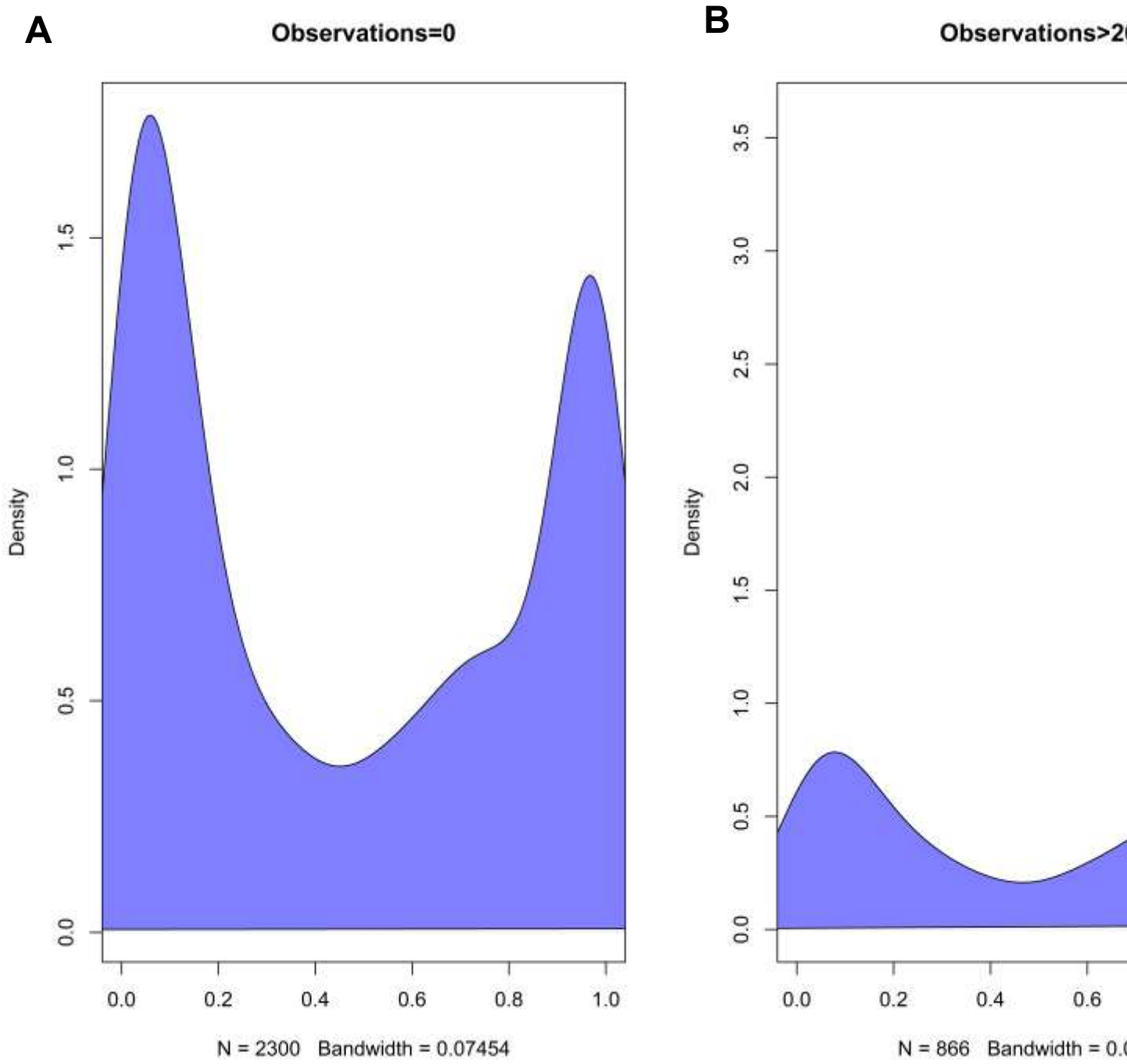

Figure 33. Logistic regression observed versus posterior plots for Mojarra Gear 3 Only Model 2C for (A) 0 observations and (B) more than 20 observations. There were no multiple observations of only 1 individual Mojarra collected by Gear 3 (otter trawl).

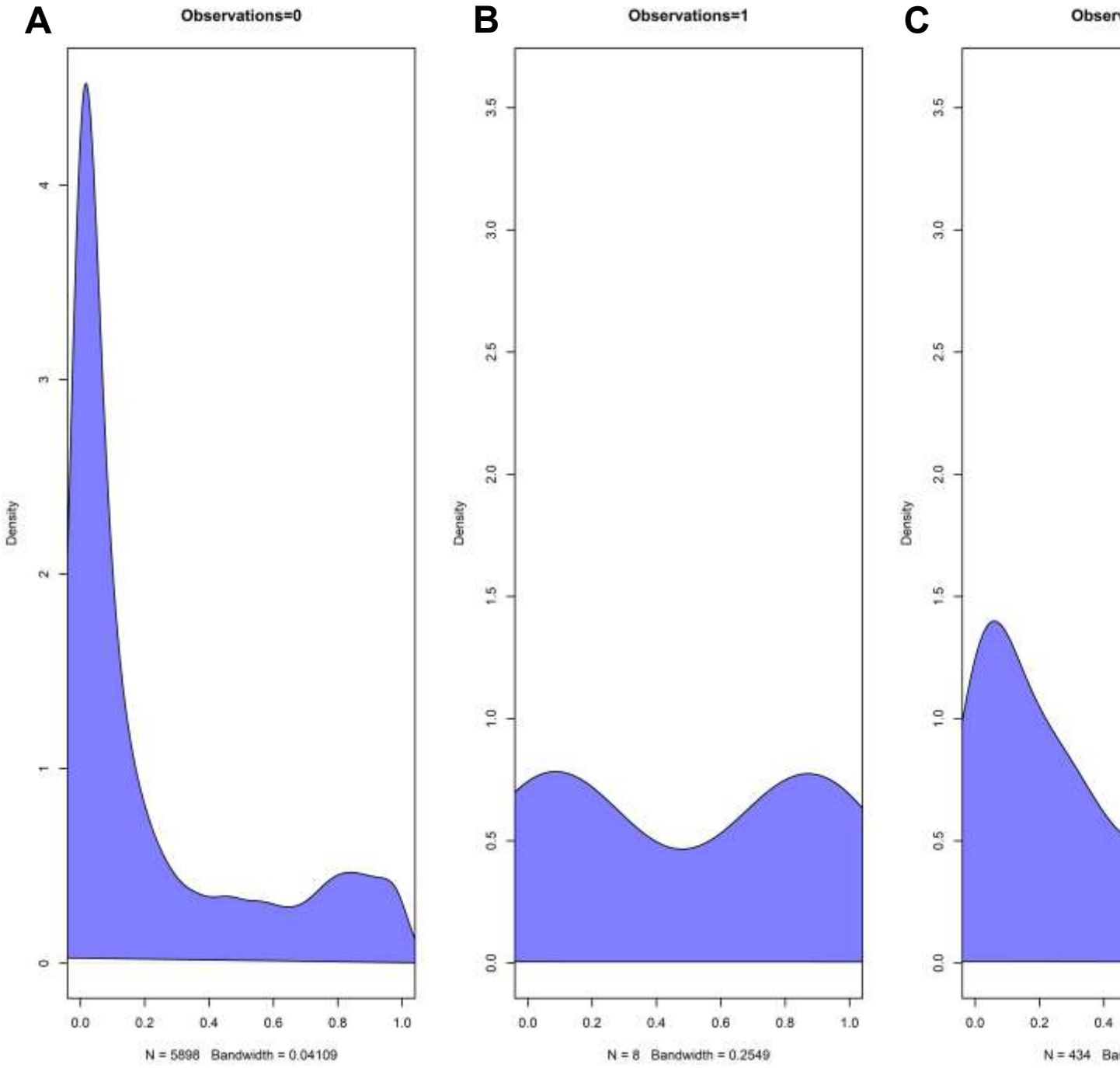

Figure 34. Logistic regression observed versus posterior plots for Silver Perch Gear 1 Only Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

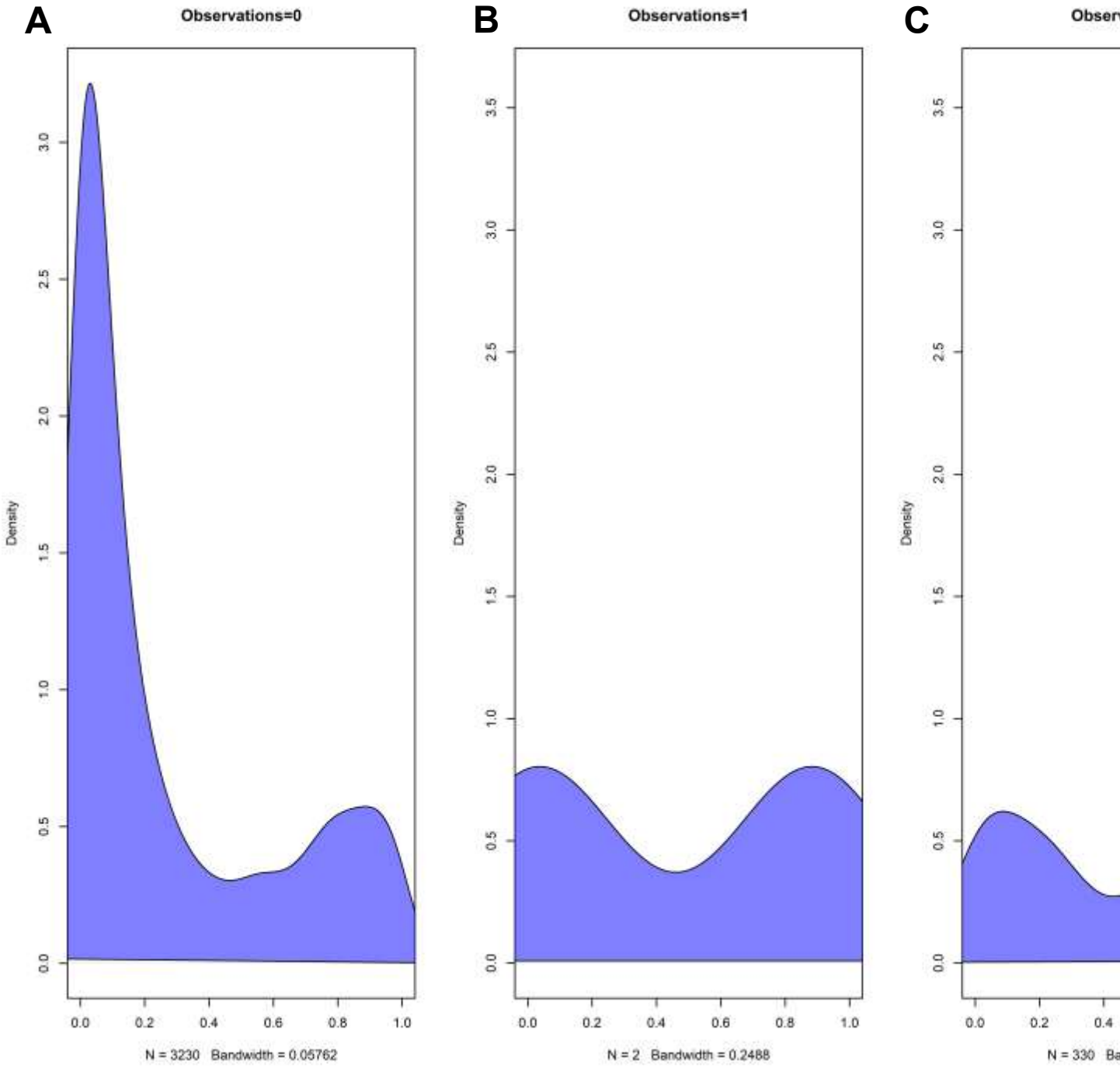

Figure 35. Logistic regression observed versus posterior plots for Silver Perch Gear 2 Only Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

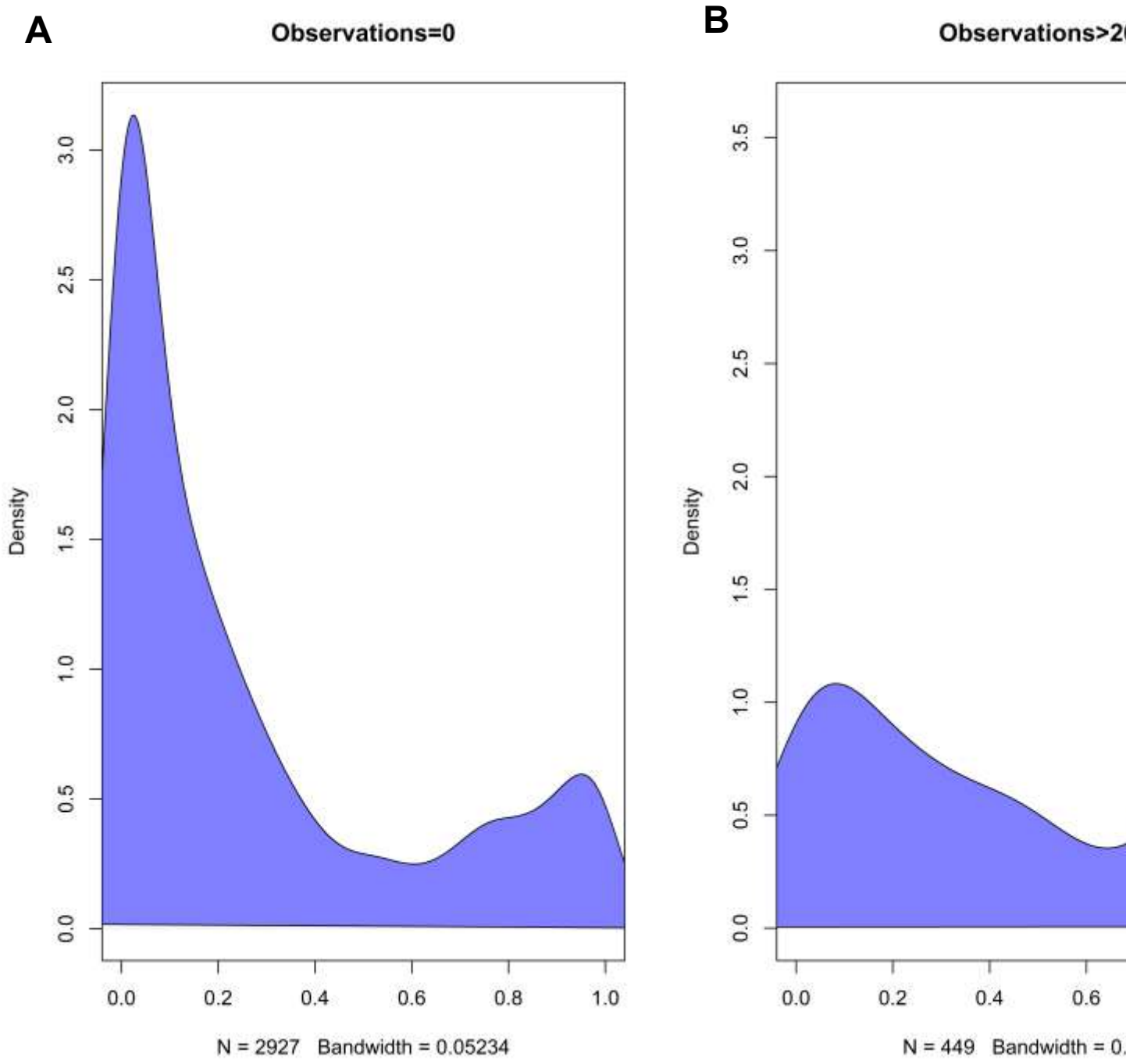

Figure 36. Logistic regression observed versus posterior plots for Silver Perch Gear 3 Only Model 2C for (A) 0 observations and (B) more than 20 observations. There were no multiple observations of only 1 individual Silver Perch collected by Gear 3 (otter trawl).

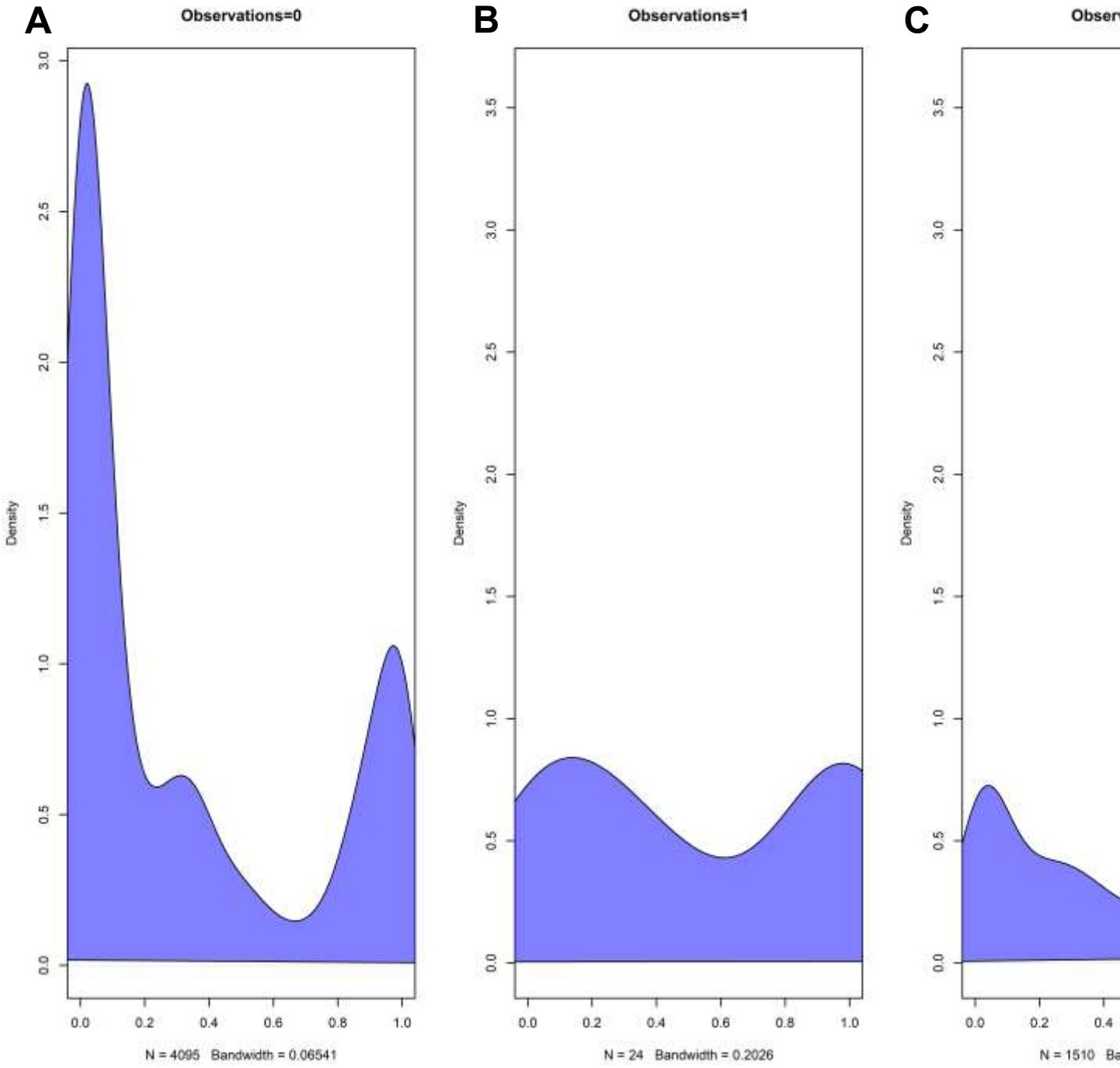

Figure 37. Logistic regression observed versus posterior plots for Pinfish Gear 1 Only Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

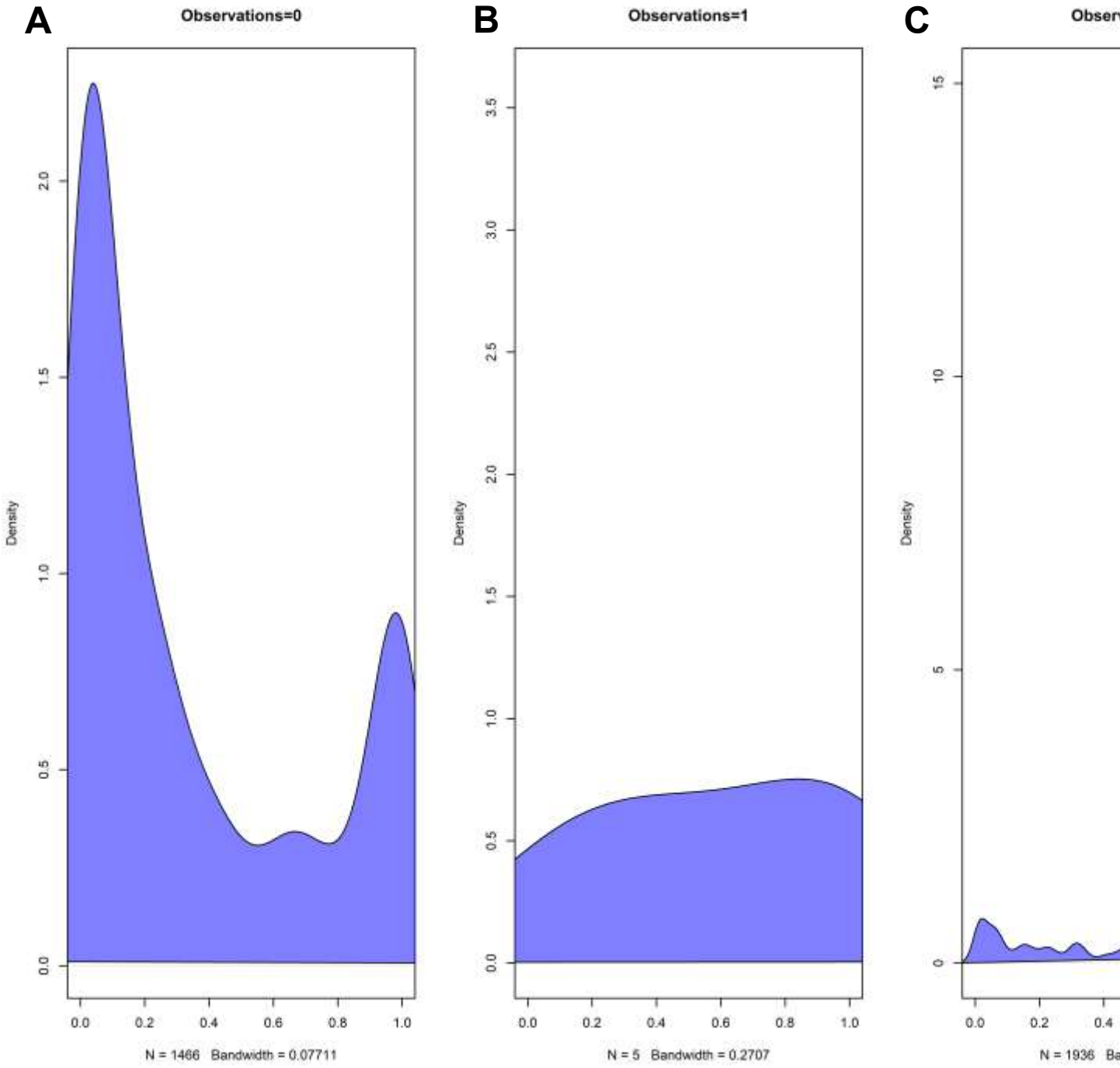

Figure 38. Logistic regression observed versus posterior plots for Pinfish Gear 2 Only Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

A

B

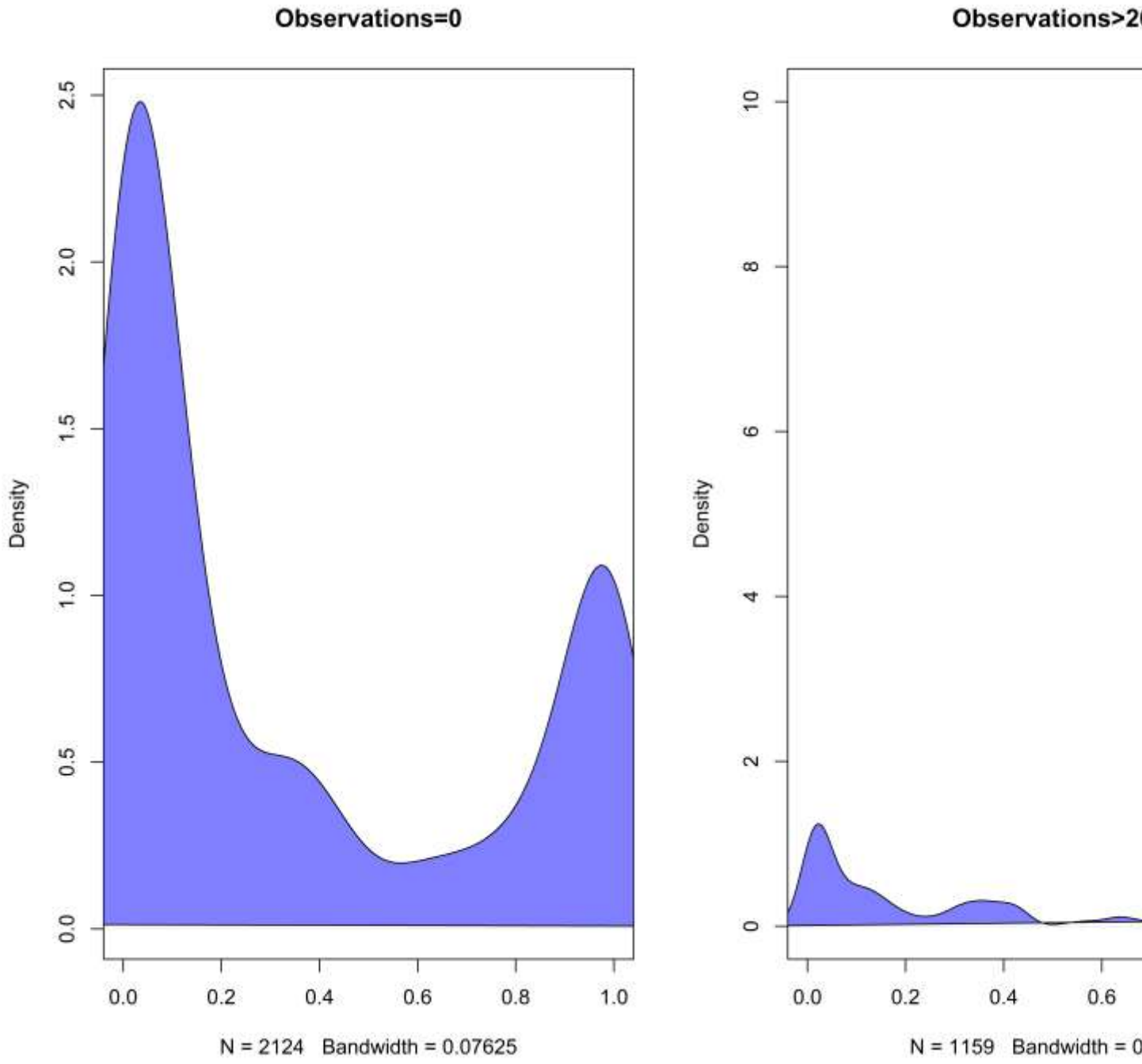

Figure 39. Logistic regression observed versus posterior plots for Pinfish Gear 3 Only Model 2C for (A) 0 observations and (B) more than 20 observations. There were no multiple observations of only 1 individual Pinfish collected by Gear 3 (otter trawl).

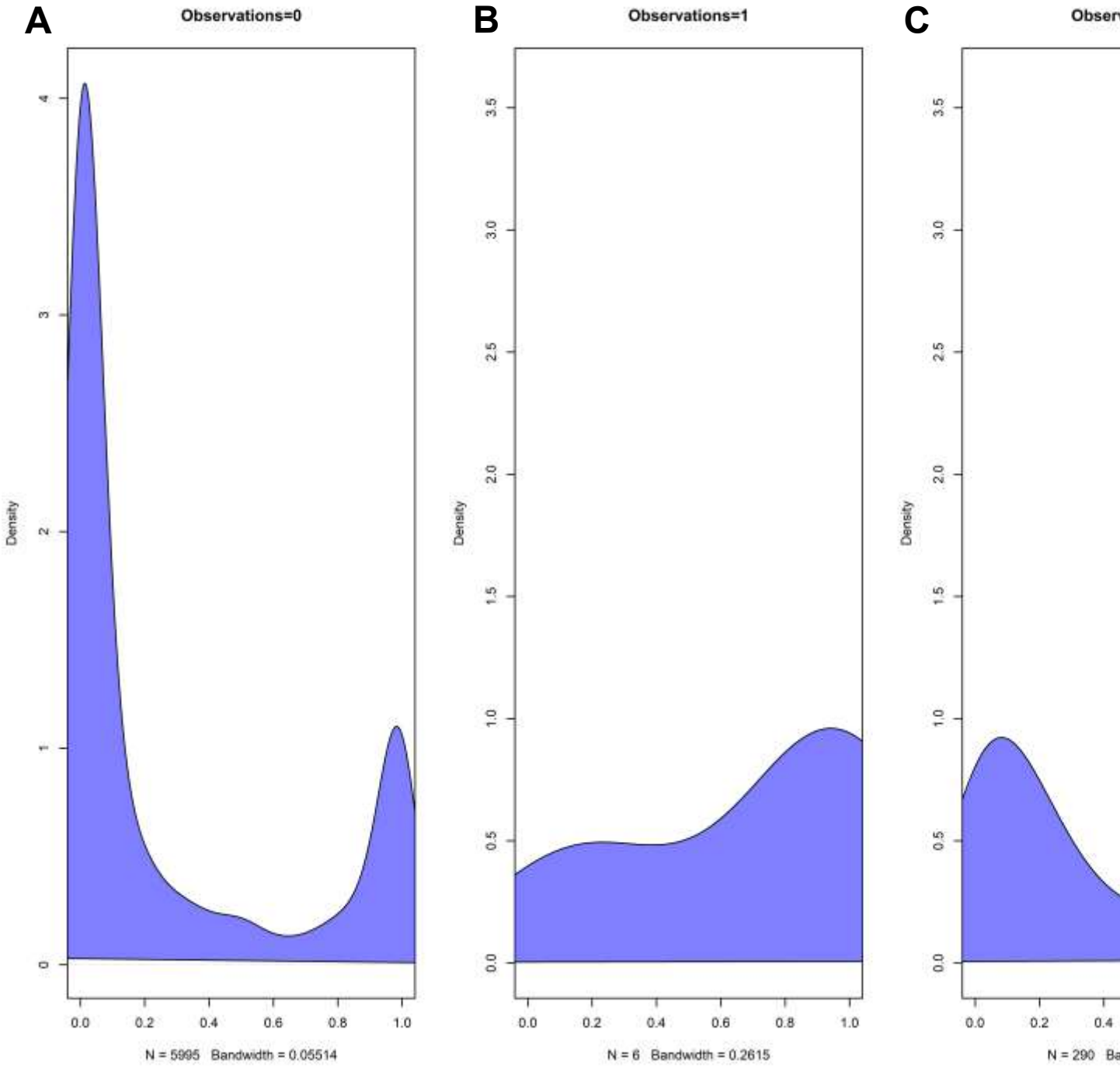

Figure 40. Logistic regression observed versus posterior plots for Pigfish Gear 1 Only Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

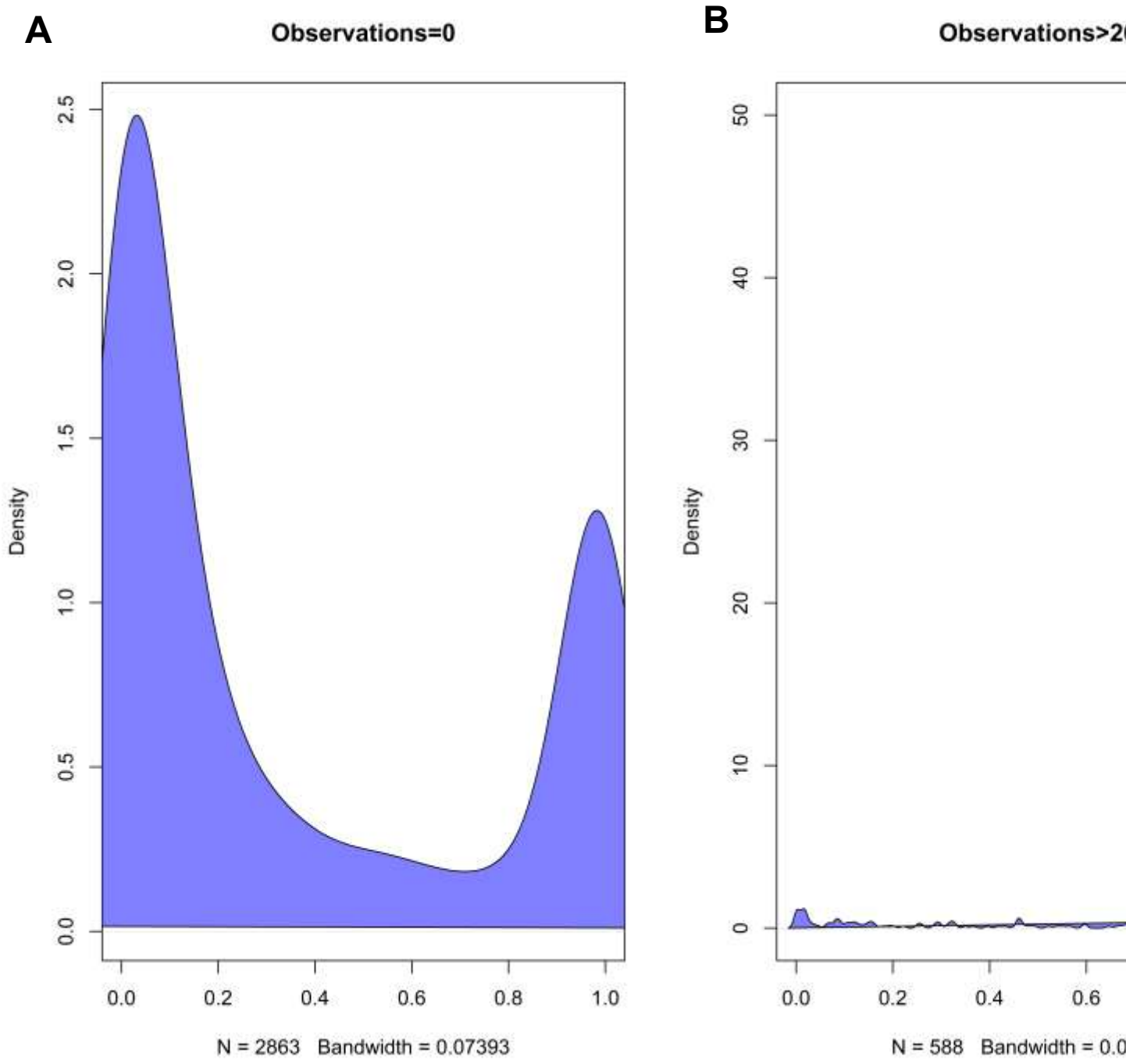

Figure 41. Logistic regression observed versus posterior plots for Pigfish Gear 2 Only Model 2C for (A) 0 observations and (B) more than 20 observations. There were no multiple observations of only 1 individual Pigfish collected by Gear 2 (large seine).

A

B

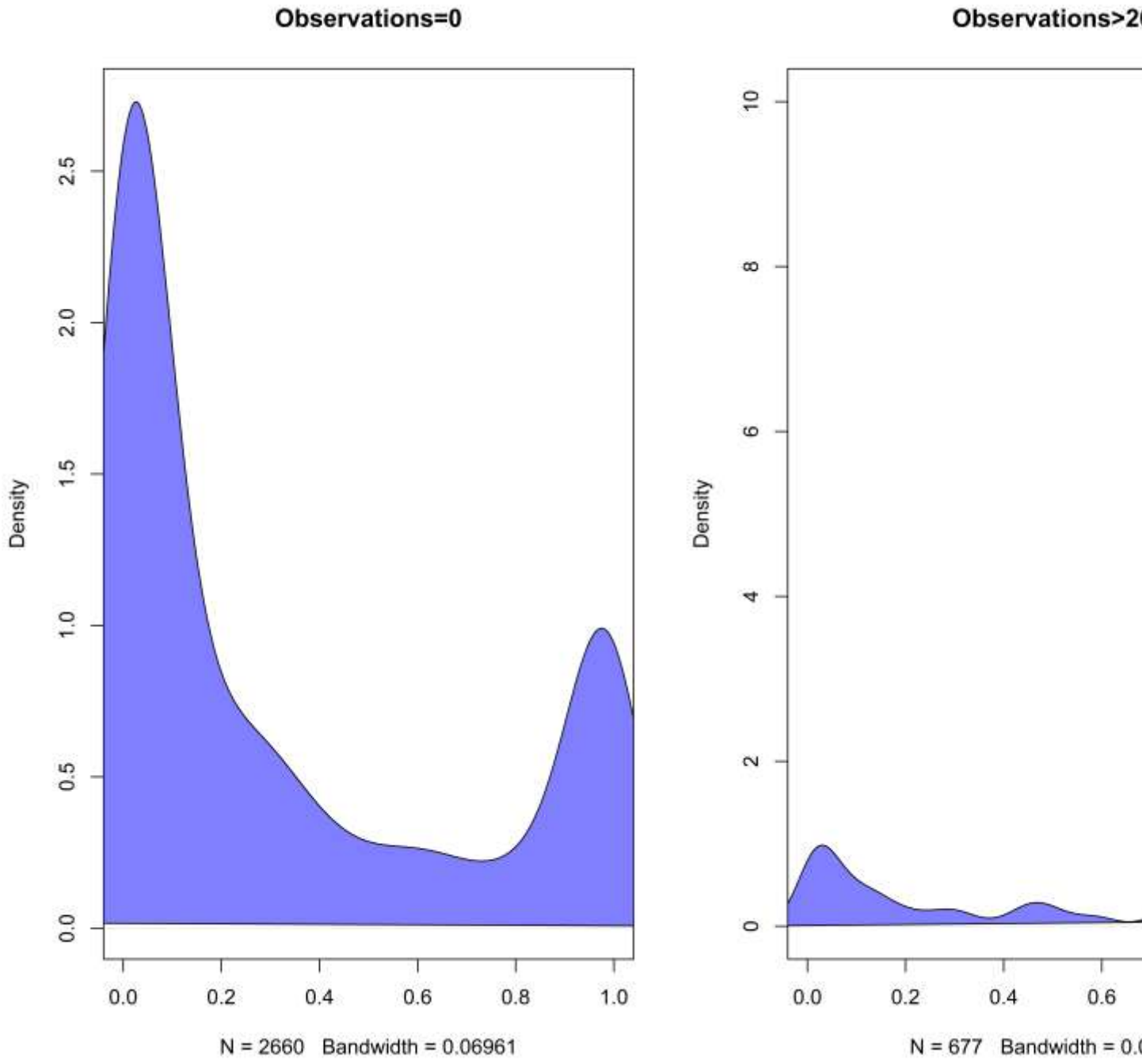

Figure 42. Logistic regression observed versus posterior plots for Pigfish Gear 3 Only Model 2C for (A) 0 observations and (B) more than 20 observations. There were no multiple observations of only 1 individual Pigfish collected by Gear 3 (otter trawl).

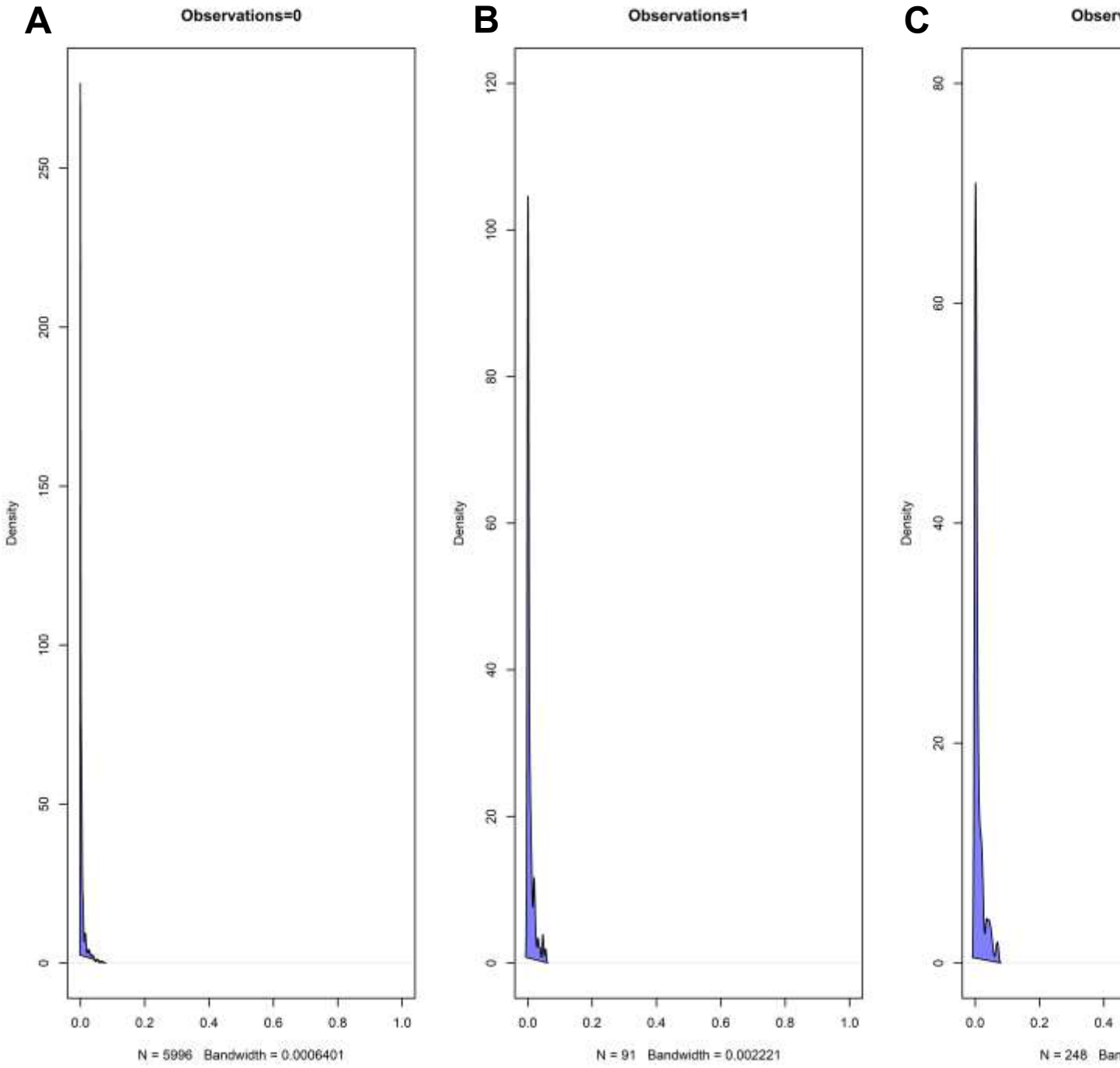

Figure 43. Logistic regression observed versus posterior plots for Silversides Gear 1 Only Model 2C for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

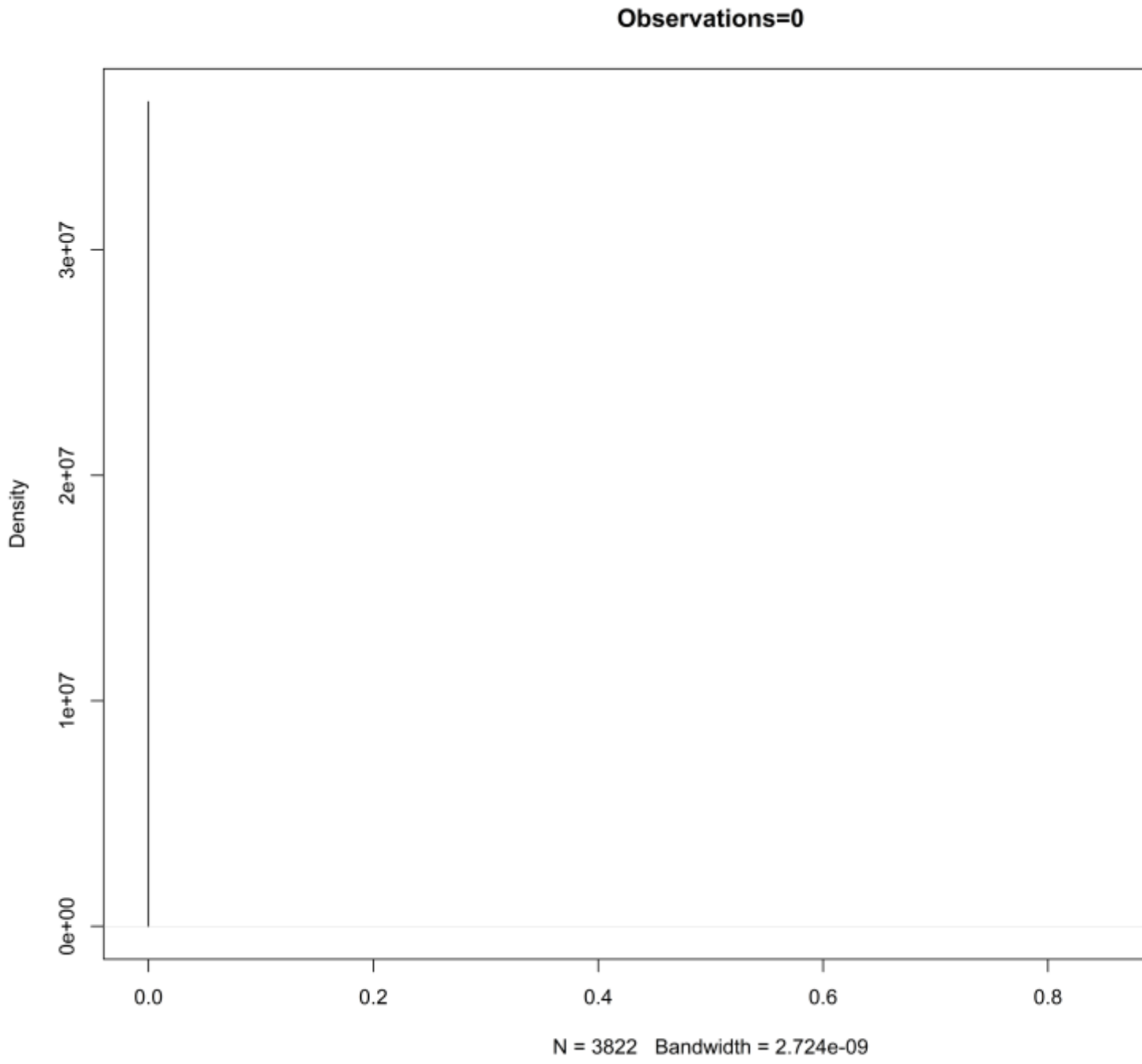

Figure 44. Logistic regression observed versus posterior plots for Silversides Gear 2 Only Model 2C for 0 observations. There were no multiple observations of only 1 individual or more than 20 individuals for Silversides collected by Gear 2 (large seine).

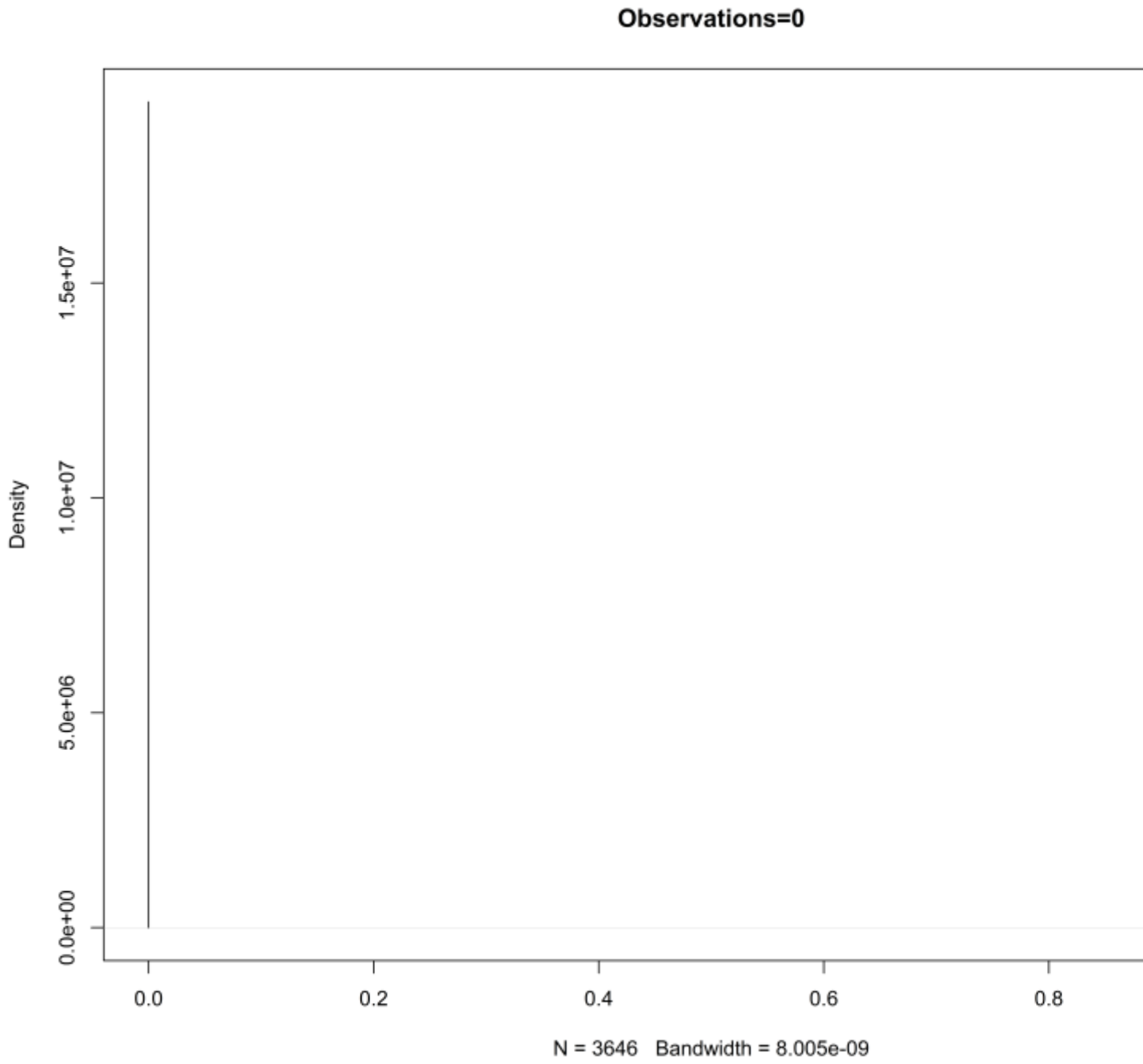

Figure 45. Logistic regression observed versus posterior plots for Silversides Gear 3 Only Model 2C 0 observations. There were no multiple observations of only 1 individual or more than 20 individuals foFigure 46. Logistic regression observed versus posterior plots for Grouped Prey Fish Gear 1 Only Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

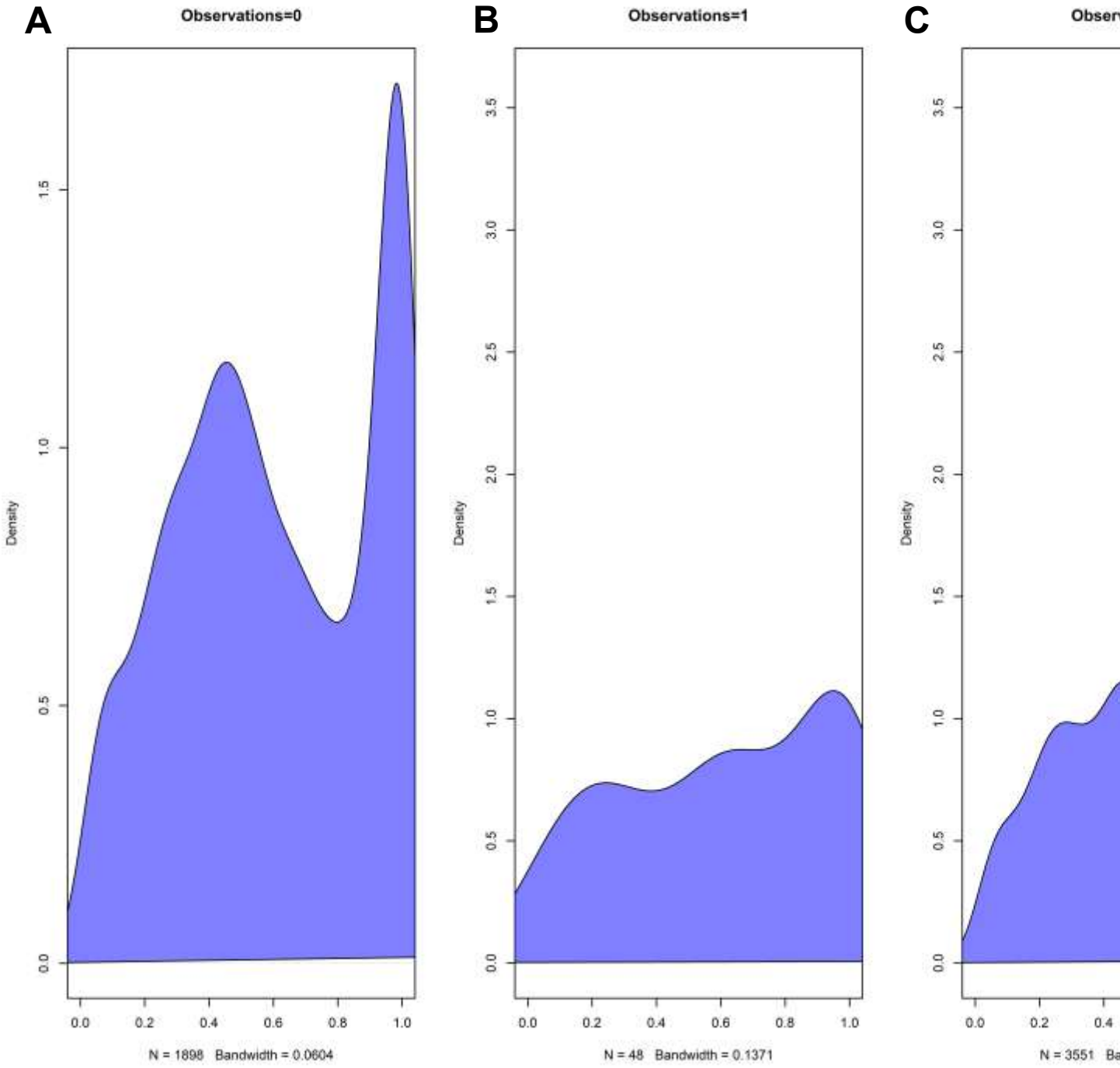

Figure 46. Logistic regression observed versus posterior plots for Grouped Prey Fish Gear 1 Only Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

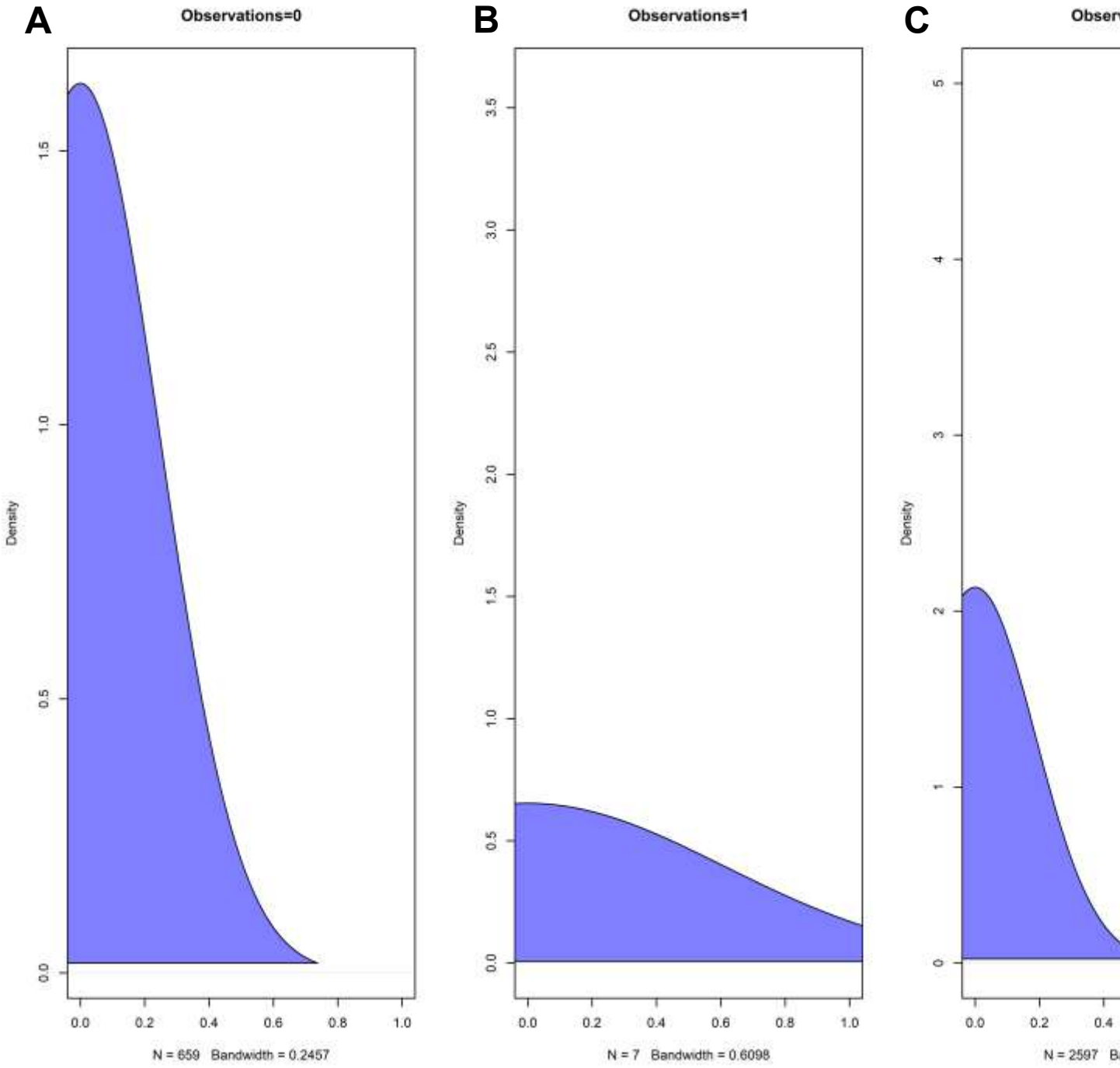

Figure 47. Logistic regression observed versus posterior plots for Grouped Prey Fish Gear 2 Only Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

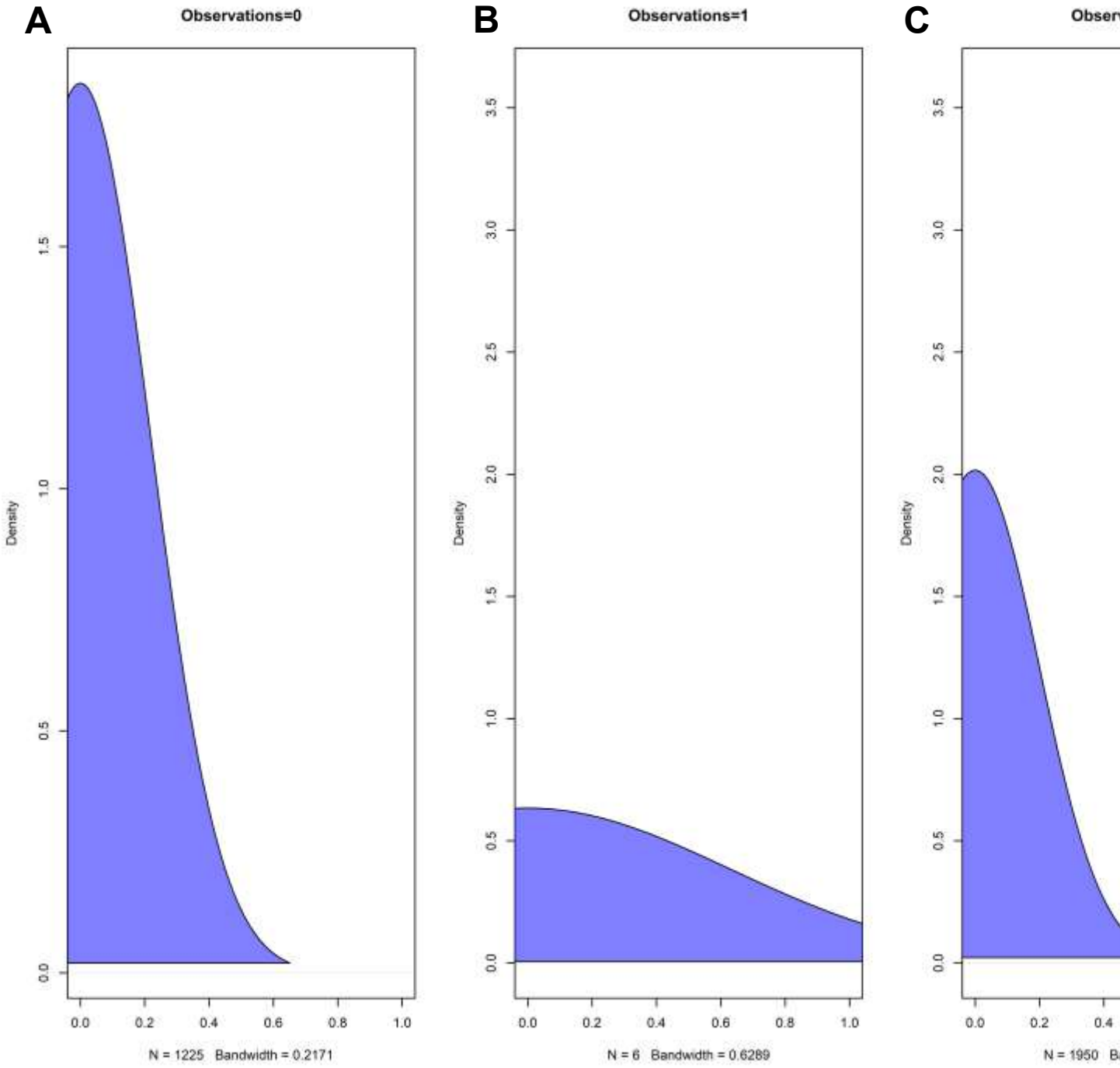

Figure 48. Logistic regression observed versus posterior plots for Grouped Prey Fish Gear 3 Only Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

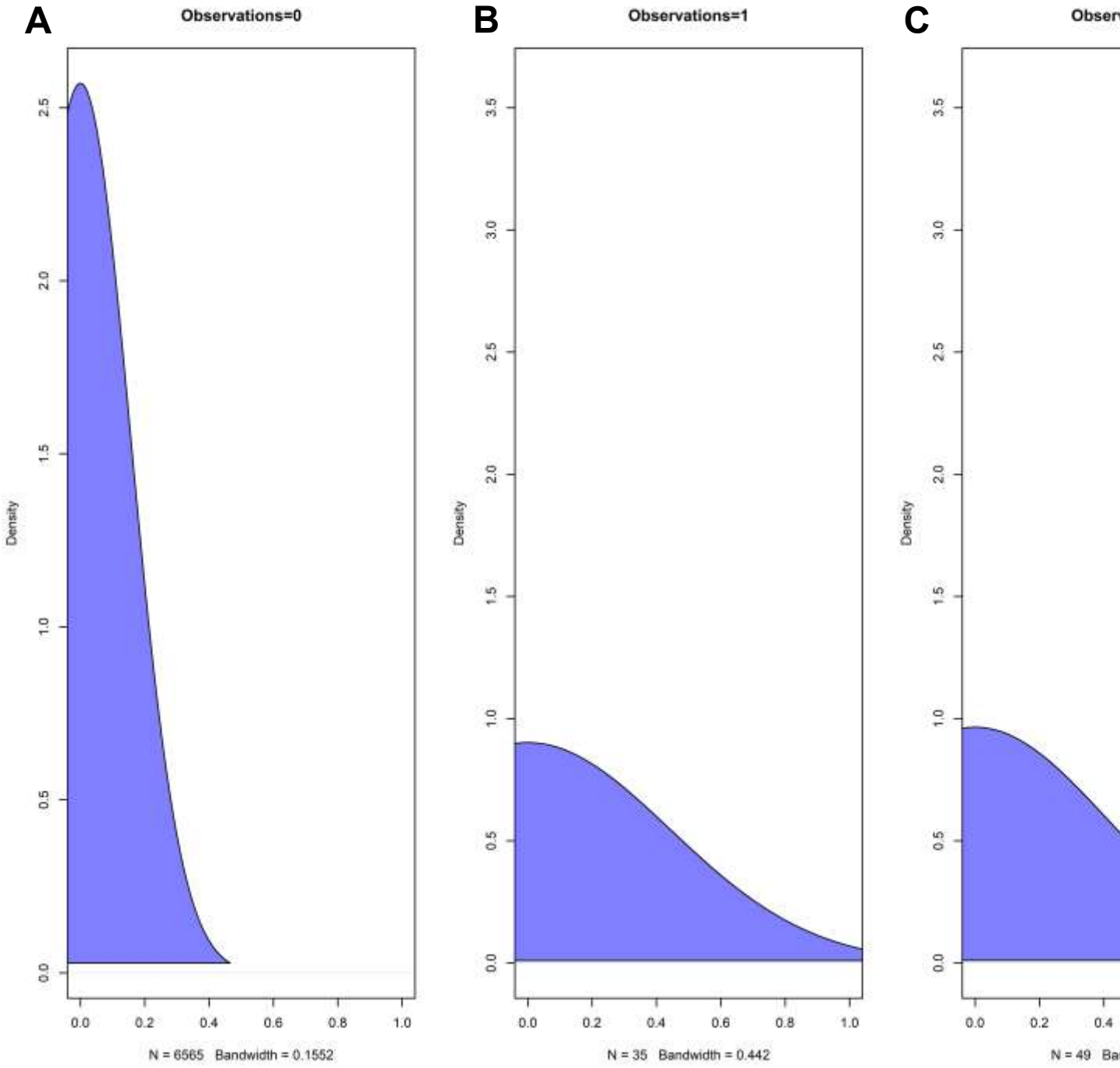

Figure 49. Logistic regression observed versus posterior plots for Herring Gear 1 Only Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

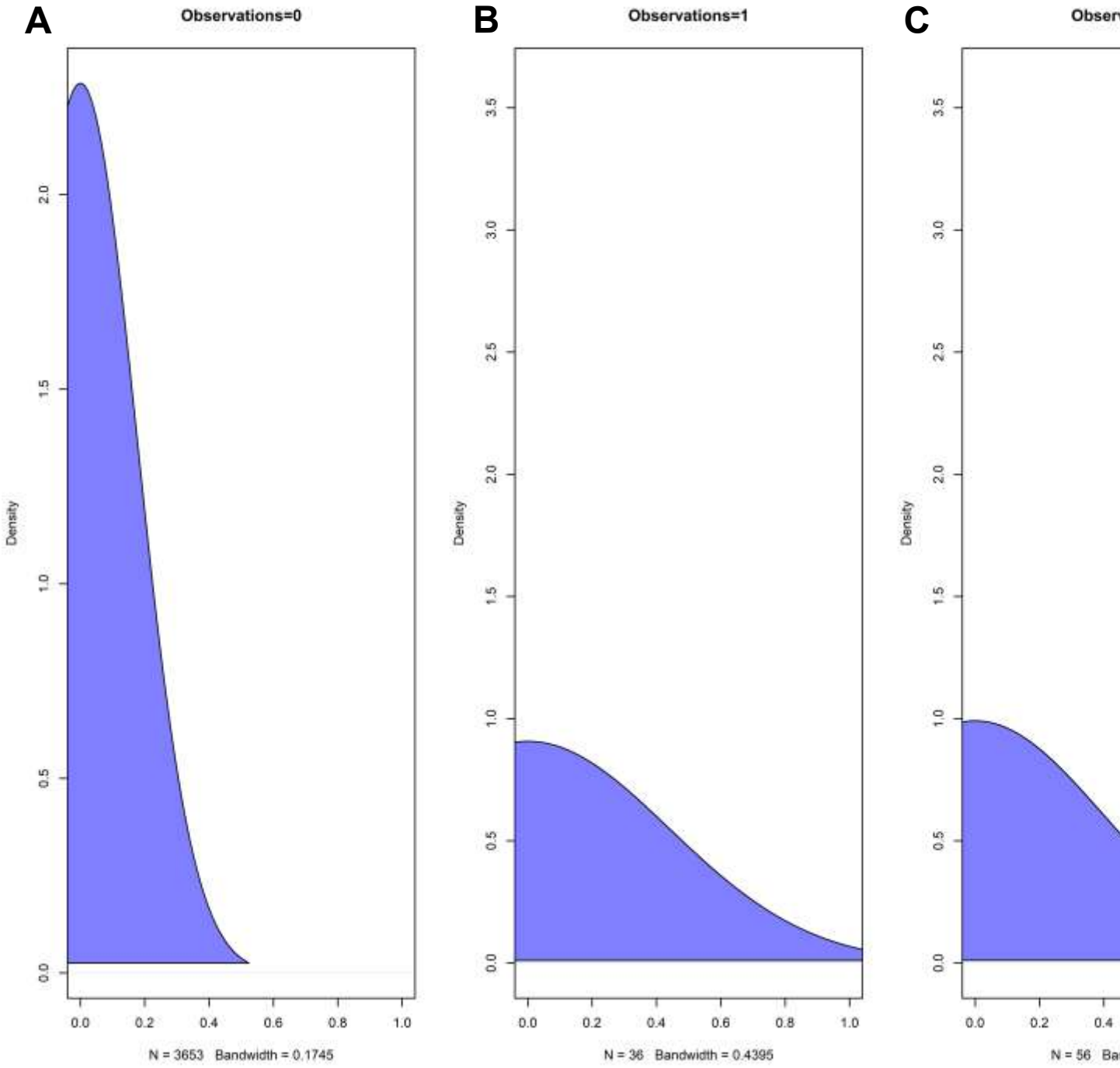

Figure 50. Logistic regression observed versus posterior plots for Herring Gear 2 Only Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

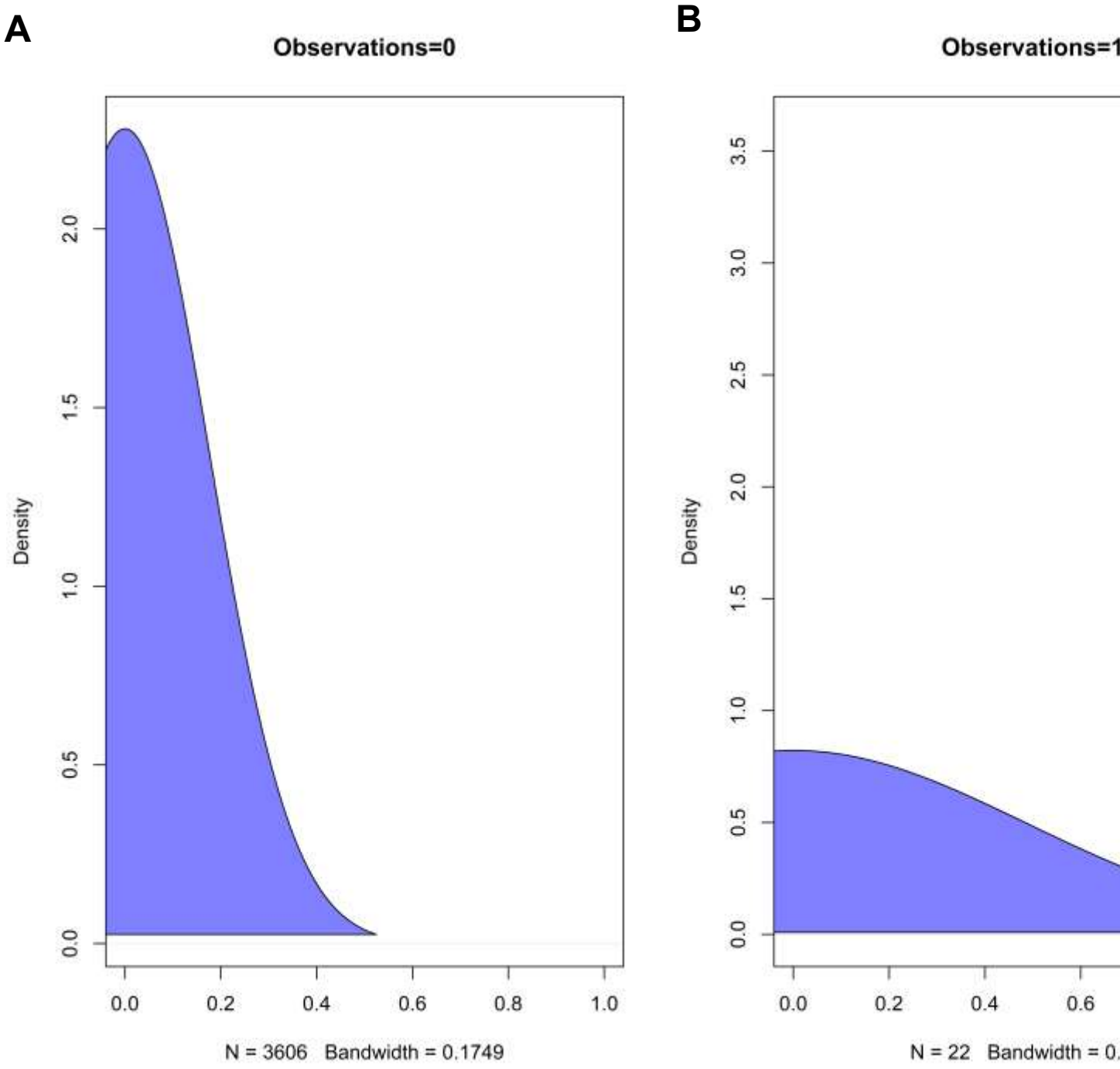

Figure 51. Logistic regression observed versus posterior plots for Herring Gear 3 Only Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

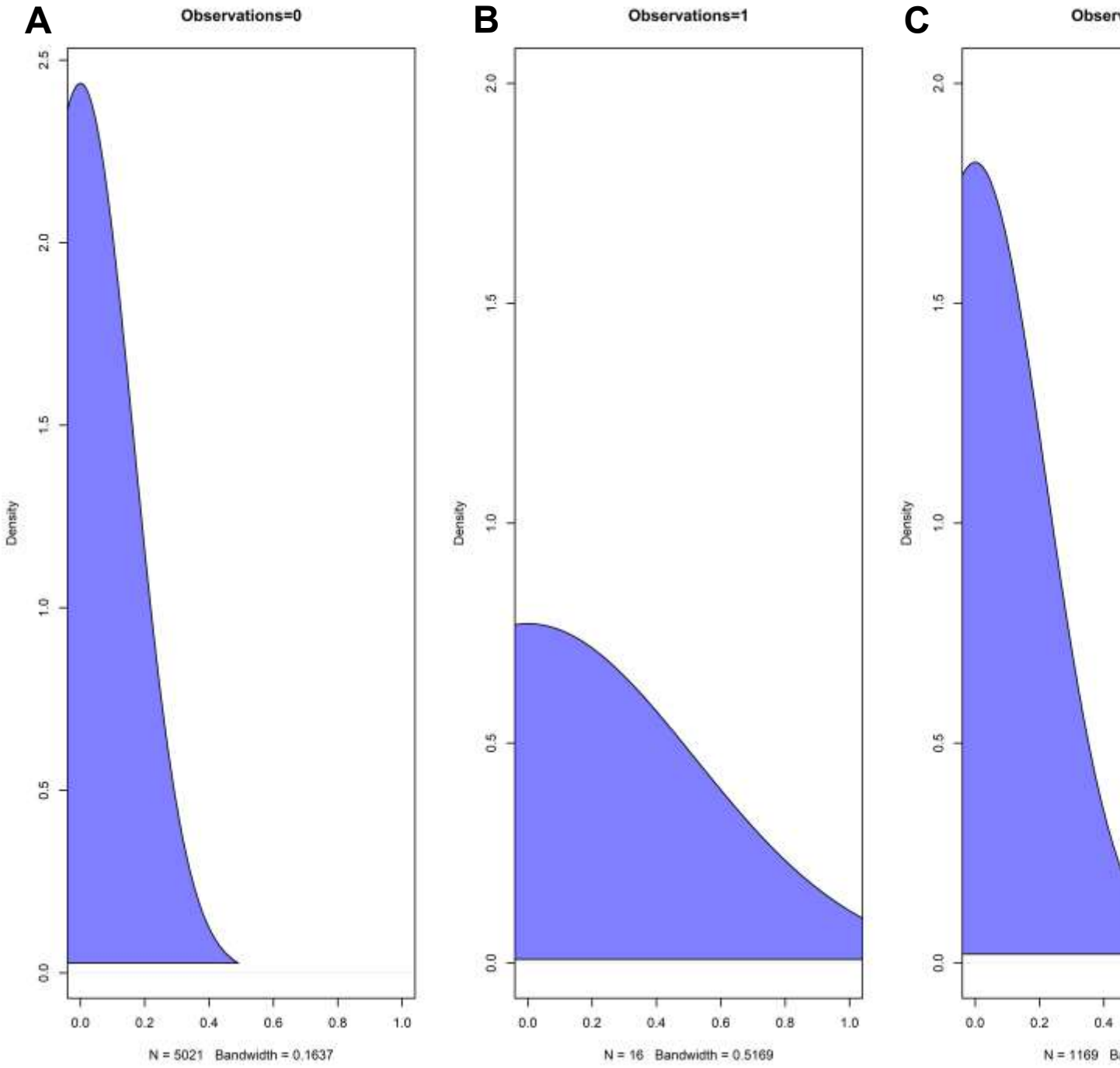

Figure 52. Logistic regression observed versus posterior plots for Anchovy Gear 1 Only Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

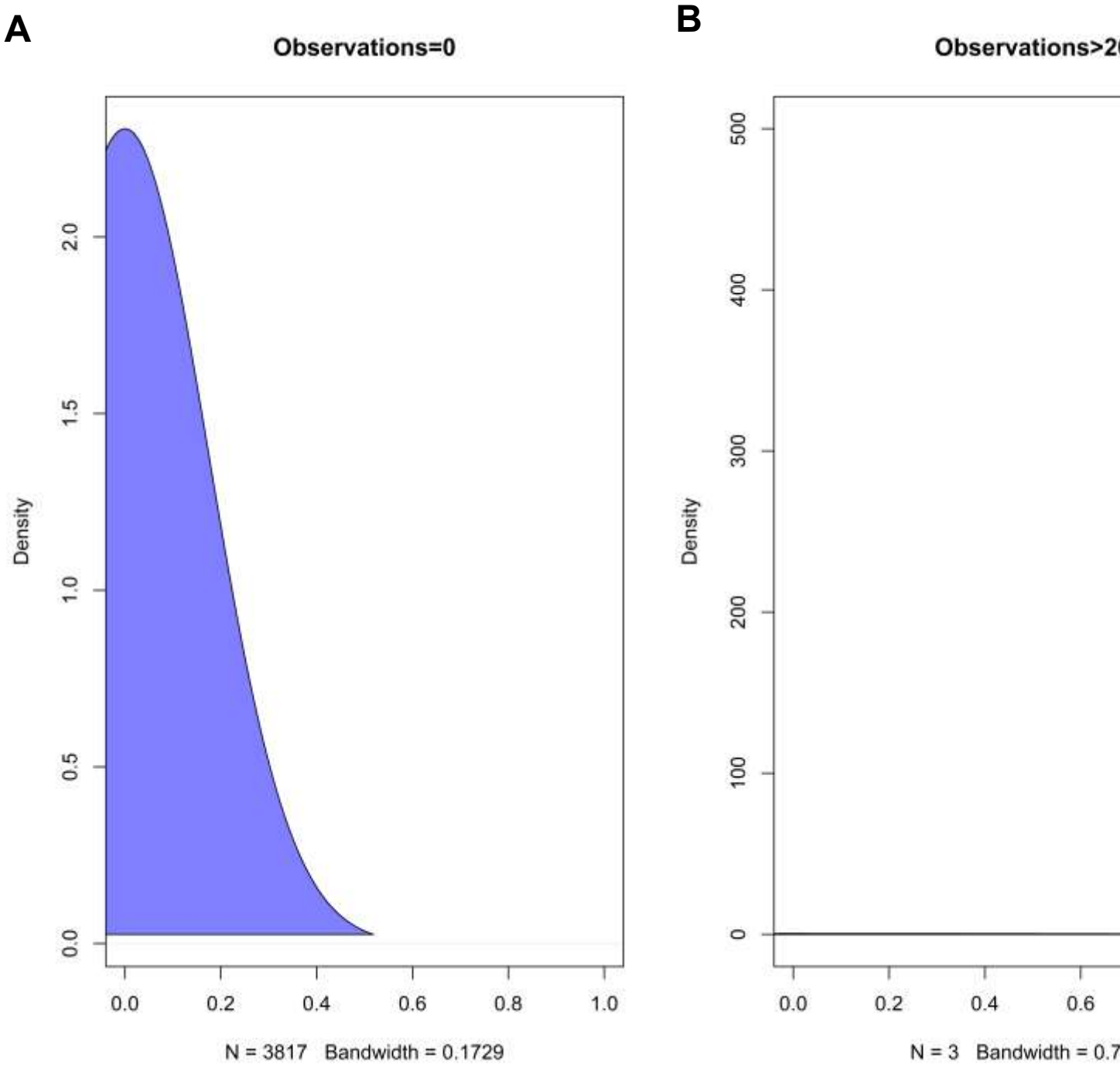

Figure 53. Logistic regression observed versus posterior plots for Anchovy Gear 2 Only Model 2G for (A) 0 observations, and (B) more than 20 observations. There were no observations of 1.

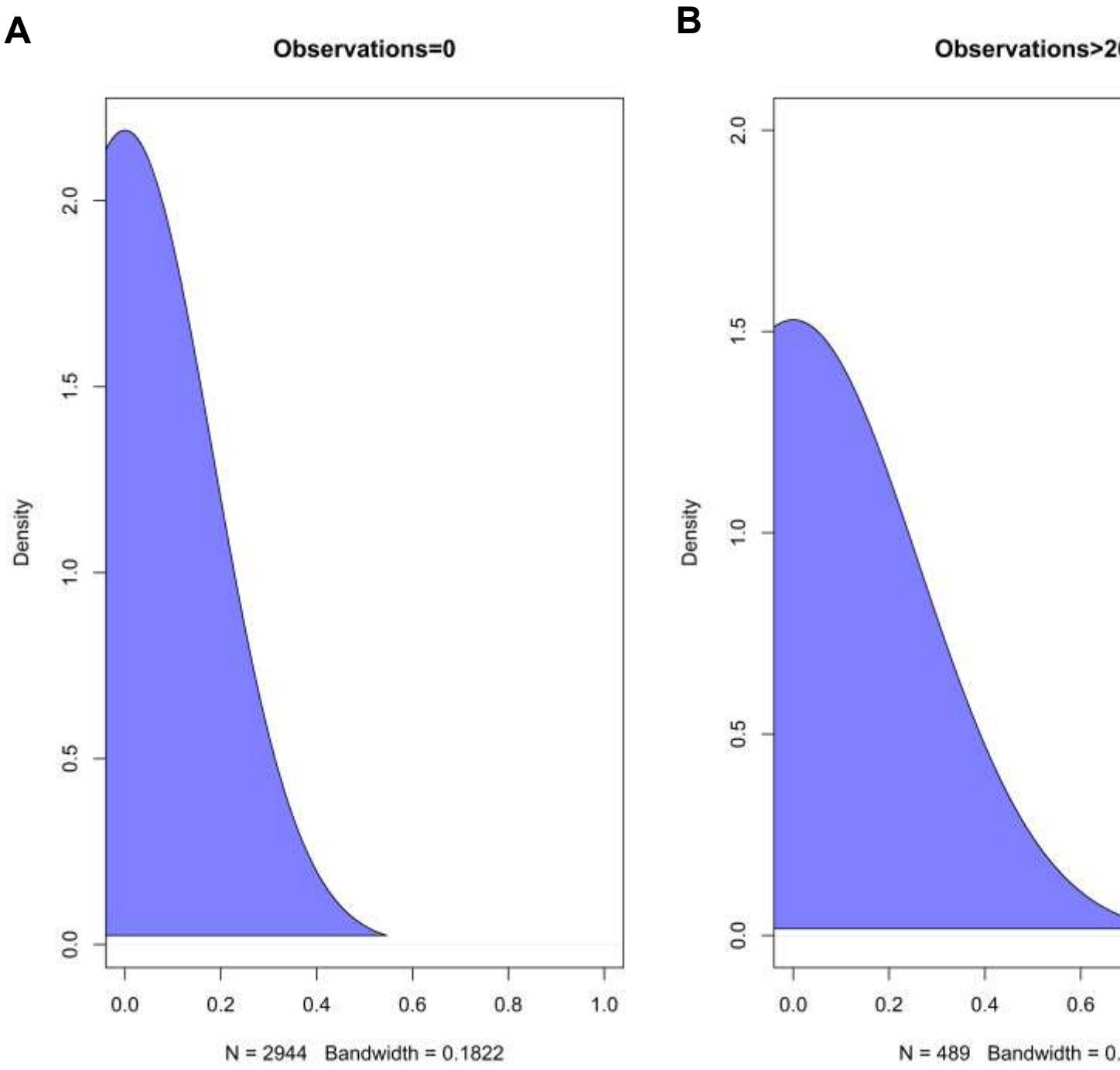

Figure 54. Logistic regression observed versus posterior plots for Anchovy Gear 3 Only Model 2G for (A) 0 observations, (B) more than 20 observations. There were no observations of 1.

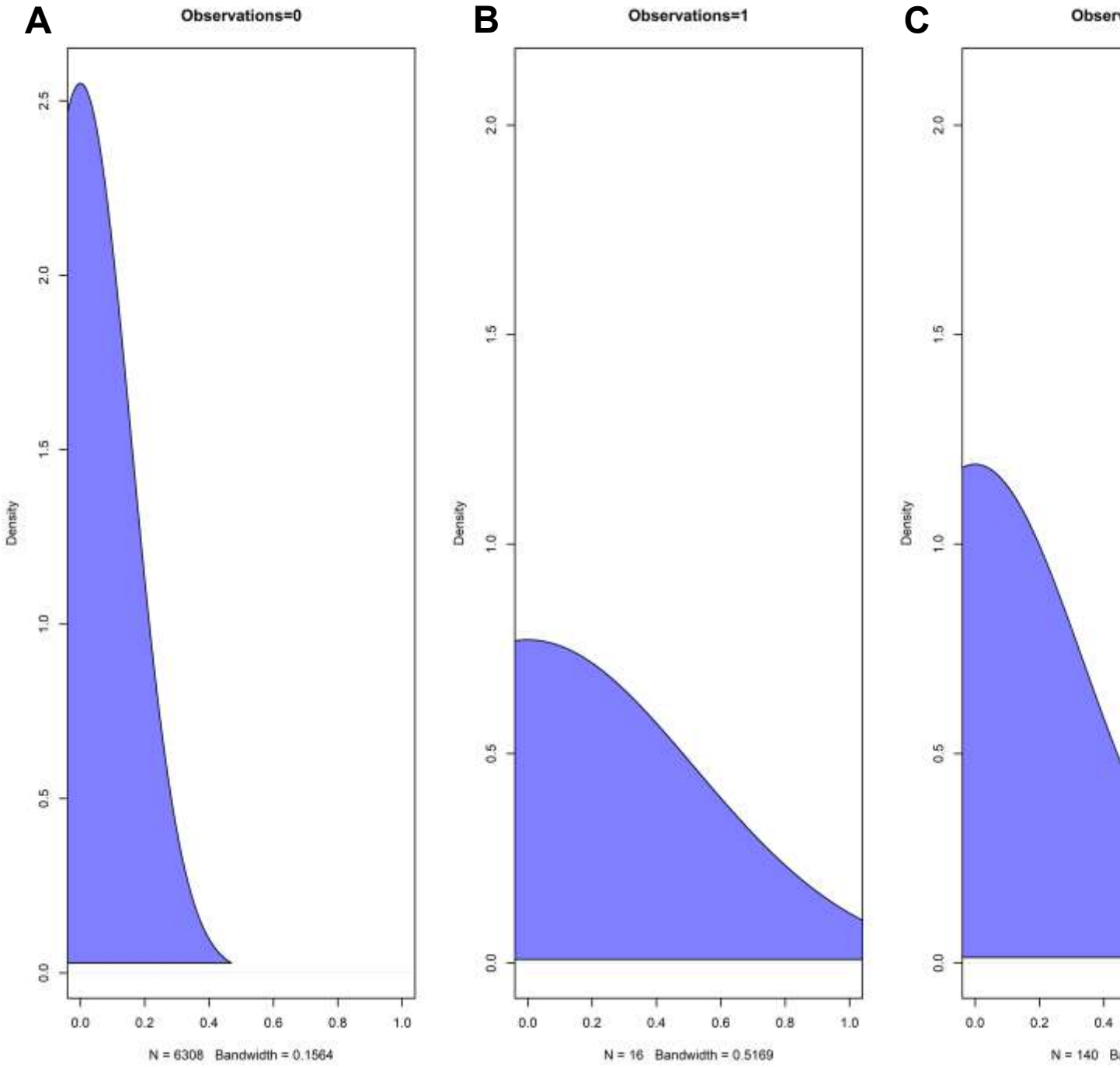

Figure 55. Logistic regression observed versus posterior plots for Mullet Gear 1 Only Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

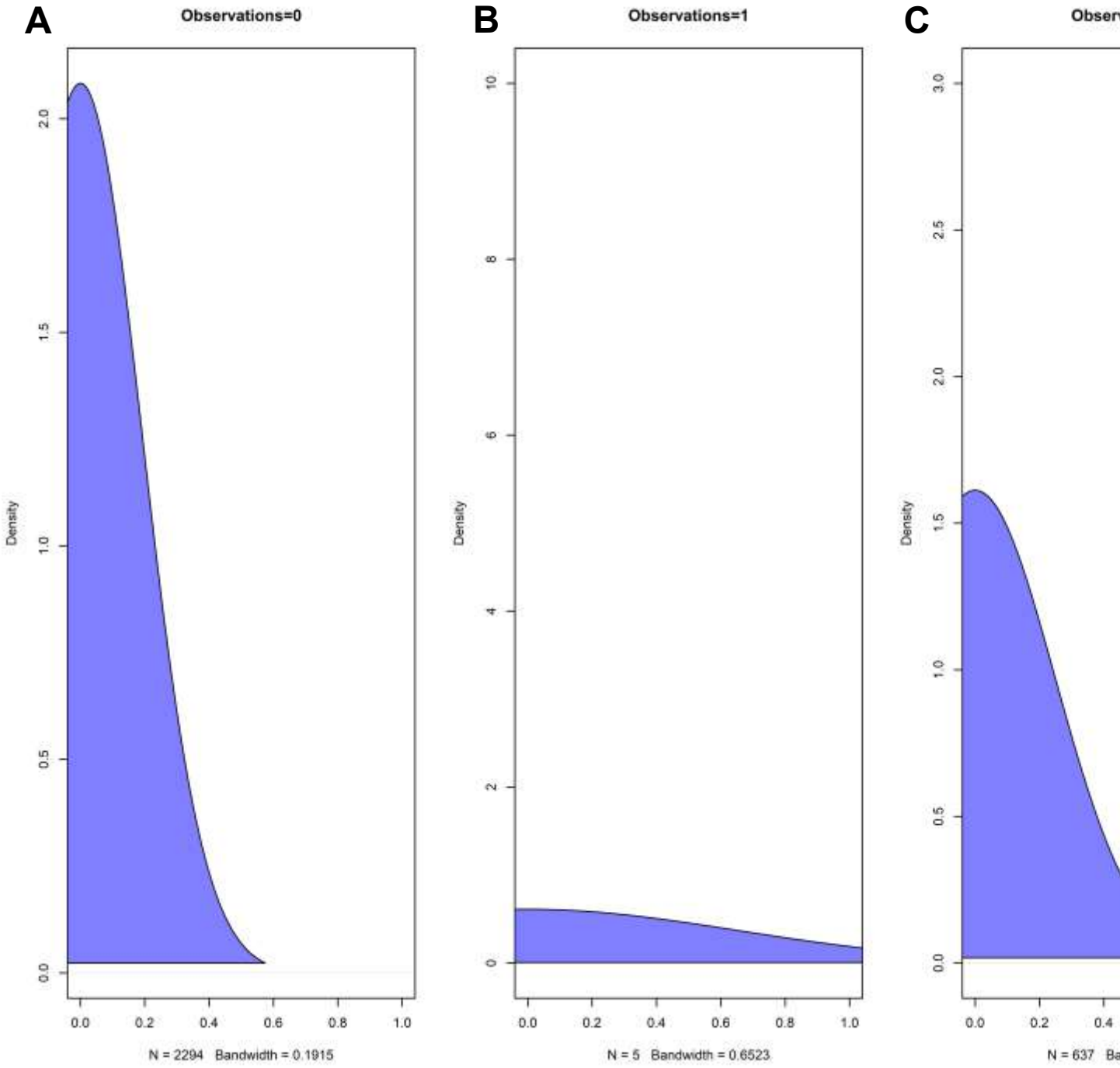

Figure 56. Logistic regression observed versus posterior plots for Mullet Gear 2 Only Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

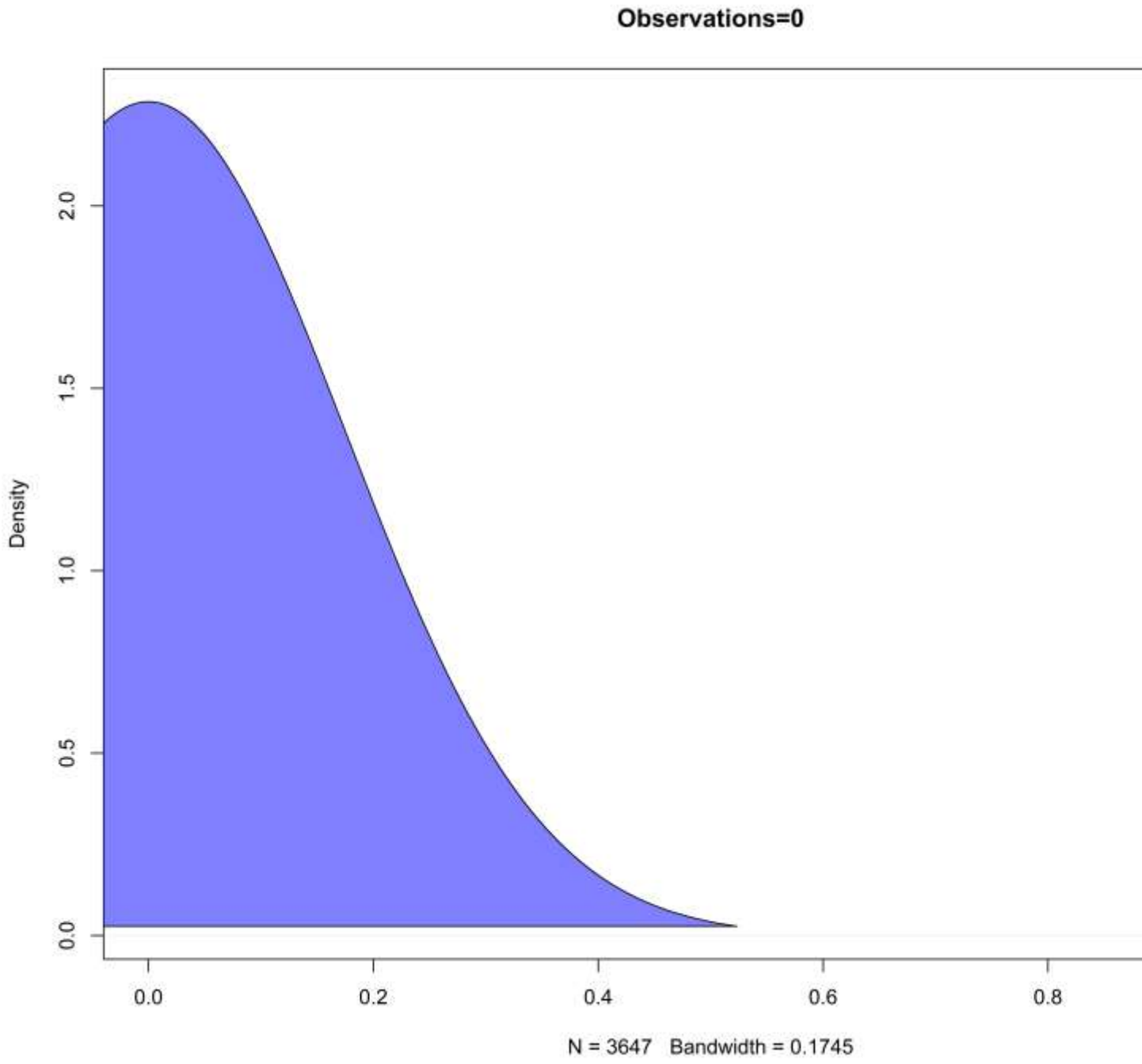

Figure 57. Logistic regression observed versus posterior plots for Mullet Gear 3 Only Model 2G for 0 observations. There were no observations of 1 or more than 20.

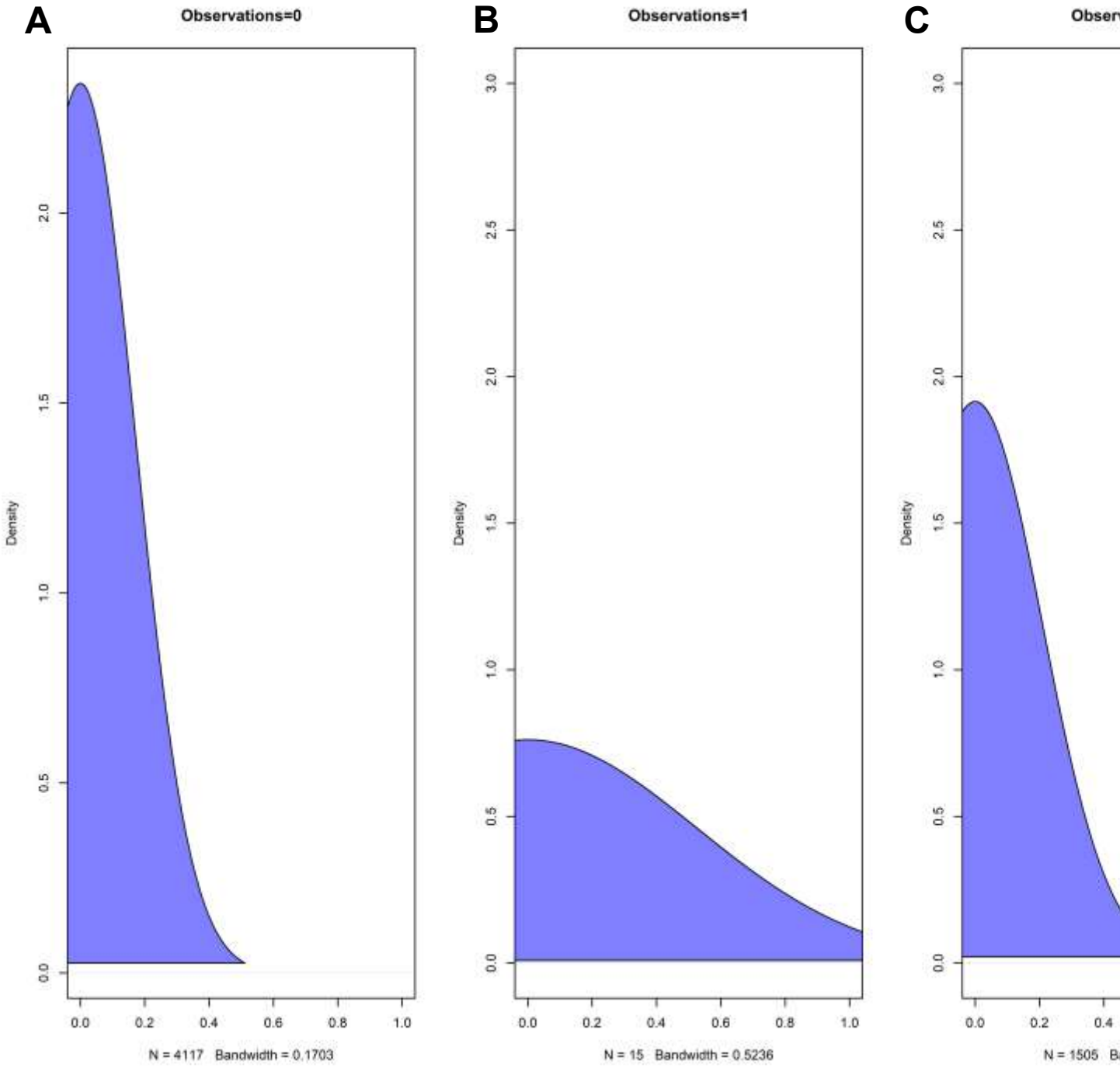

Figure 58. Logistic regression observed versus posterior plots for Mojarra Gear 1 Only Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

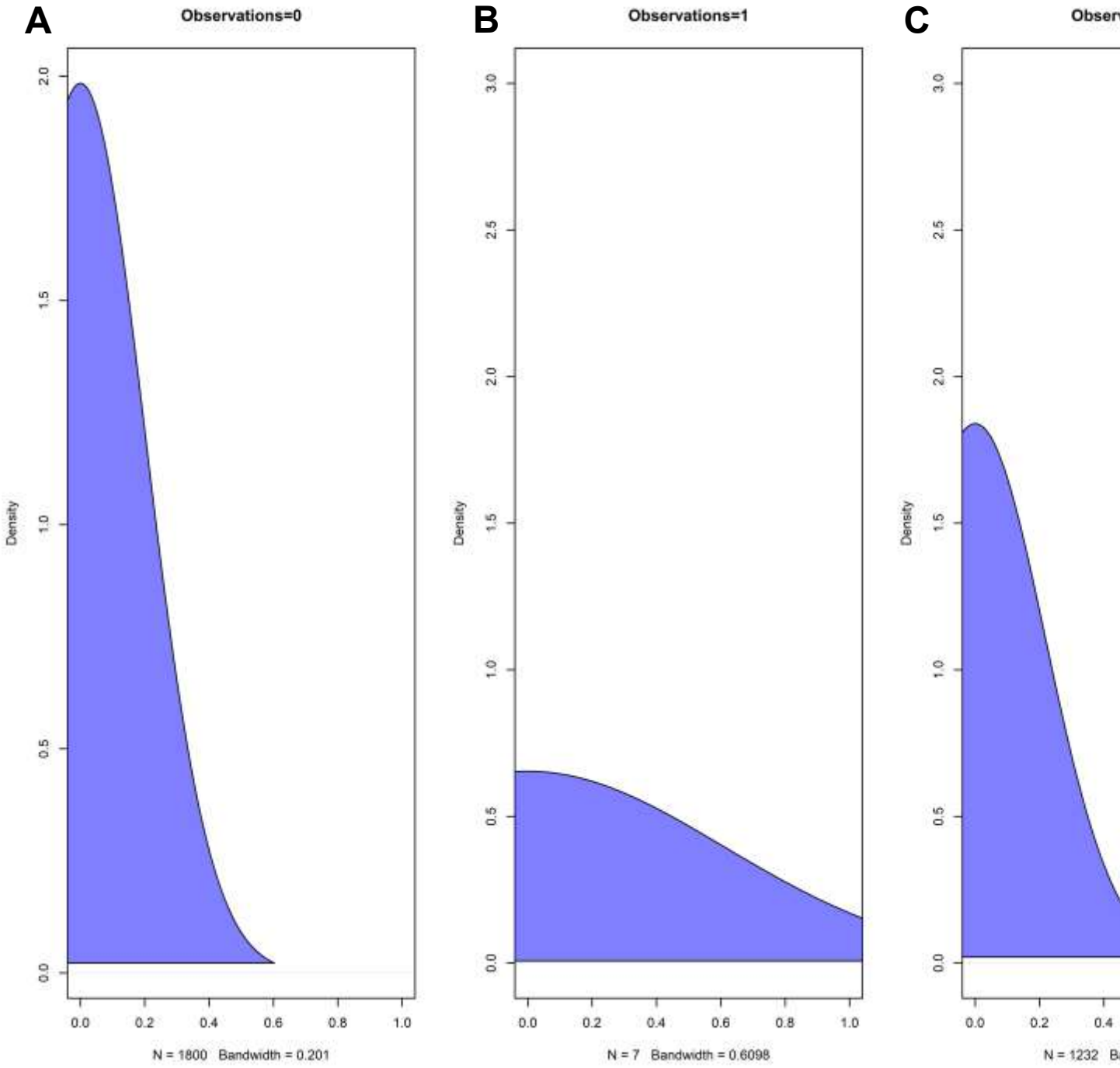

Figure 59. Logistic regression observed versus posterior plots for Mojarra Gear 2 Only Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

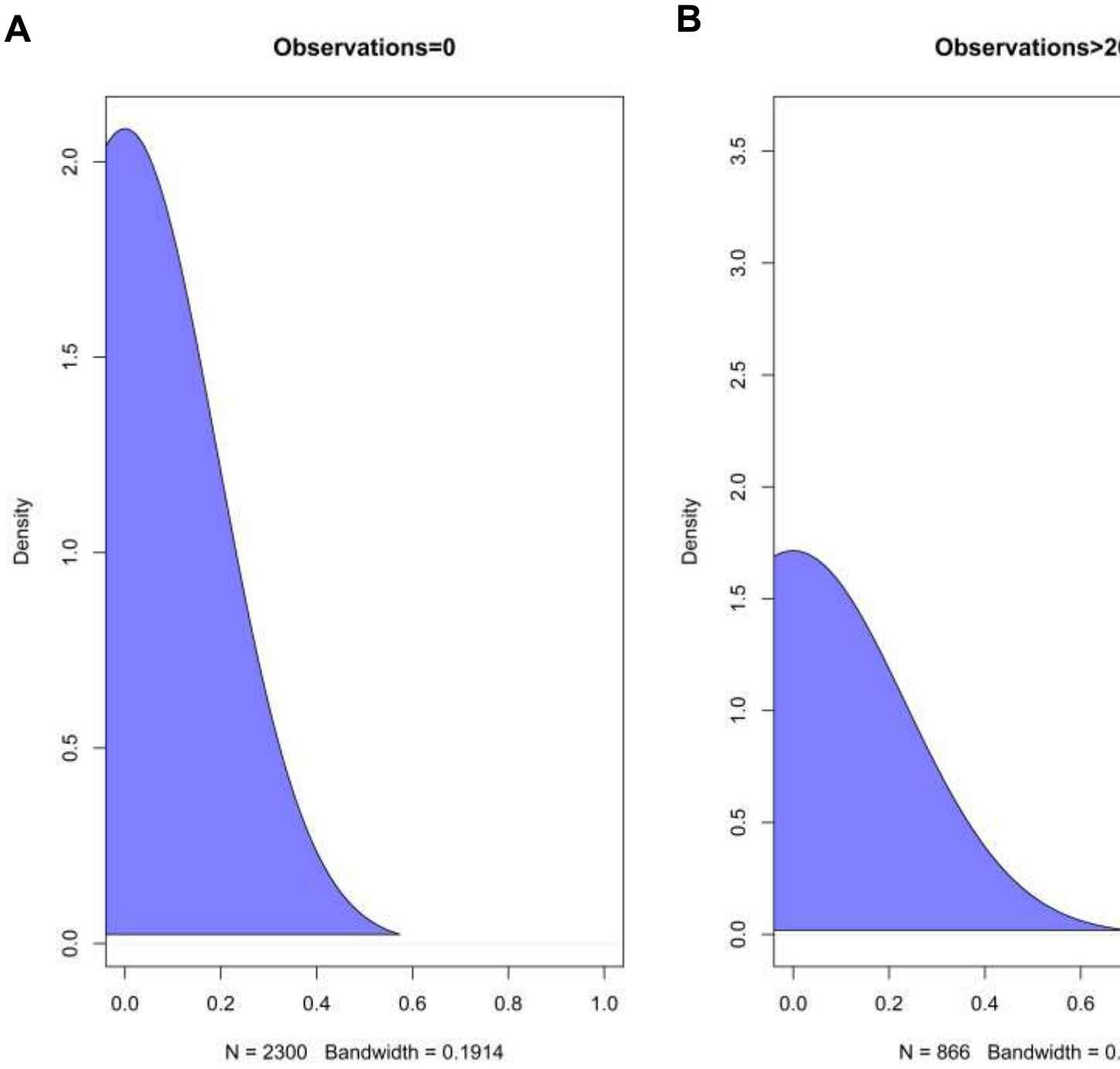

Figure 60. Logistic regression observed versus posterior plots for Mojarra Gear 3 Only Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

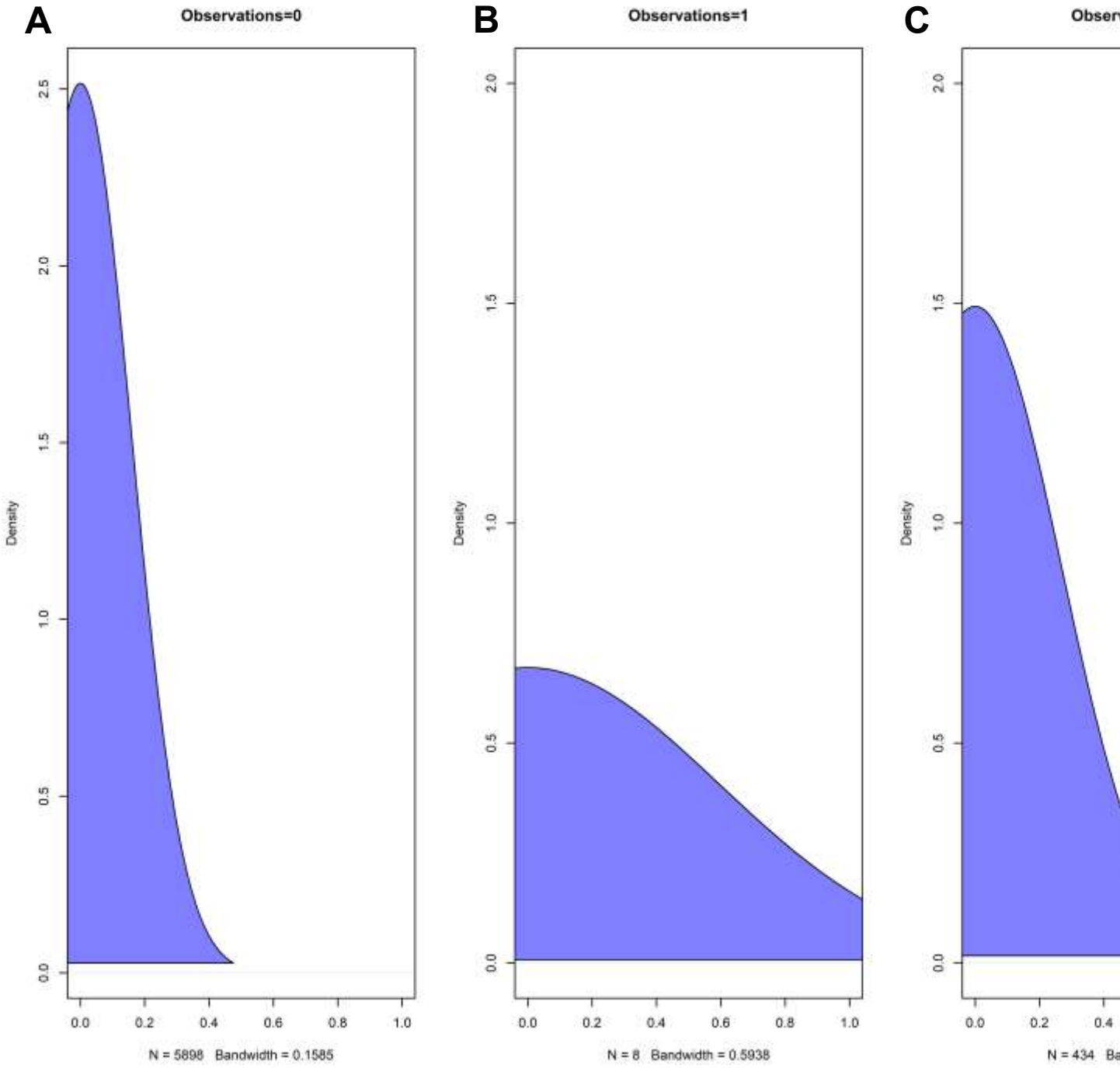

Figure 61. Logistic regression observed versus posterior plots for Silver Perch Gear 1 Only Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

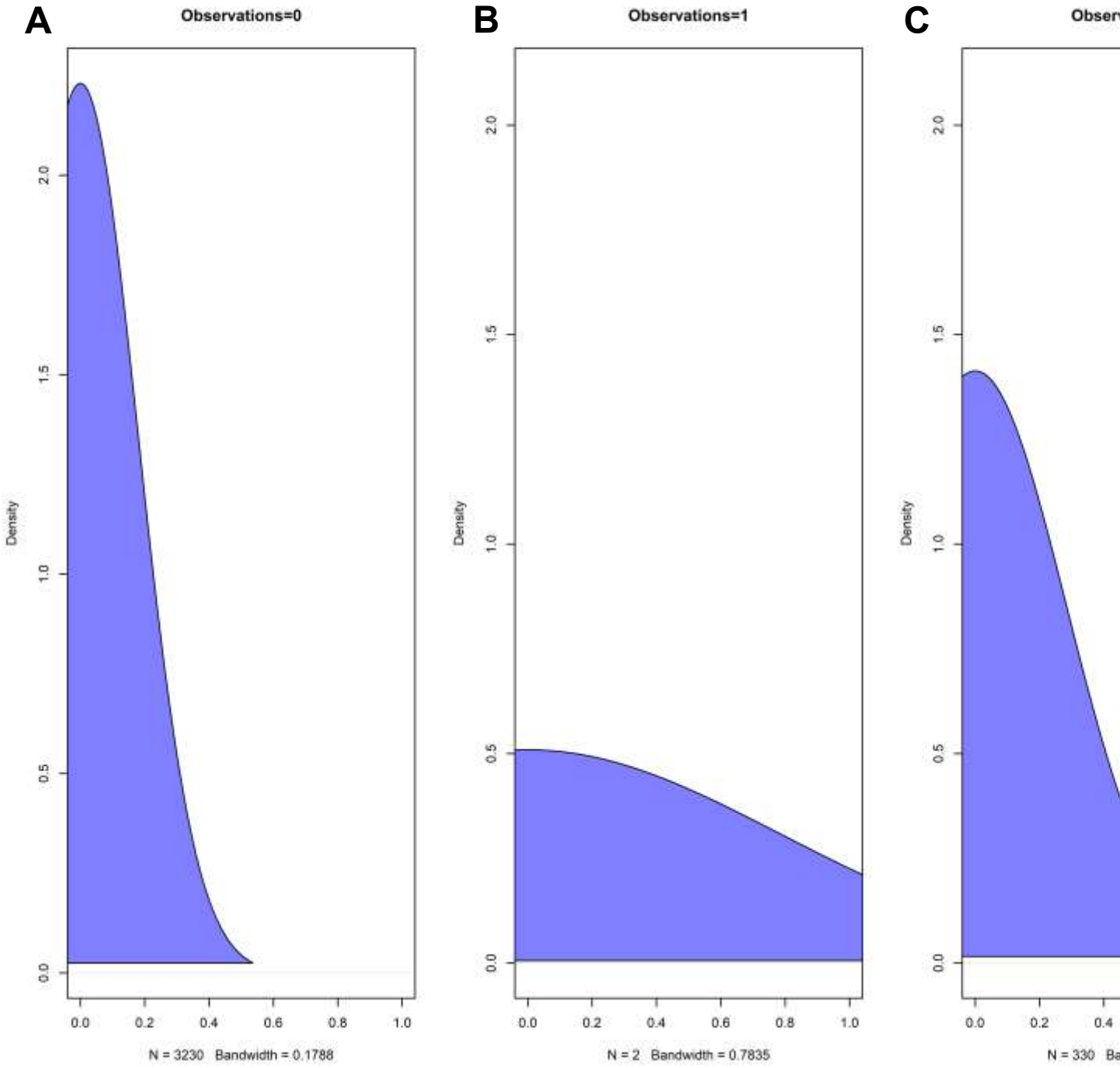

Figure 62. Logistic regression observed versus posterior plots for Silver Perch Gear 2 Only Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

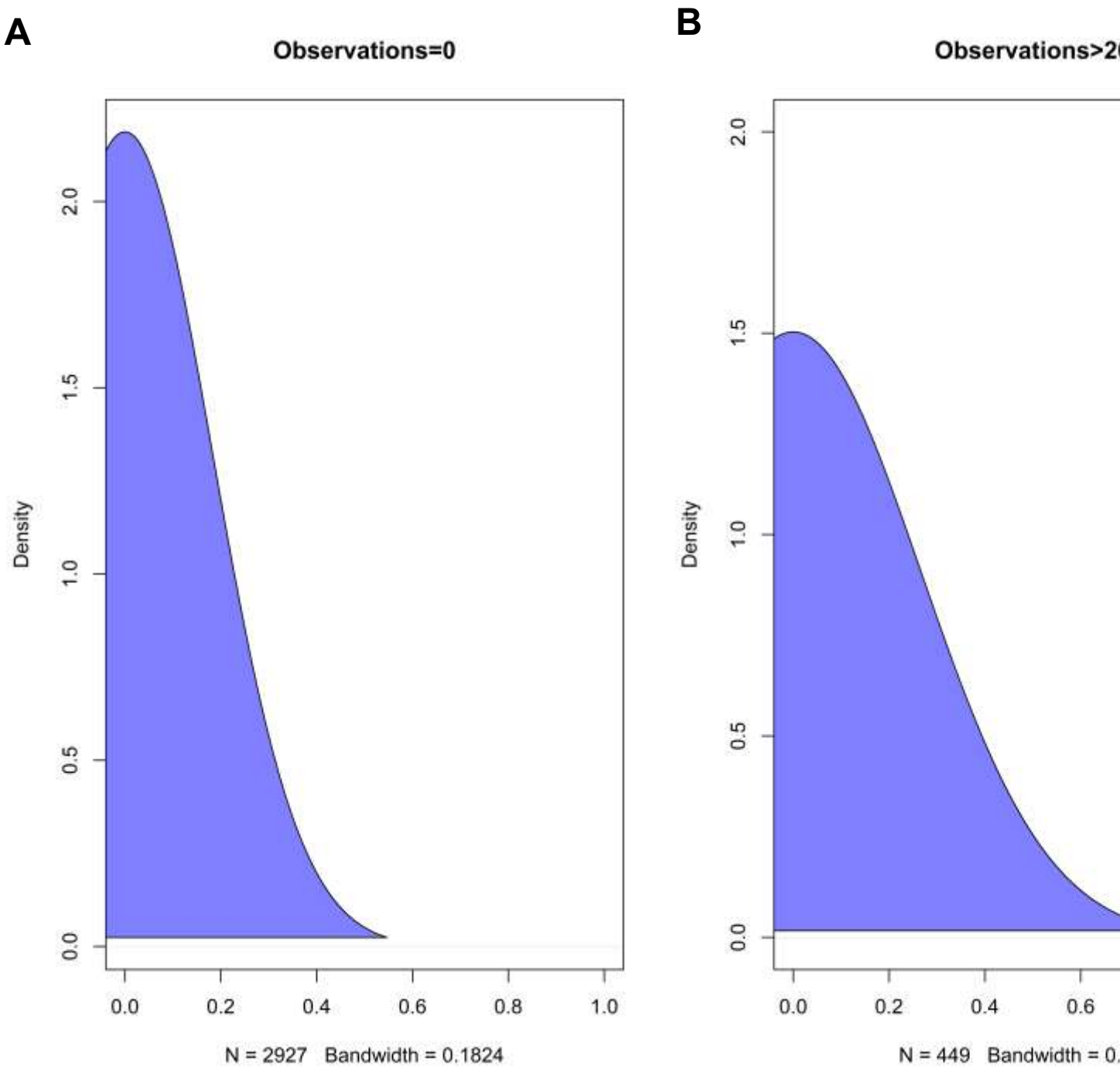

Figure 63. Logistic regression observed versus posterior plots for Silver Perch Gear 3 Only Model 2Gfor (A) 0 observations, and (B) more than 20 observations. There were no observations of 1.

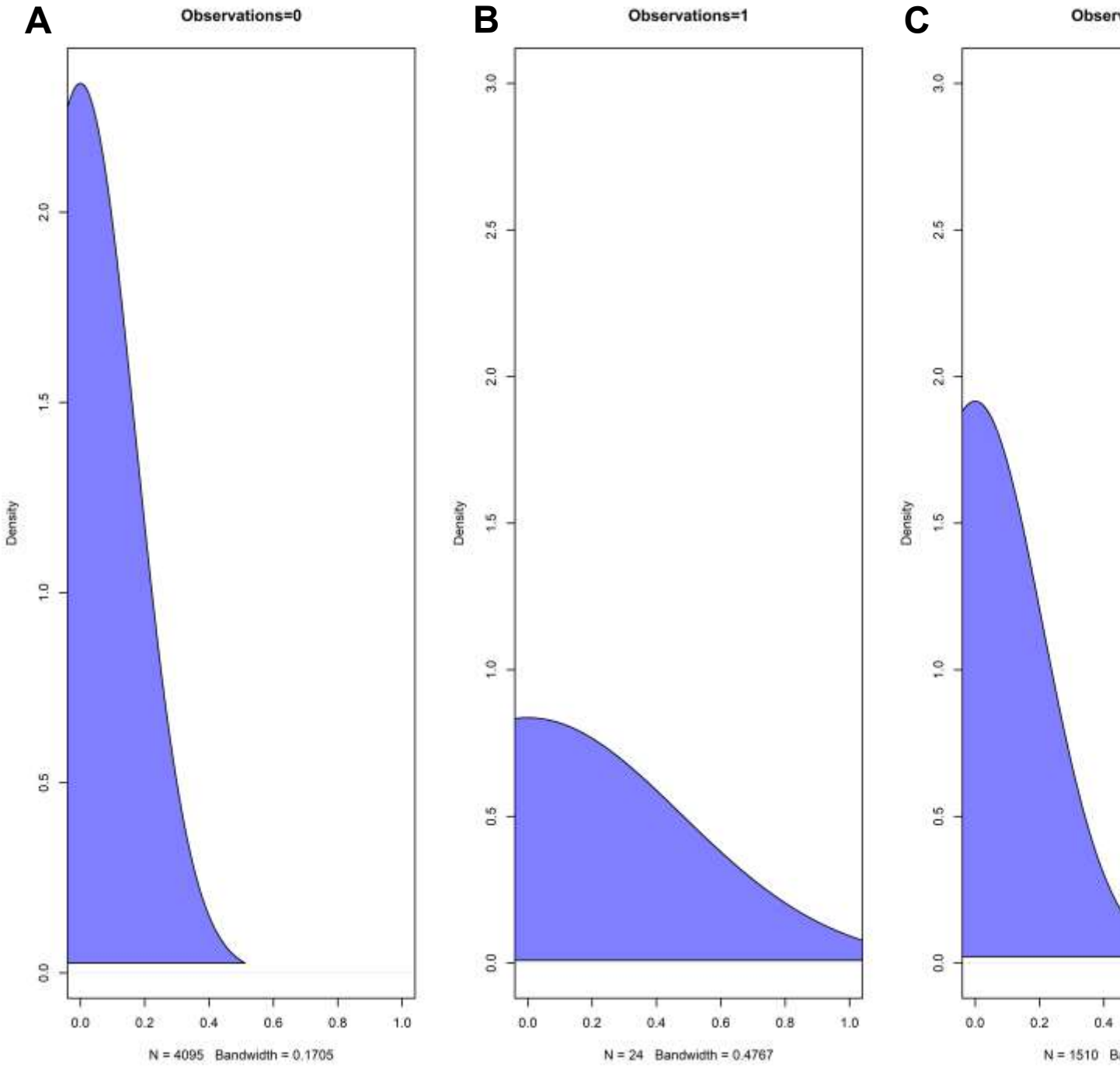

Figure 64. Logistic regression observed versus posterior plots for Pinfish Gear 1 Only Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

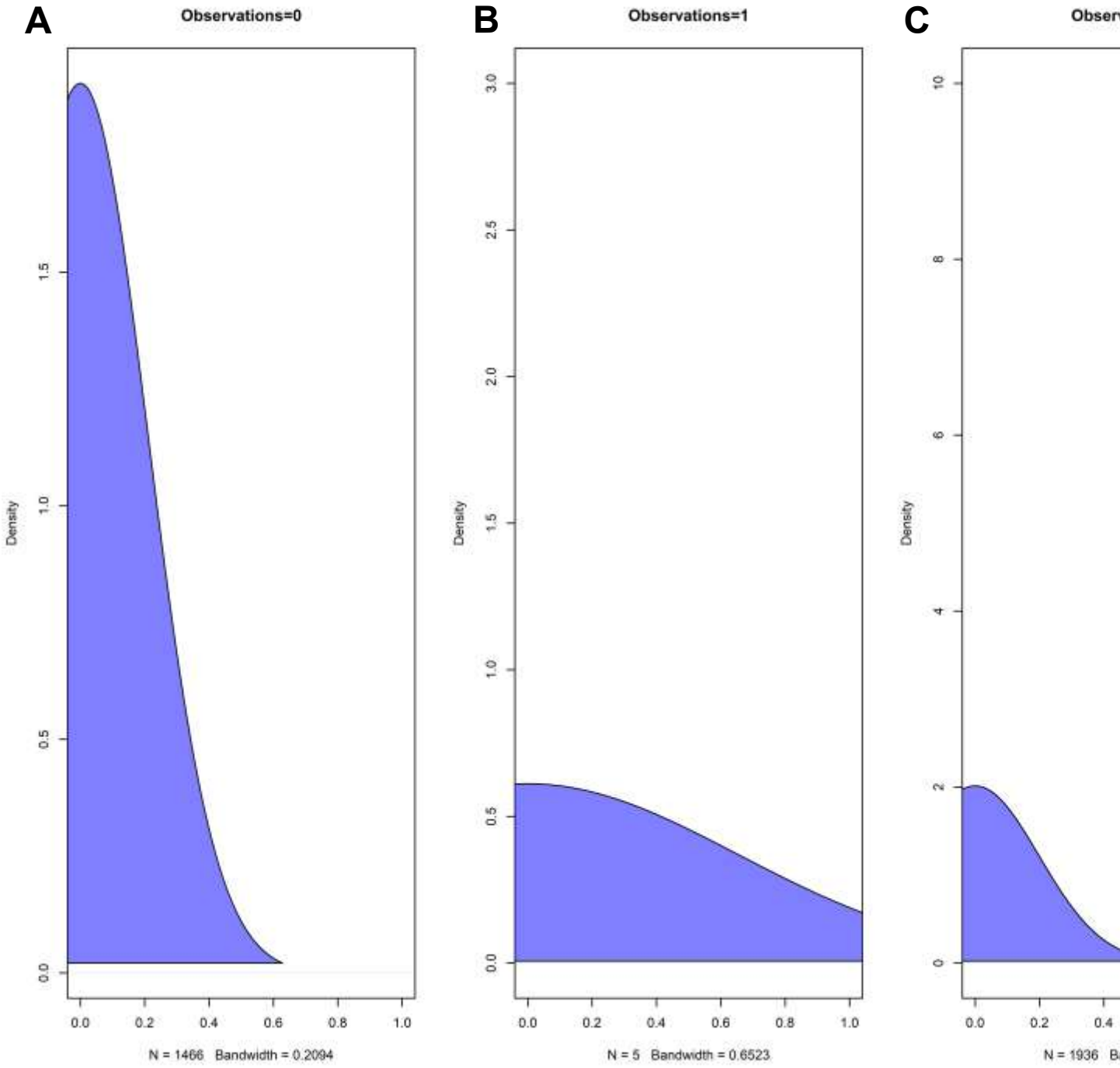

Figure 65. Logistic regression observed versus posterior plots for Pinfish Gear 2 Only Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

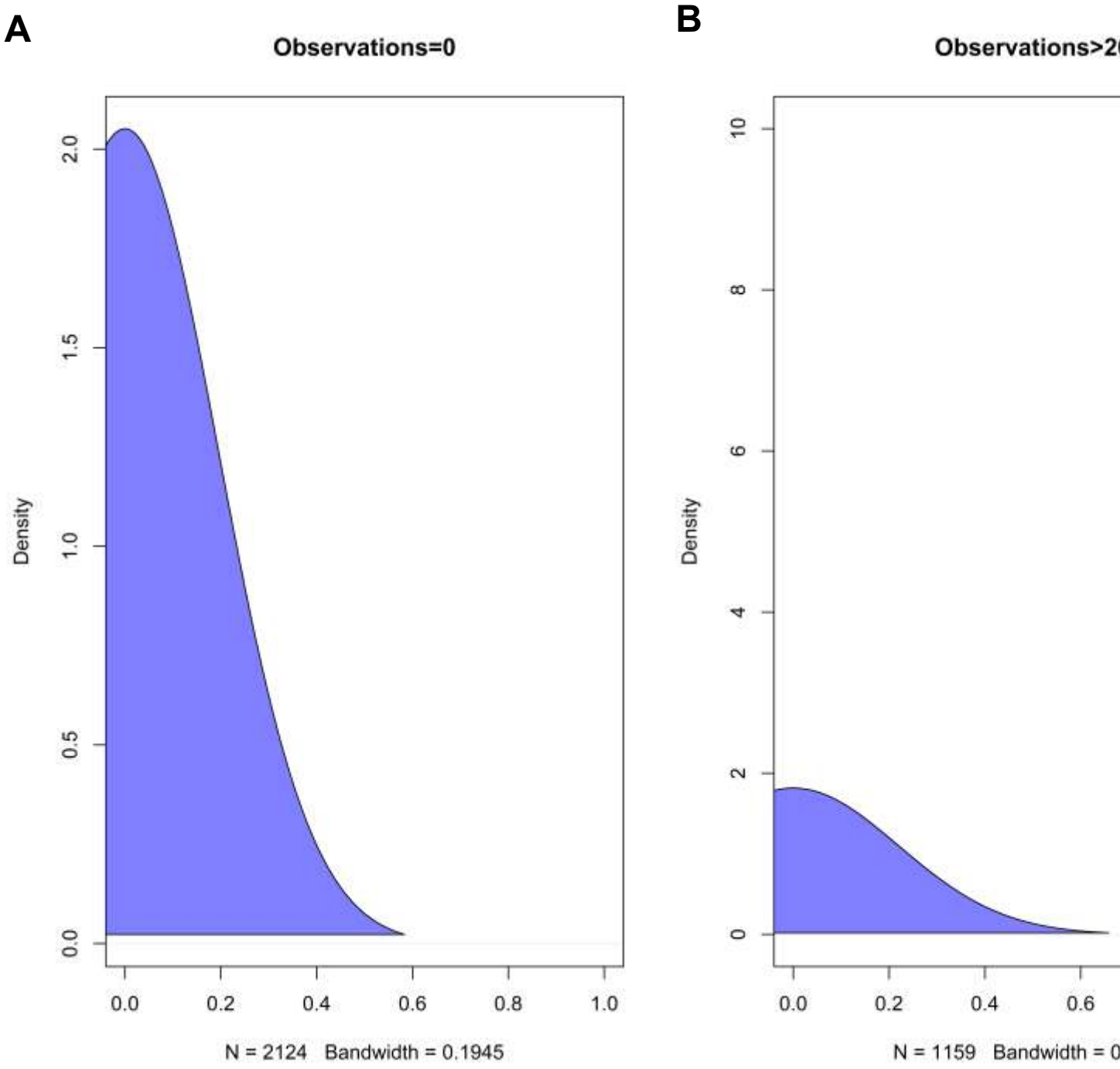

Figure 66. Logistic regression observed versus posterior plots for Pinfish Gear 3 Only Model 2G for (A) 0 observations, and (B) more than 20 observations. There were no observations of 1.

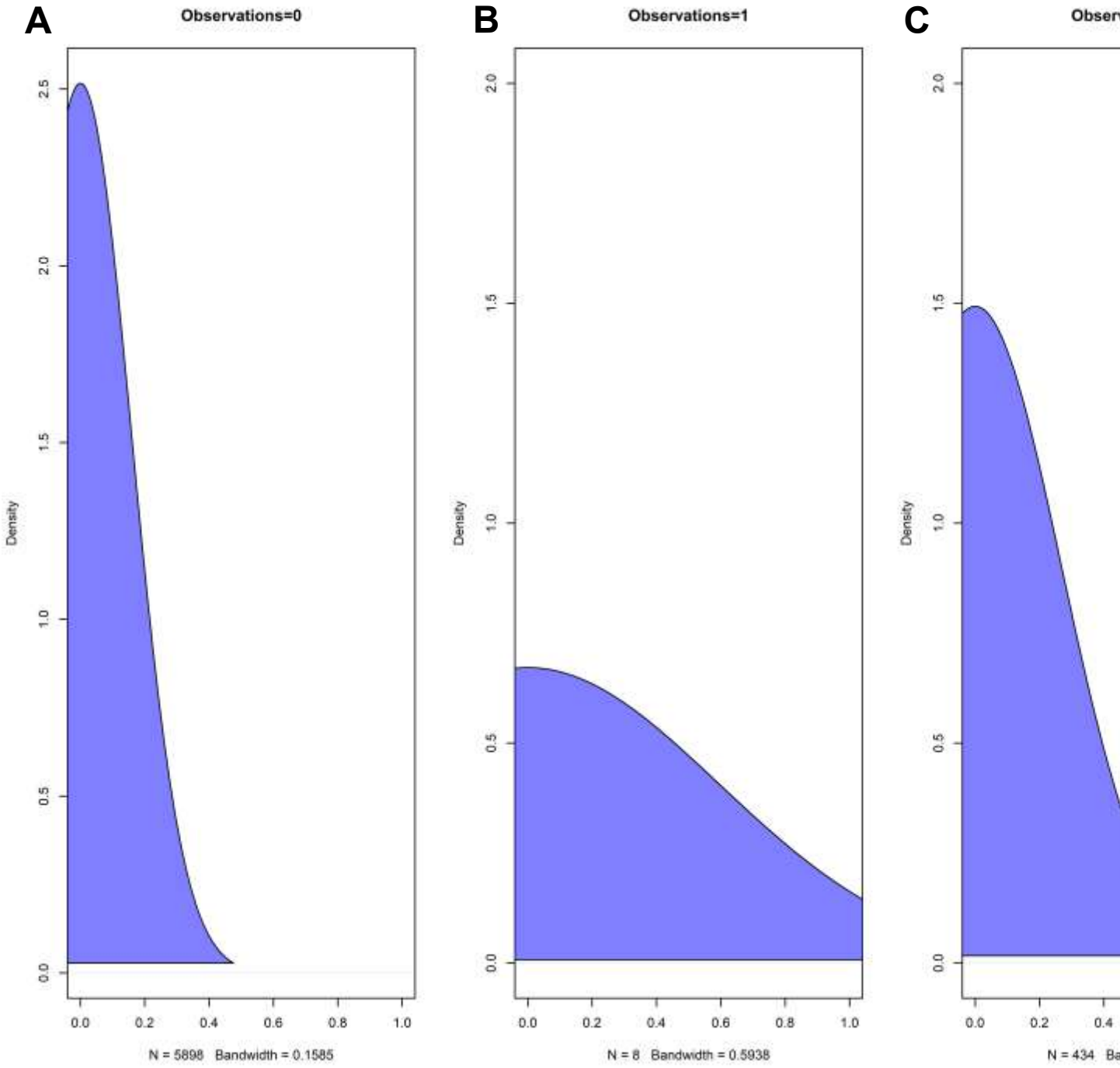

Figure 67. Logistic regression observed versus posterior plots for Pigfish Gear 1 Only Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

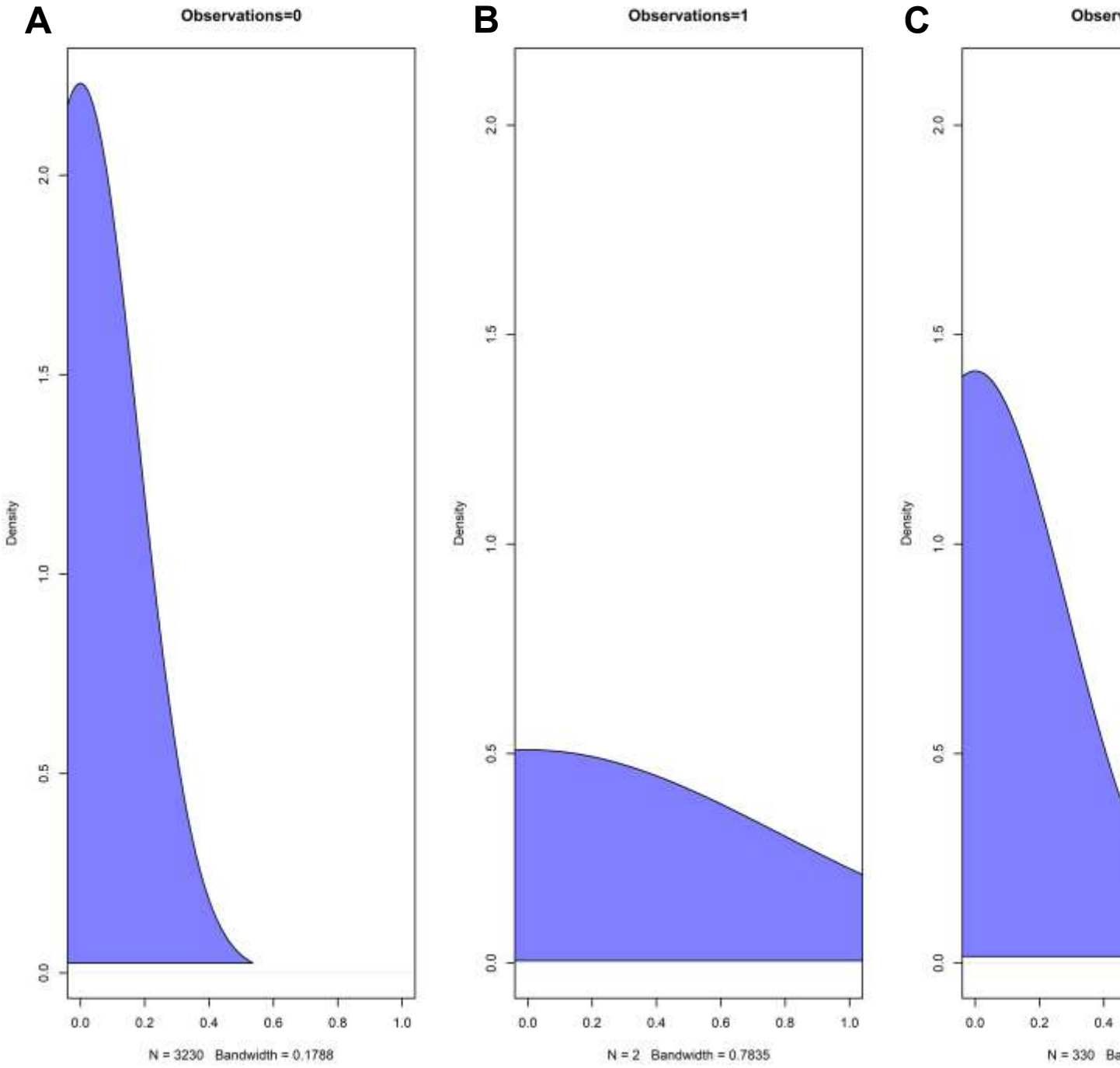

Figure 68. Logistic regression observed versus posterior plots for Pigfish Gear 2 Only Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

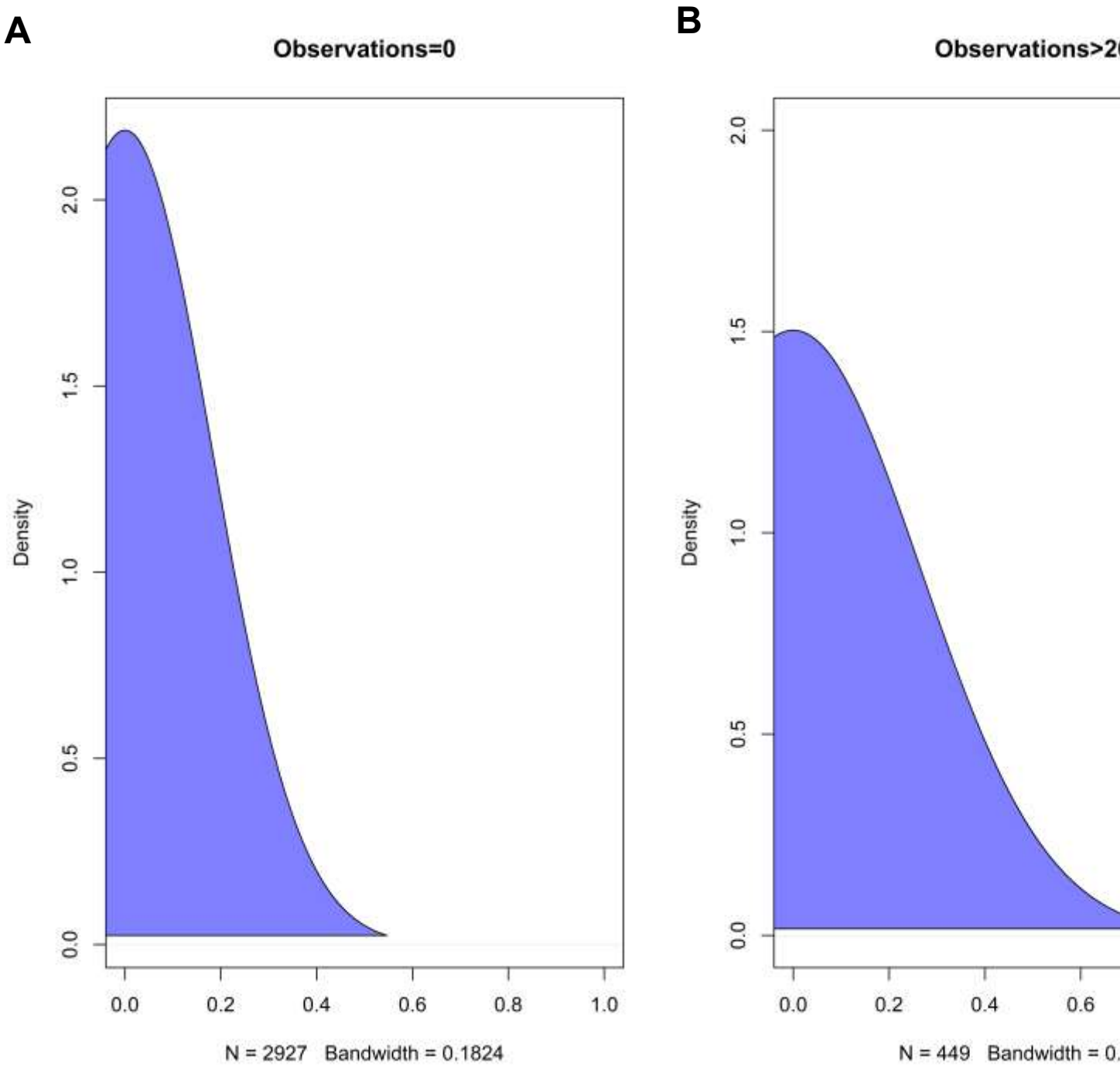

Figure 69. Logistic regression observed versus posterior plots for Pigfish Gear 3 Only Model 2G for (A) 0 observations, and (B) more than 20 observations. There were no observations of 1.

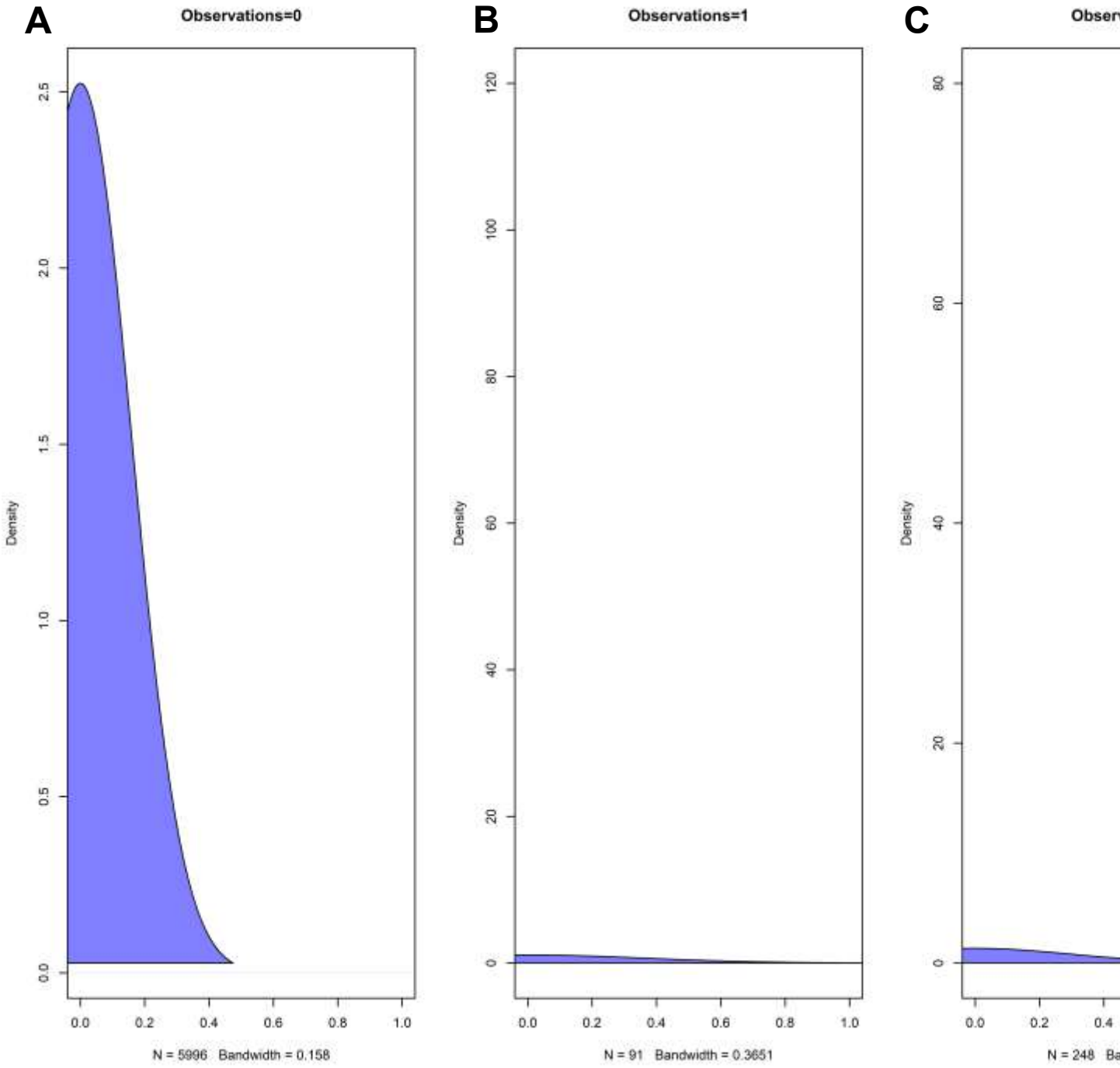

Figure 70. Logistic regression observed versus posterior plots for Silversides Gear 1 Only Model 2G for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

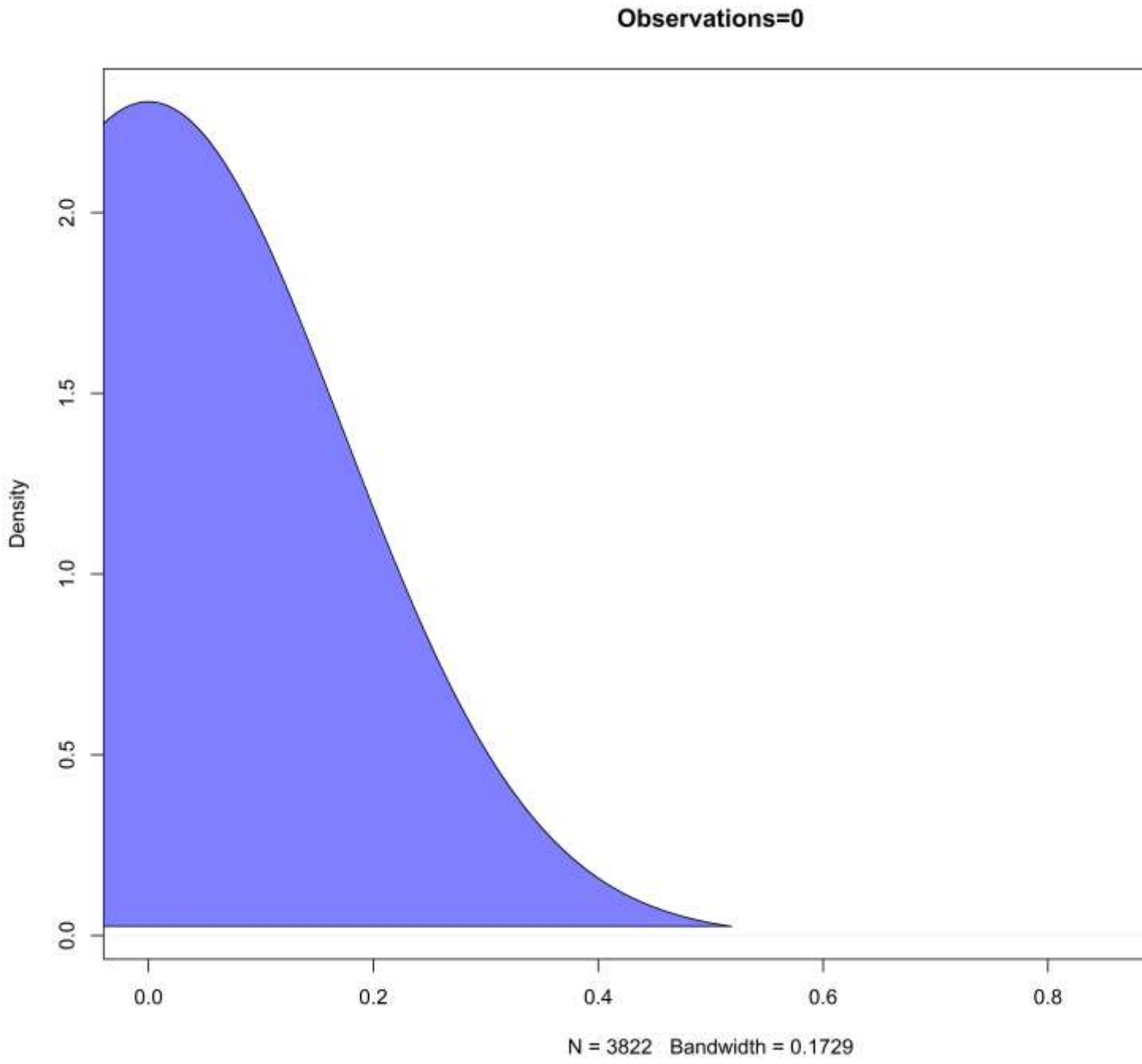

Figure 71. Logistic regression observed versus posterior plots for Silversides Gear 2 Only Model 2G for 0 observations. There were no observations of 1 or more than 20.

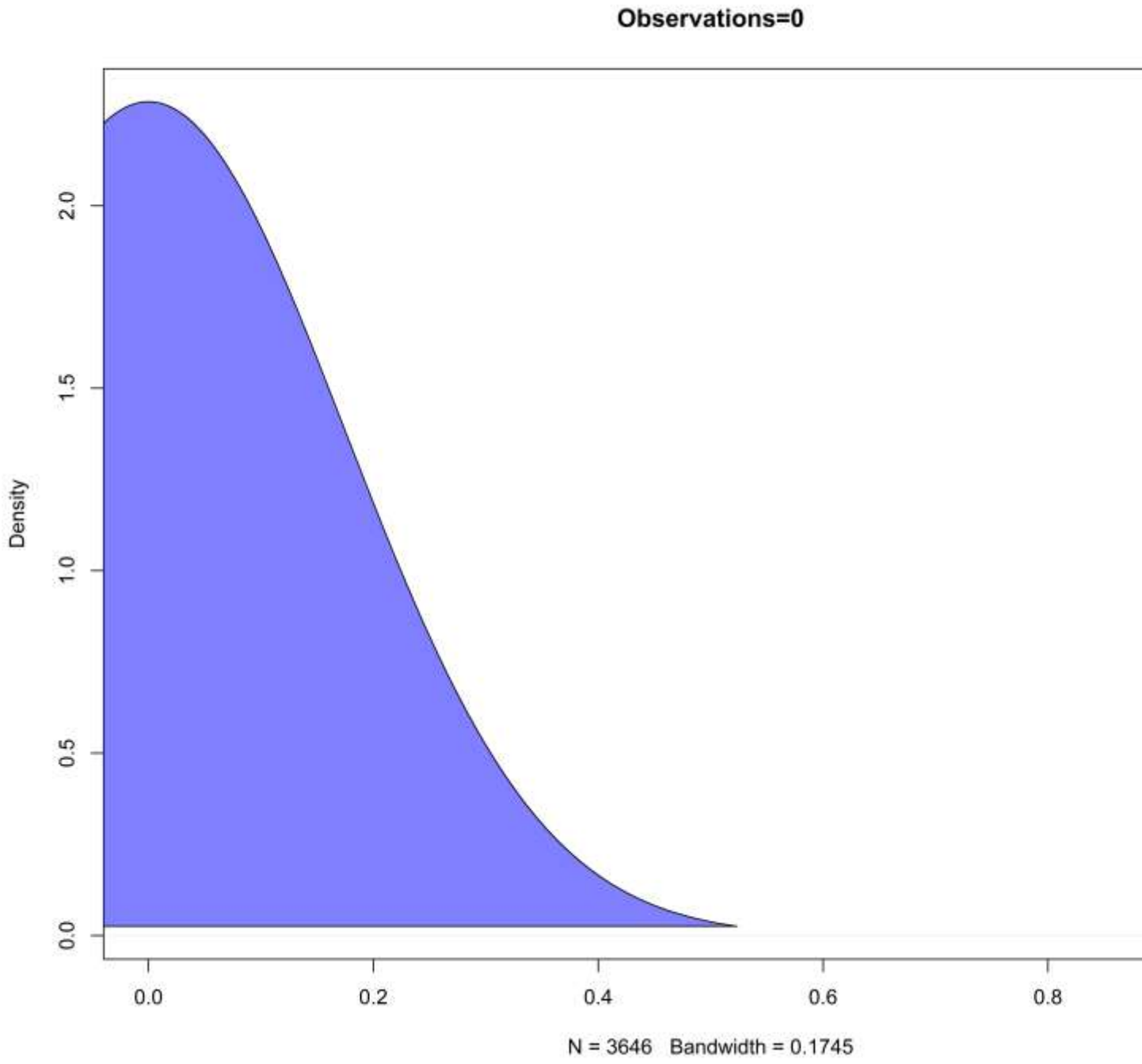

Figure 72. Logistic regression observed versus posterior plots for Silversides Gear 3 Only Model 2G for 0 observations. There were no observations of 1 or more than 20.

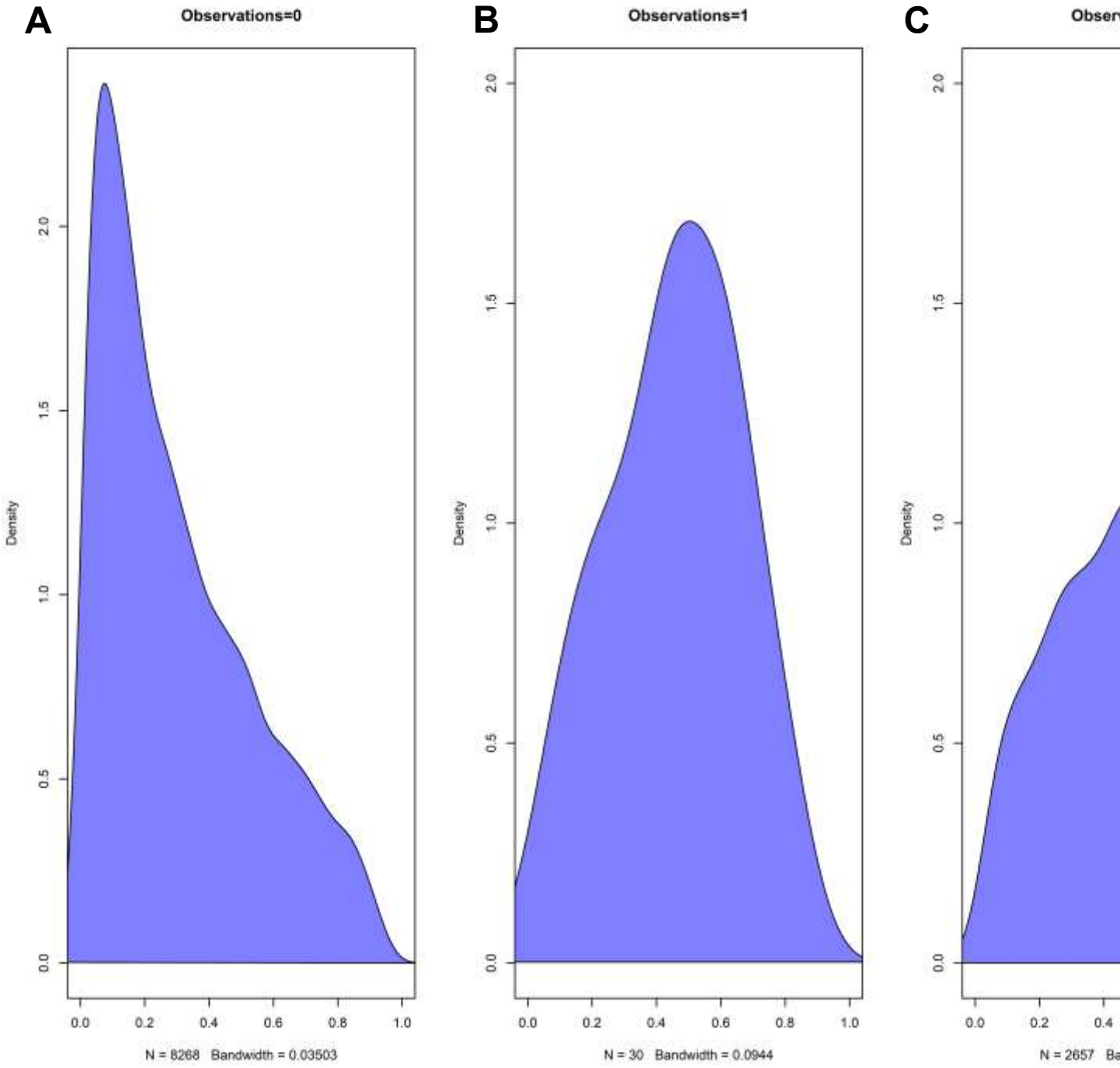

Figure 73. Logistic regression observed versus posterior plots for Grouped Predator Fish Model 2I for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

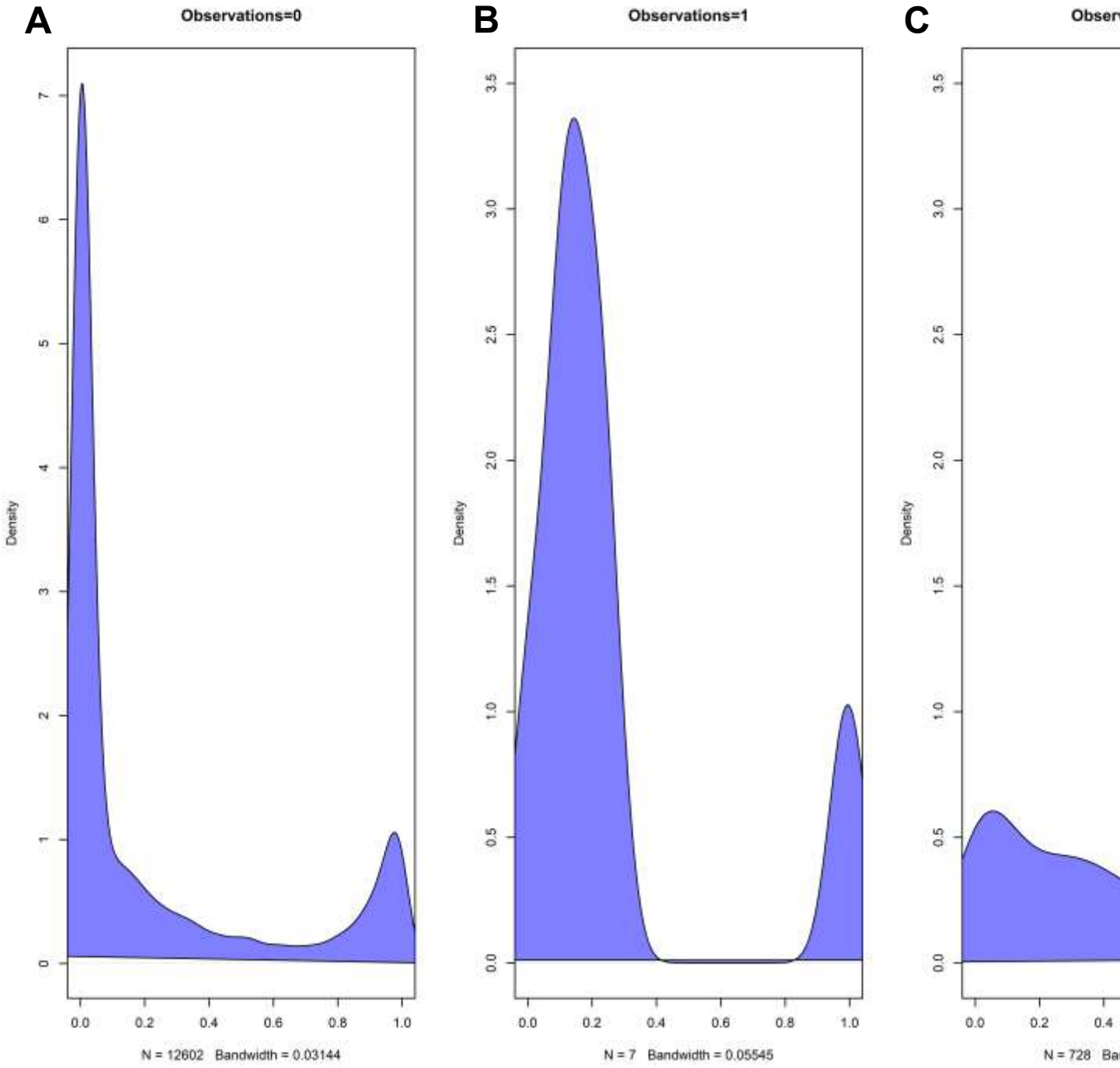

Figure 74. Logistic regression observed versus posterior plots for Common Snook Model 2I for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

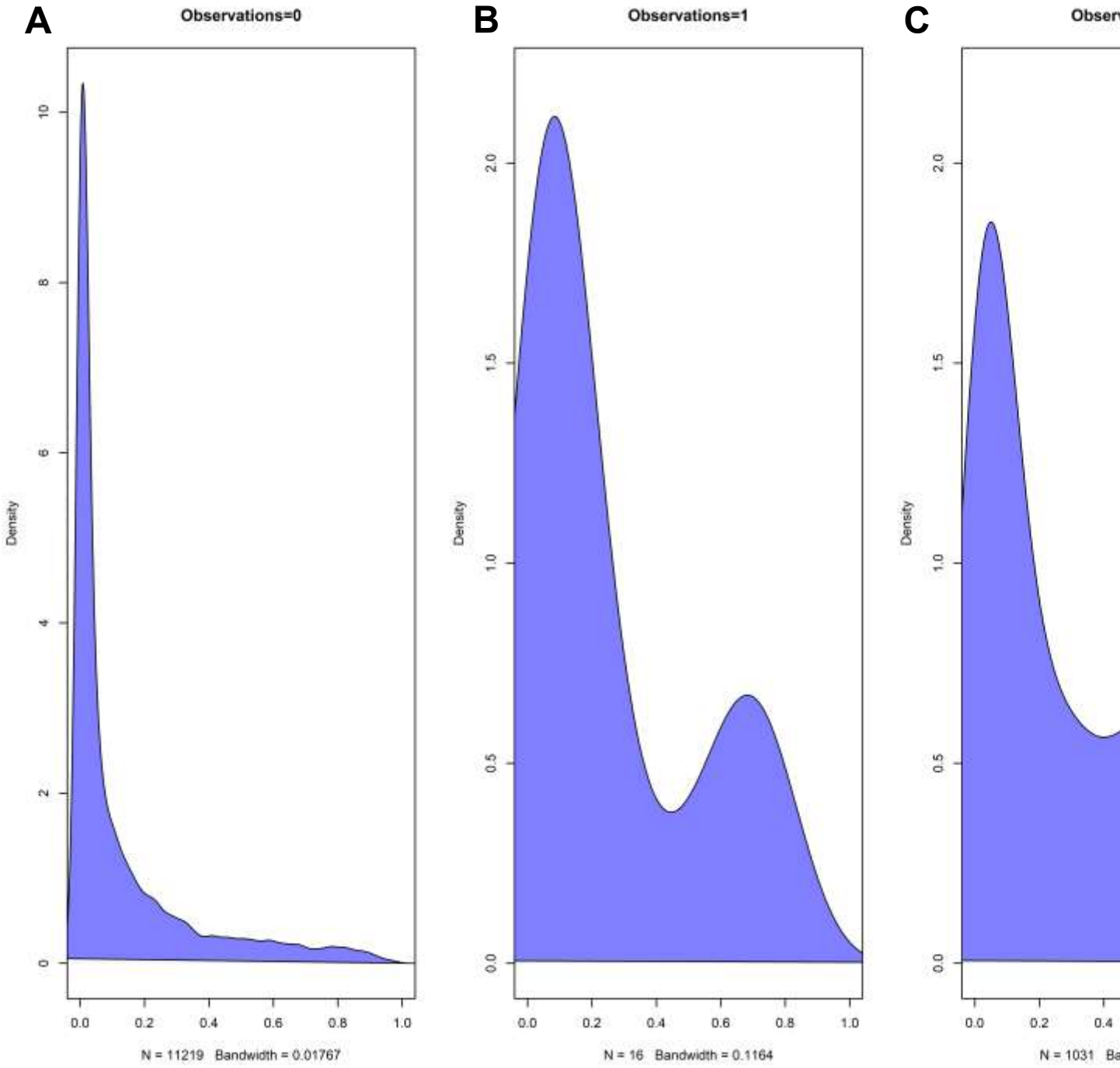

Figure 75. Logistic regression observed versus posterior plots for Trouts Model 2I for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

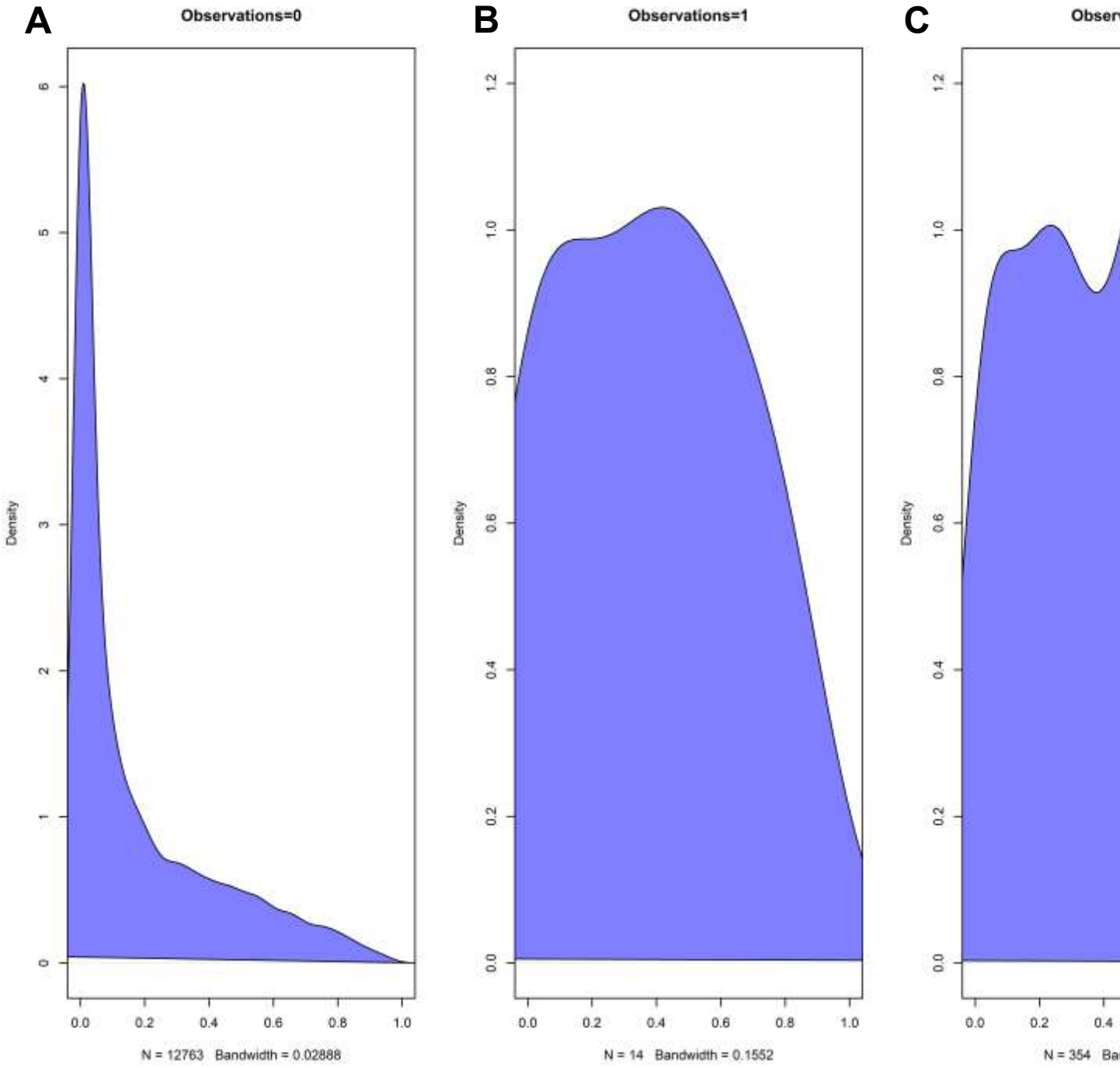

Figure 76. Logistic regression observed versus posterior plots for Red Drum Model 2I for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

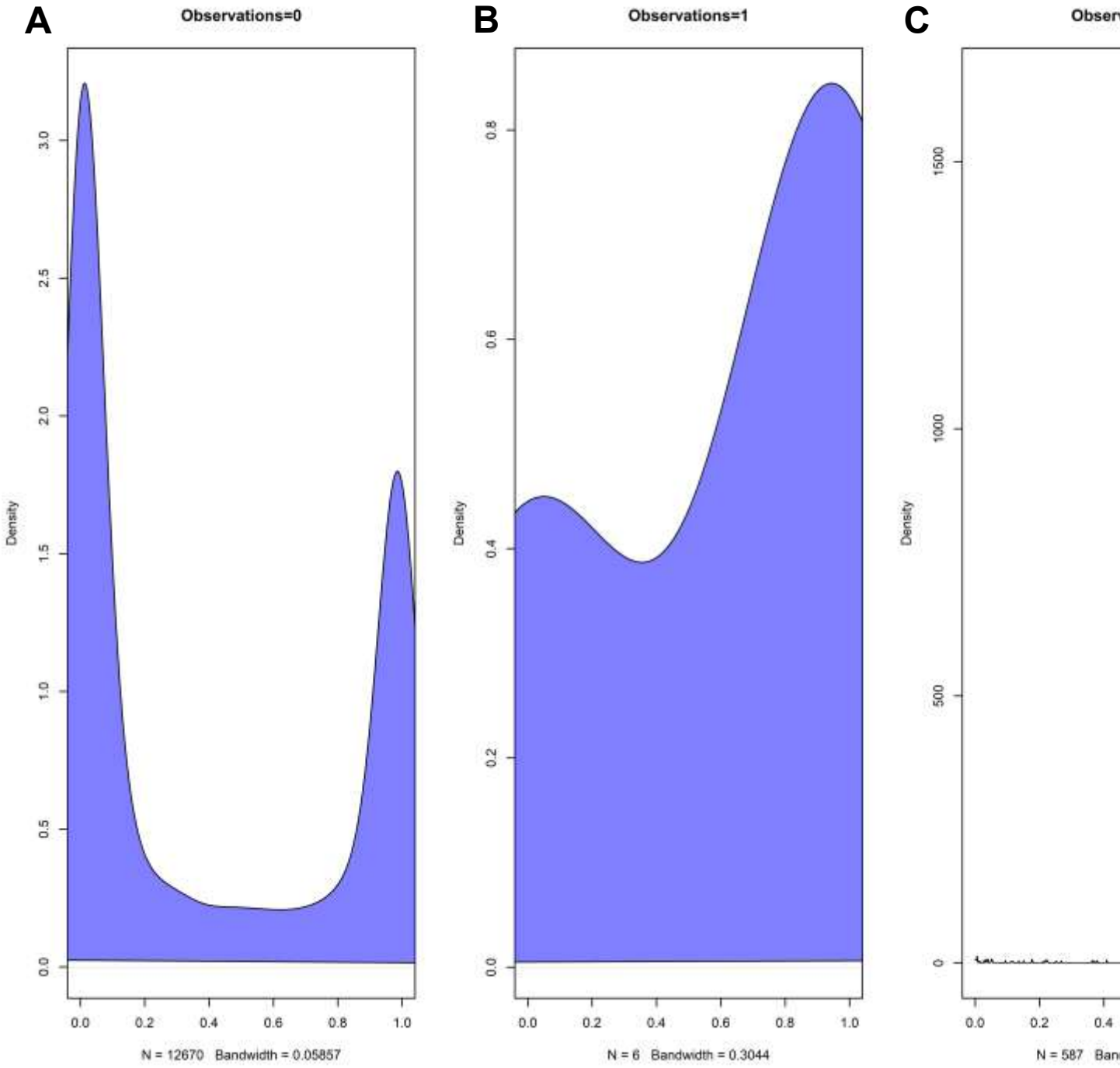

Figure 77. Logistic regression observed versus posterior plots for Snappers Model 2I for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

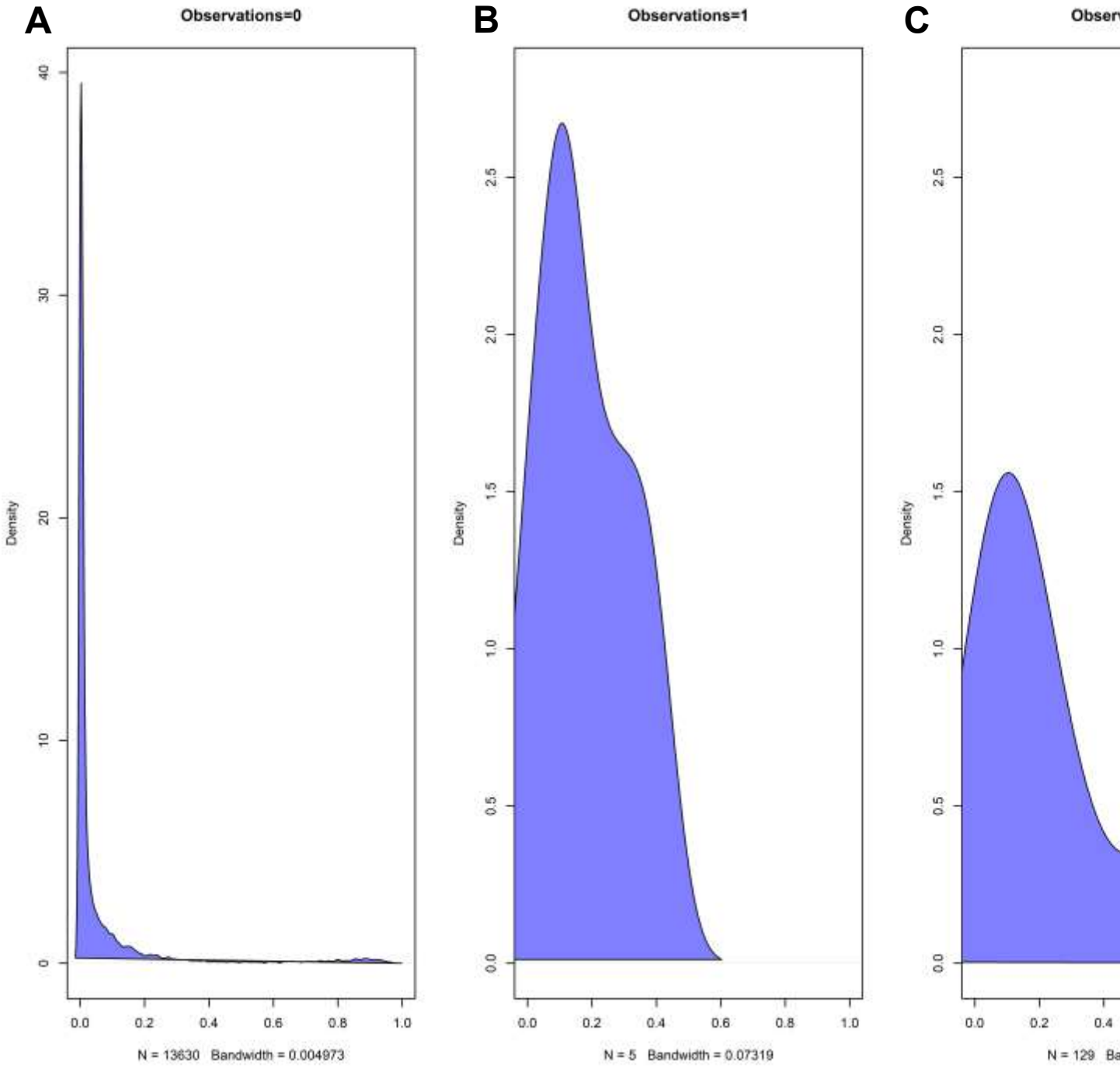

Figure 78. Logistic regression observed versus posterior plots for Crevalle Jack Model 2I for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

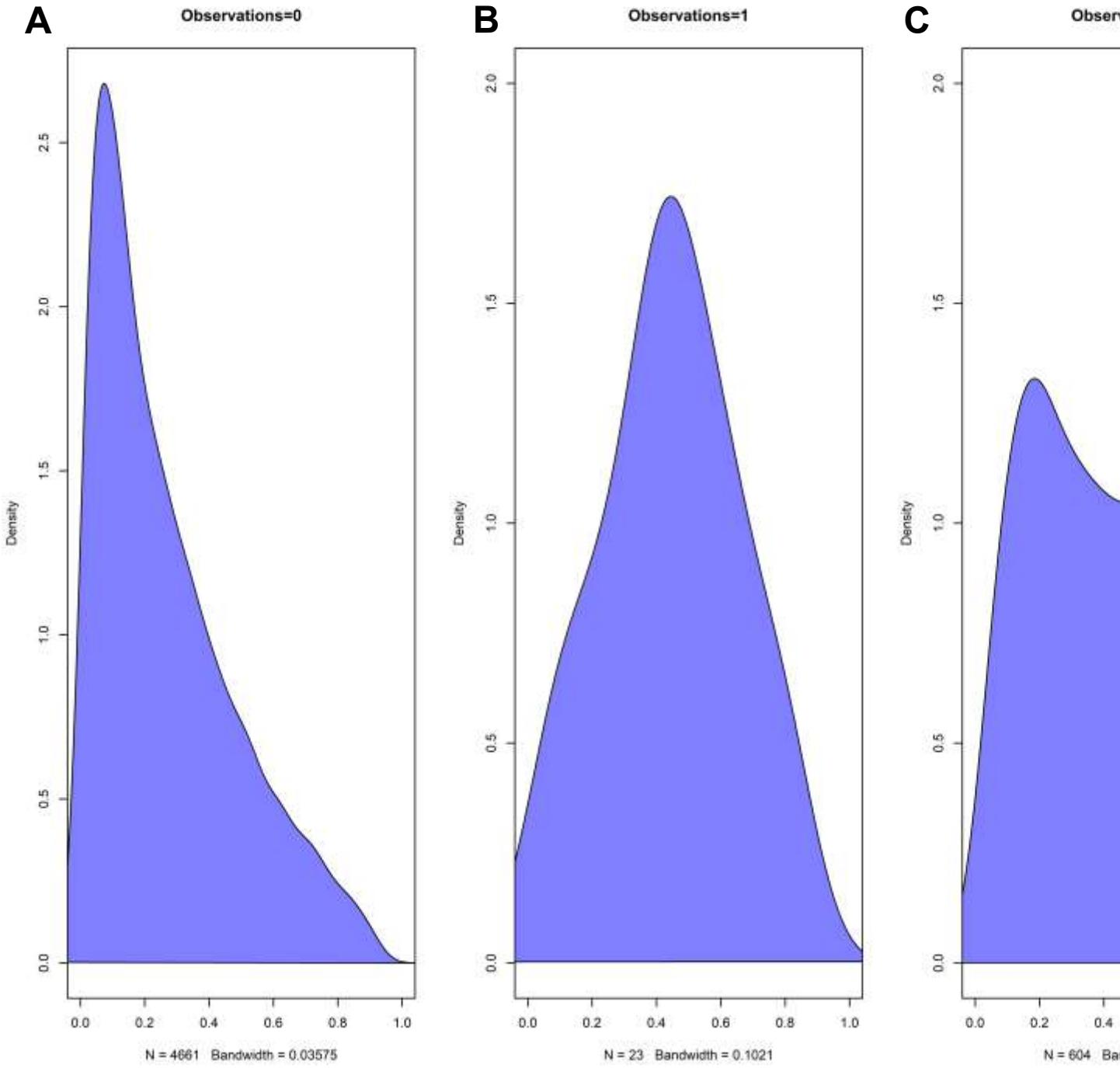

Figure 79. Logistic regression observed versus posterior plots for Grouped Predator Fish Gear 1 Only Model 2I for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

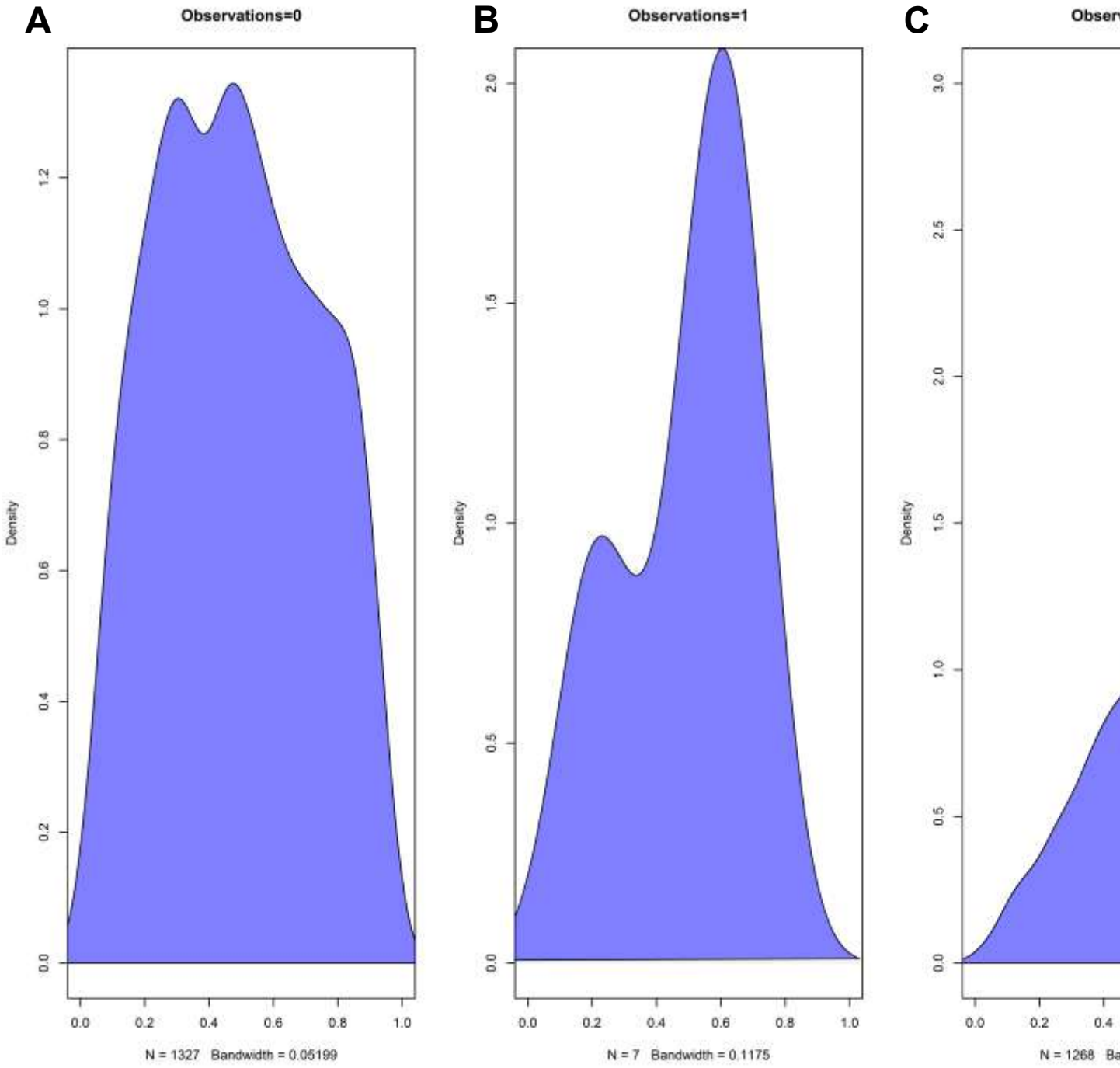

Figure 80. Logistic regression observed versus posterior plots for Grouped Predator Fish Gear 2 Only Model 2I for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

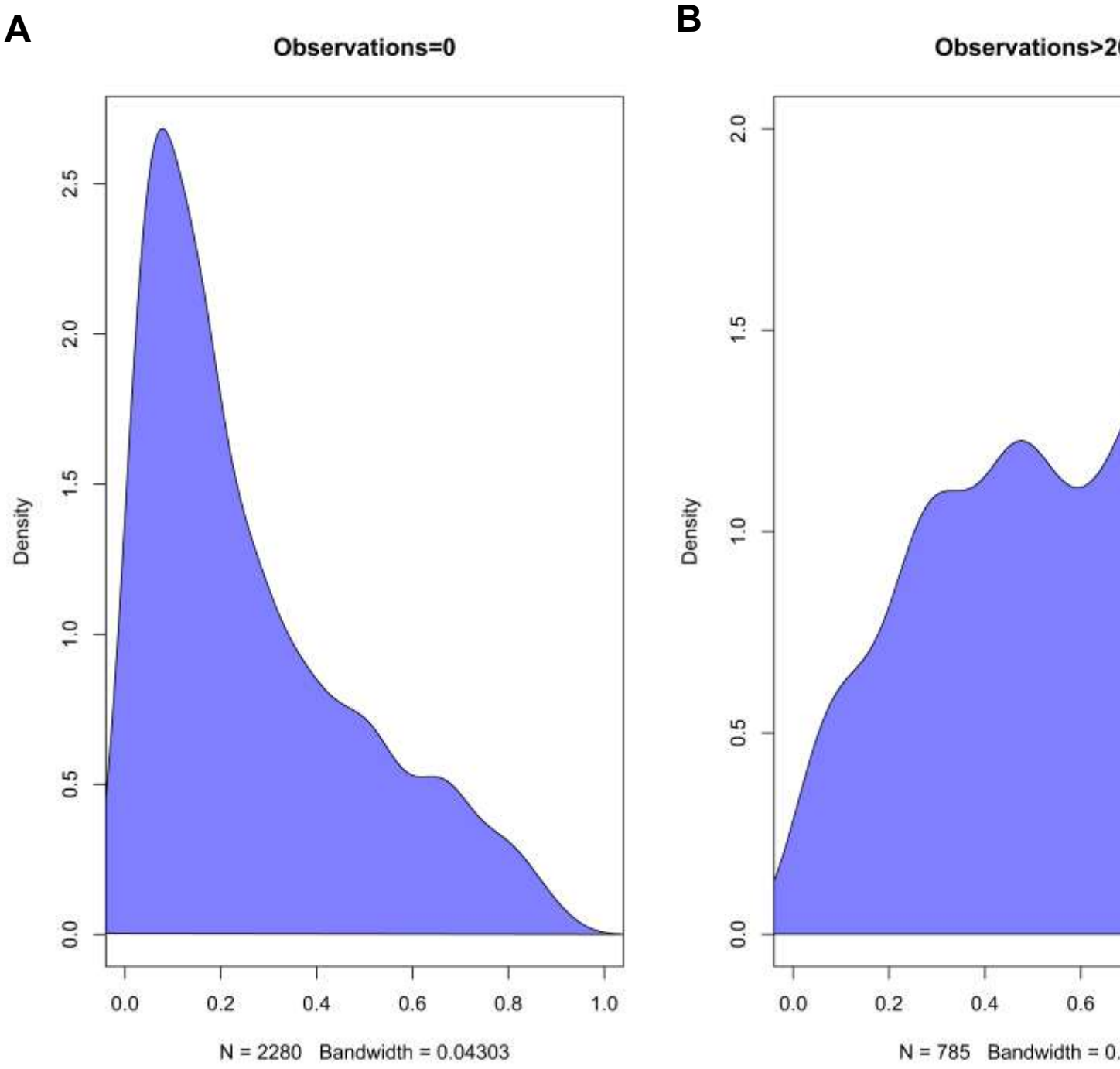

Figure 81. Logistic regression observed versus posterior plots for Grouped Predator Fish Gear 3 Only Model 2I for (A) 0 observations, and (B) more than 20 observations. There were no observations of 1.

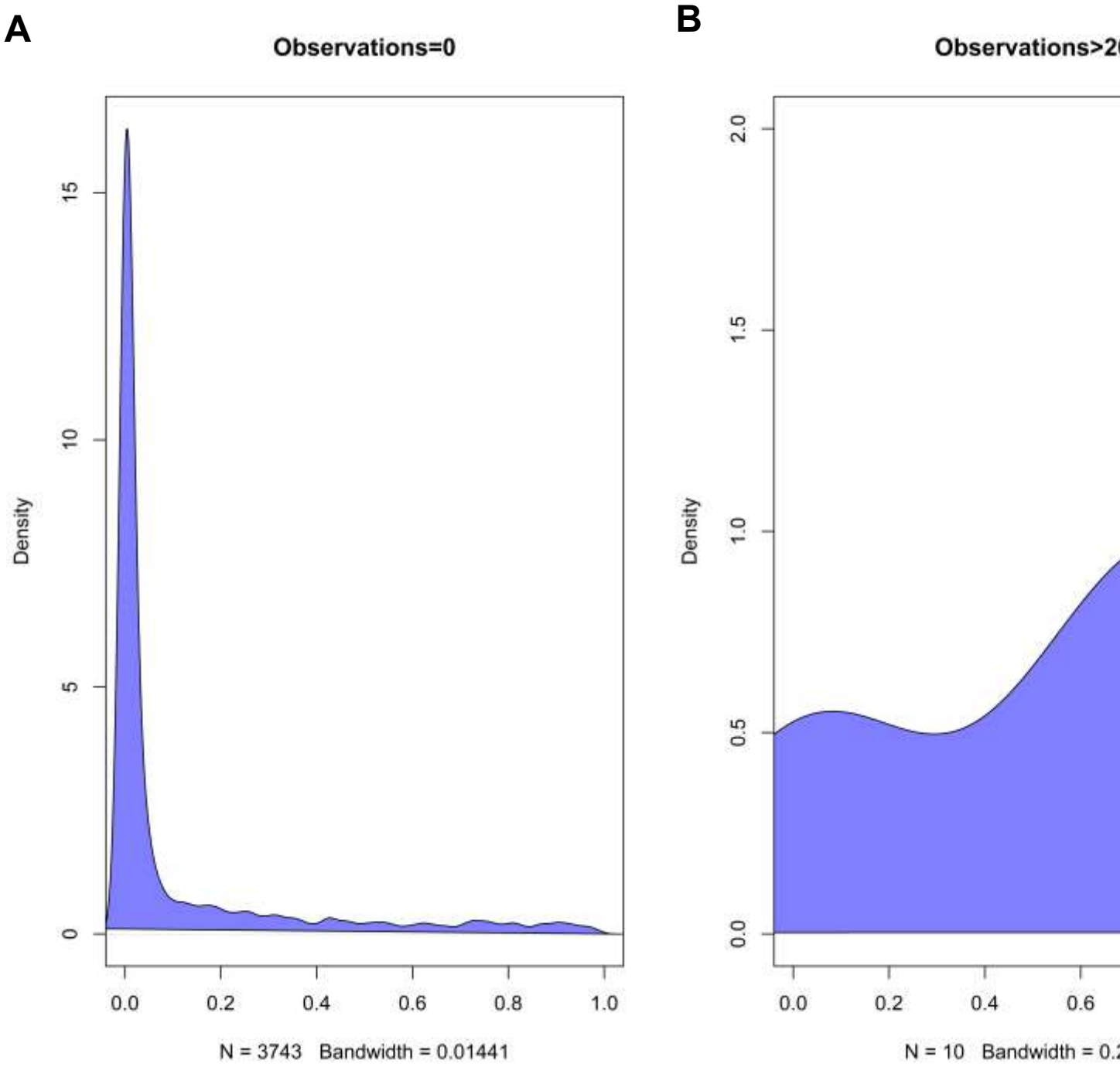

Figure 82. Logistic regression observed versus posterior plots for Common Snook Gear 1 Only Model 2I for (A) 0 observations, and (B) more than 20 observations. There were no observations of 1 or more than 20.

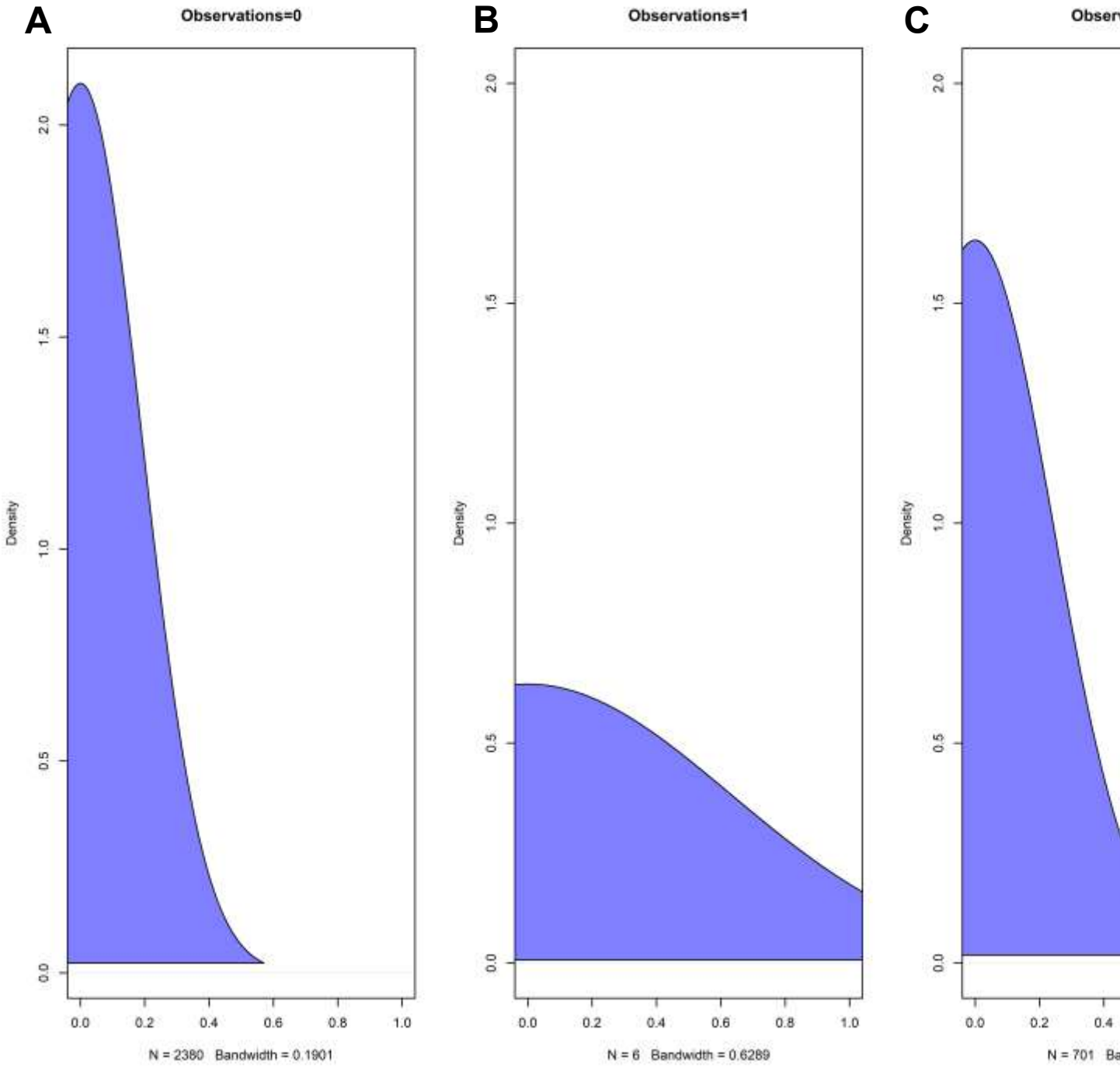

Figure 83. Logistic regression observed versus posterior plots for Common Snook Gear 2 Only Model 2I for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

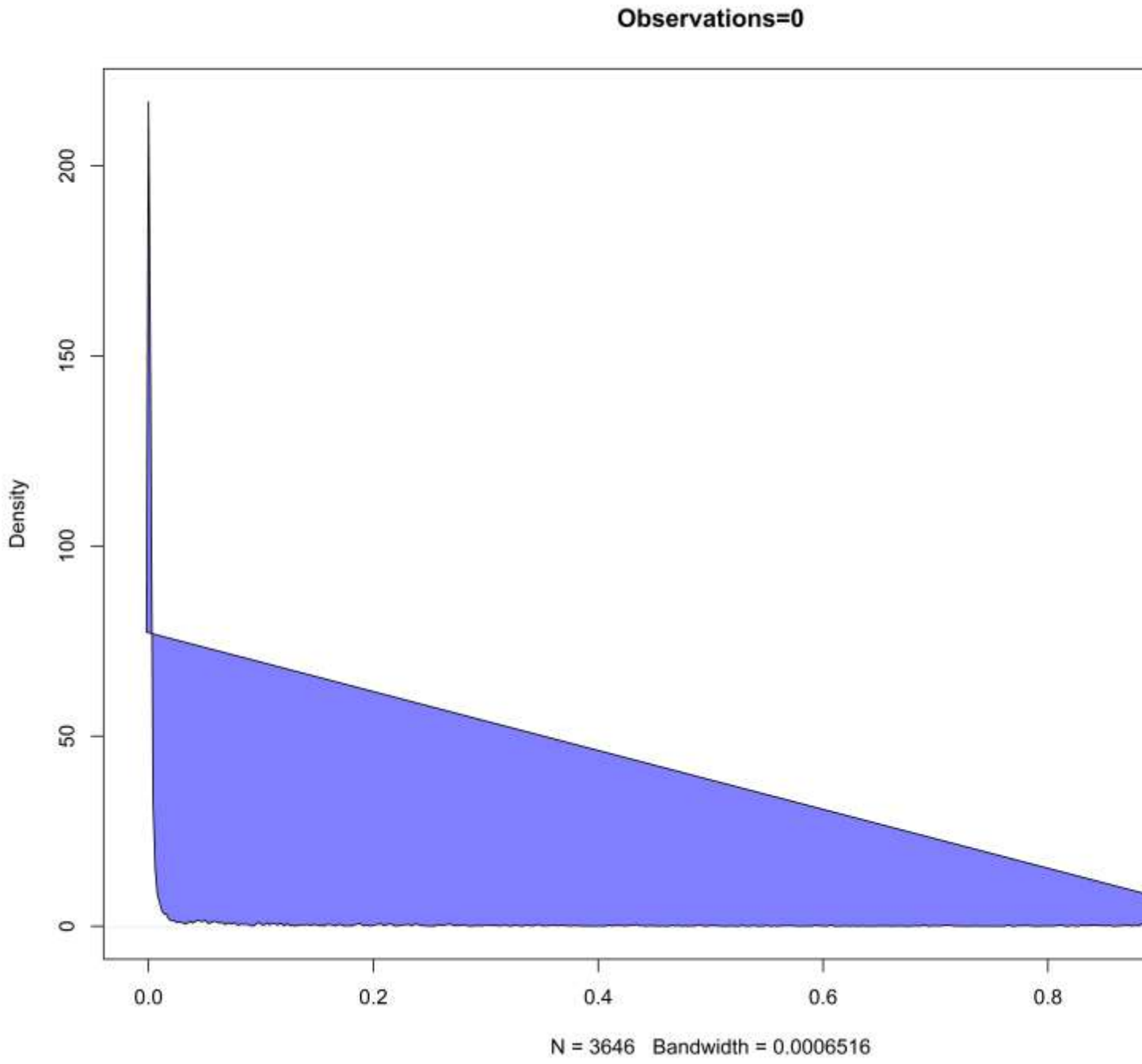

Figure 84. Logistic regression observed versus posterior plots for Common Snook Gear 3 Only Model 2I for 0 observations. There were no observations of 1 or more than 20.

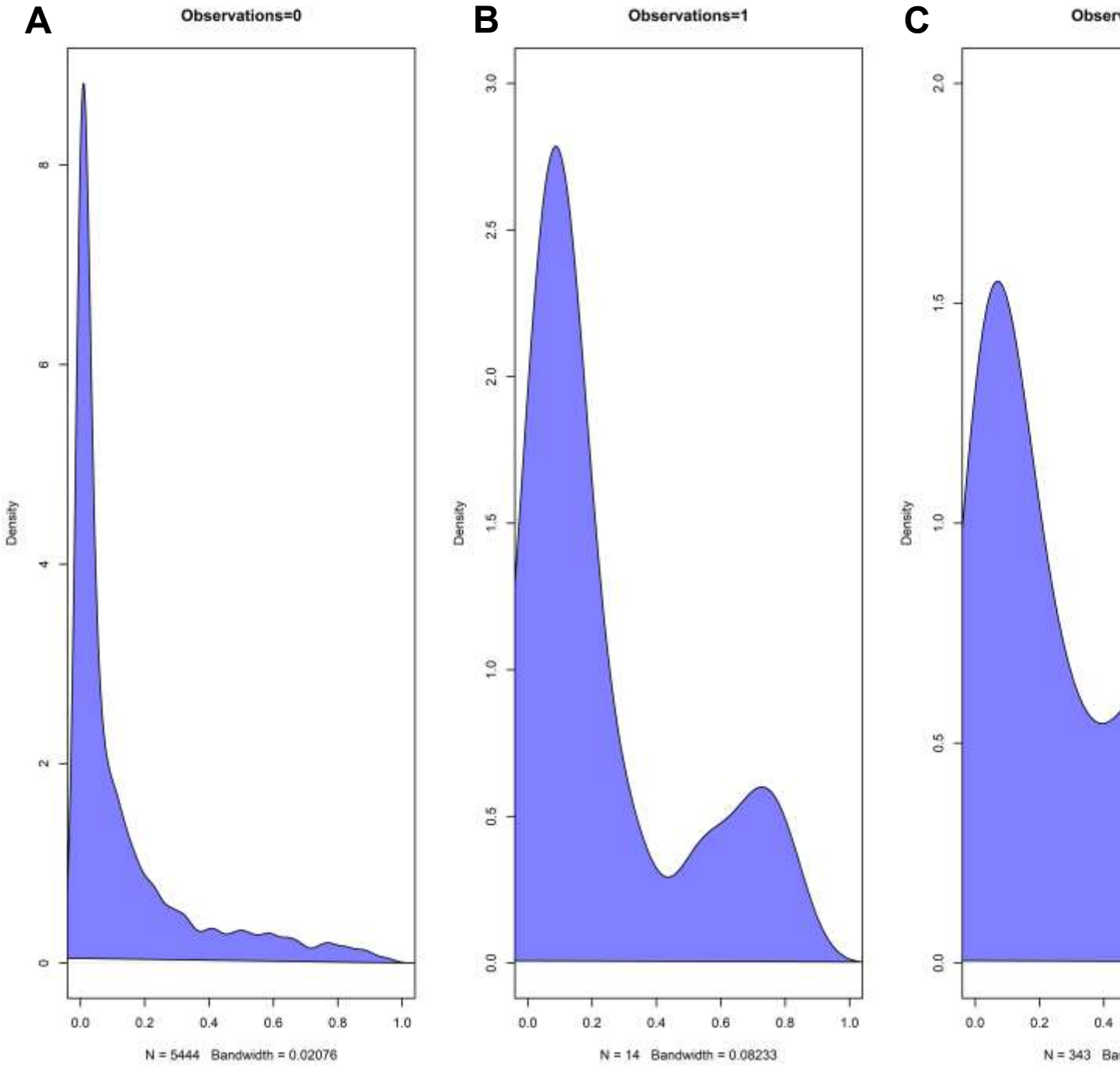

Figure 85. Logistic regression observed versus posterior plots for Trouts Gear 1 Only Model 2I for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

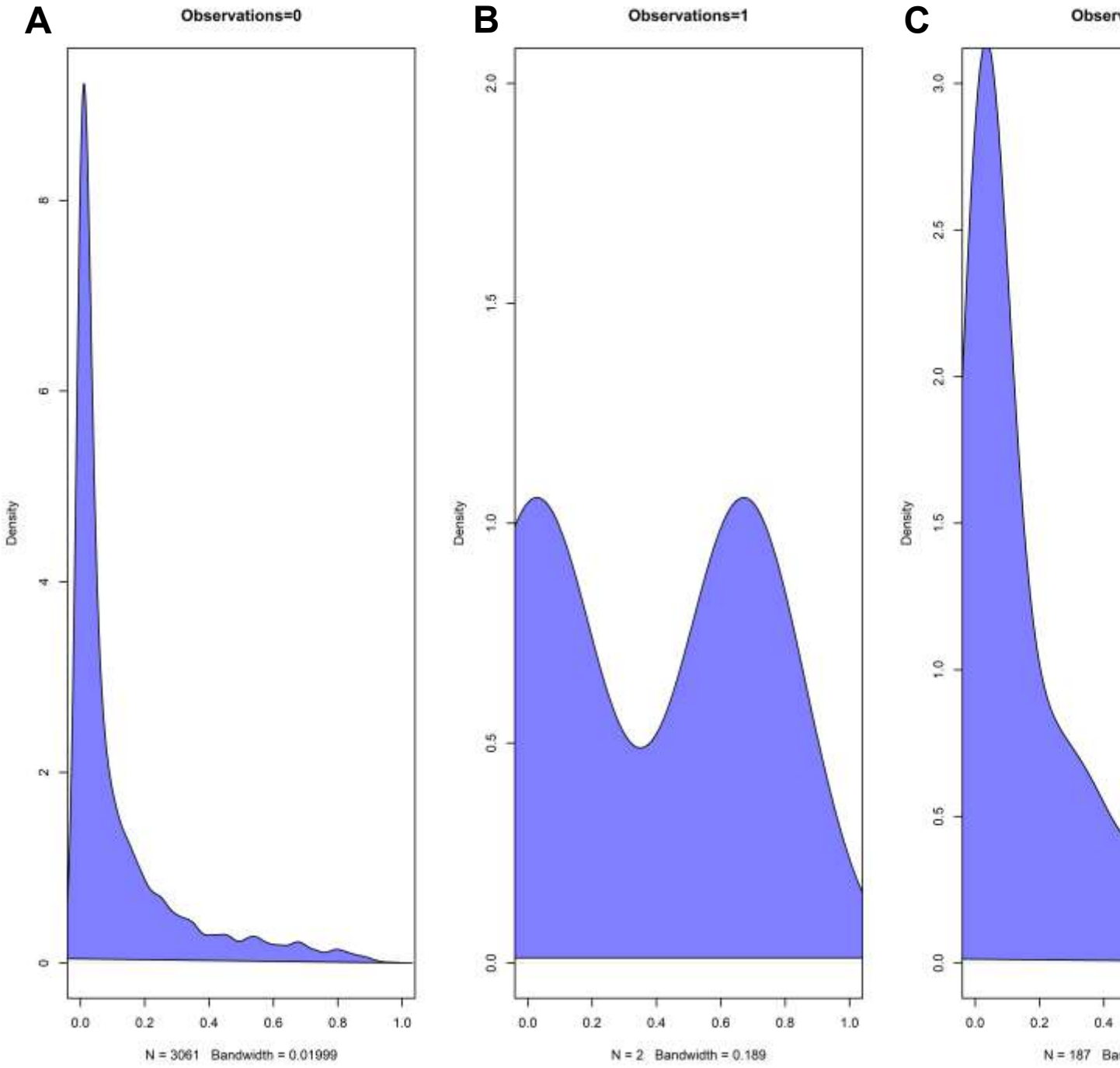

Figure 86. Logistic regression observed versus posterior plots for Trouts Gear 2 Only Model 2I for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

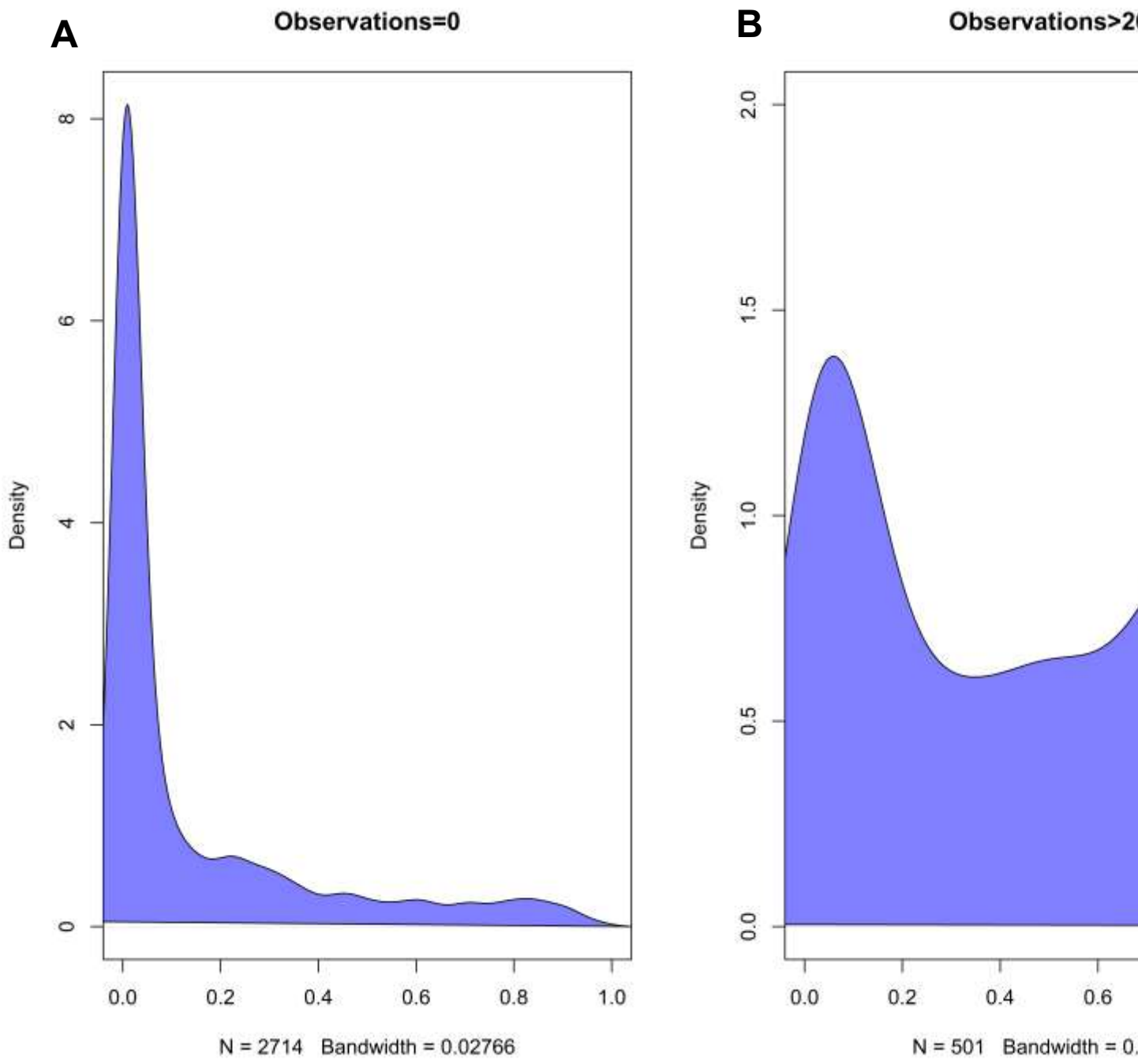

Figure 87. Logistic regression observed versus posterior plots for Trouts Gear 3 Only Model 2I for (A) 0 observations, and (B) more than 20 observations. There were no observations of 1.

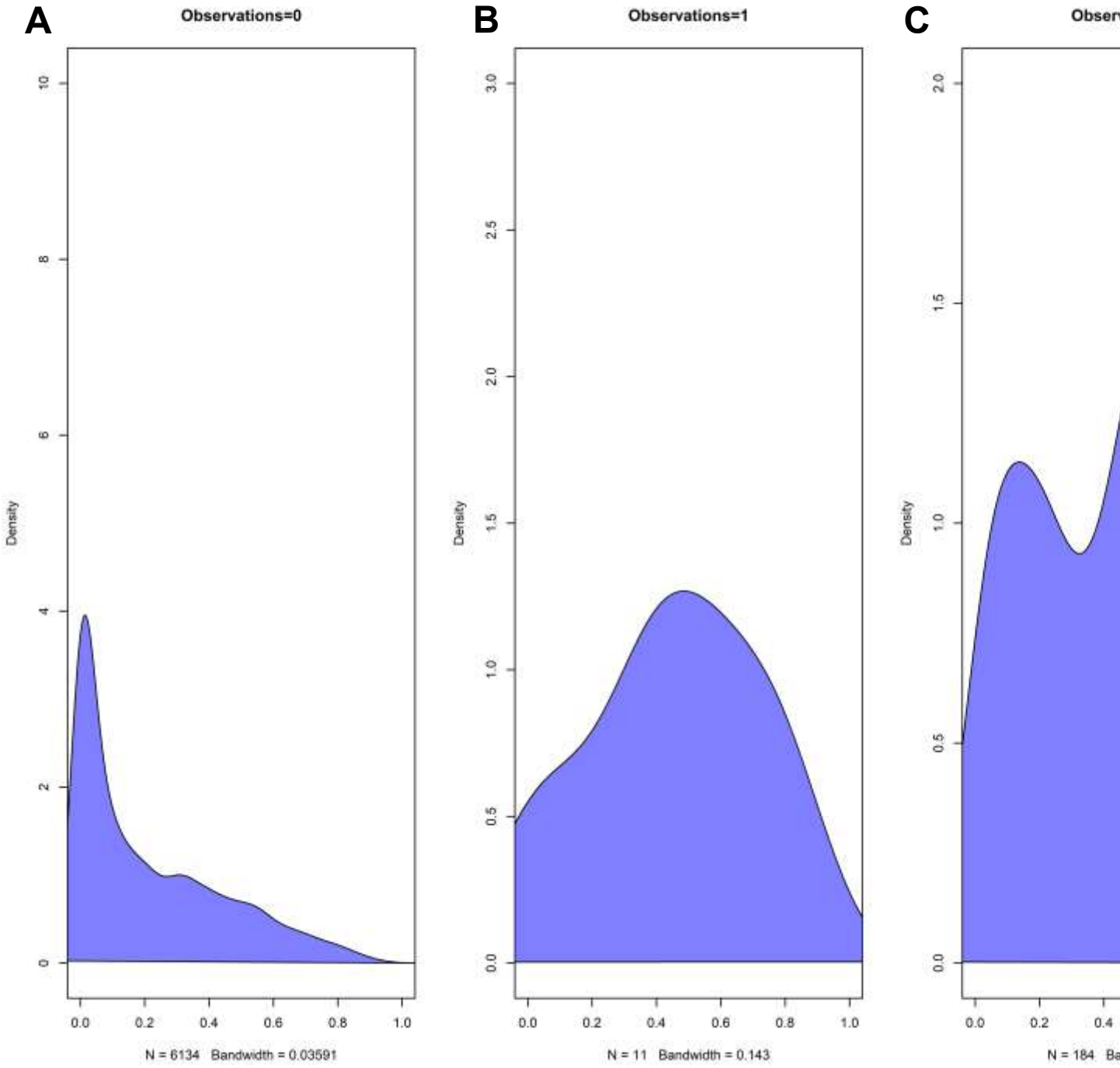

Figure 88. Logistic regression observed versus posterior plots for Red Drum Gear 1 Only Model 2I for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

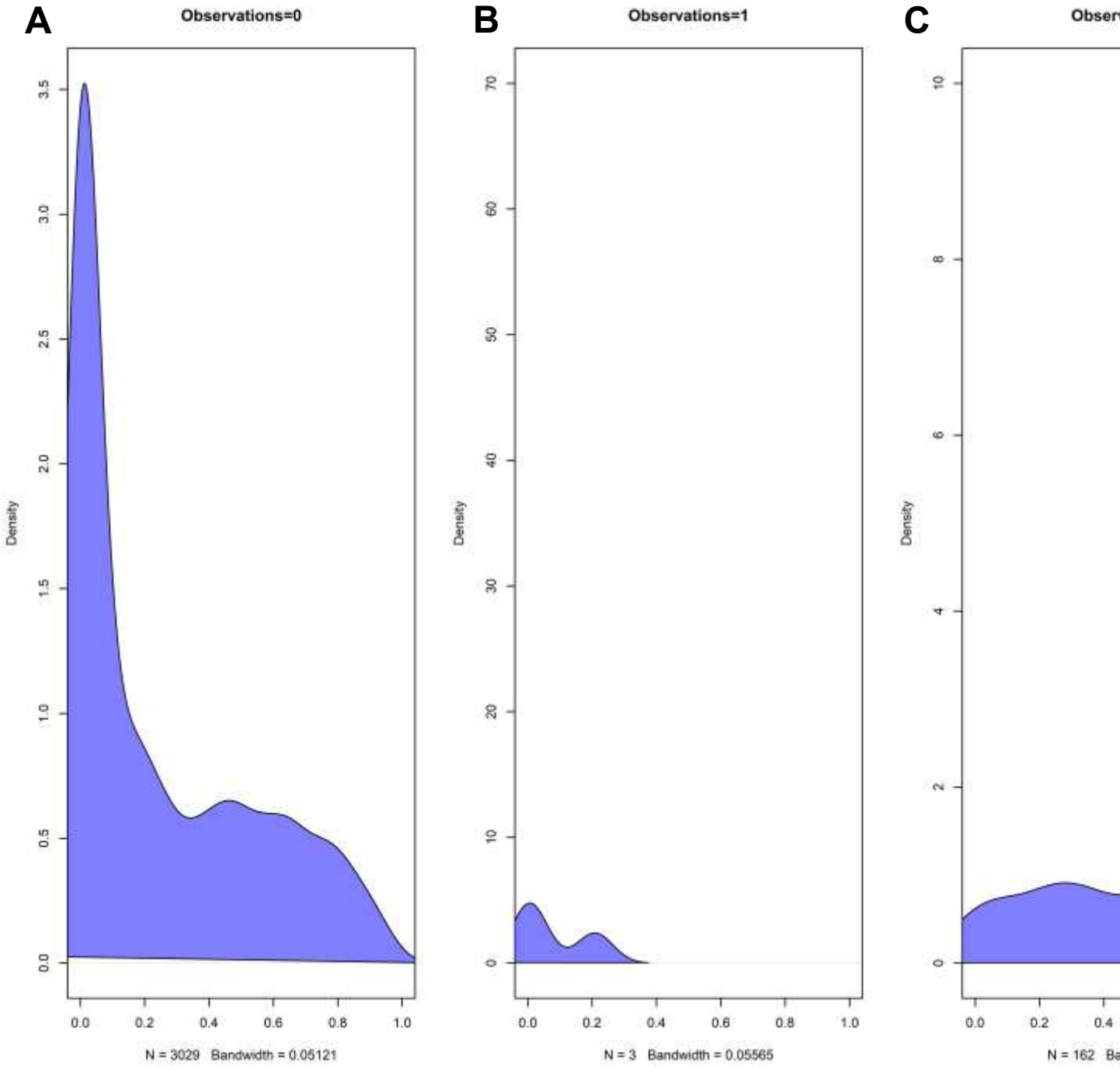

Figure 89. Logistic regression observed versus posterior plots for Red Drum Gear 2 Only Model 2I for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

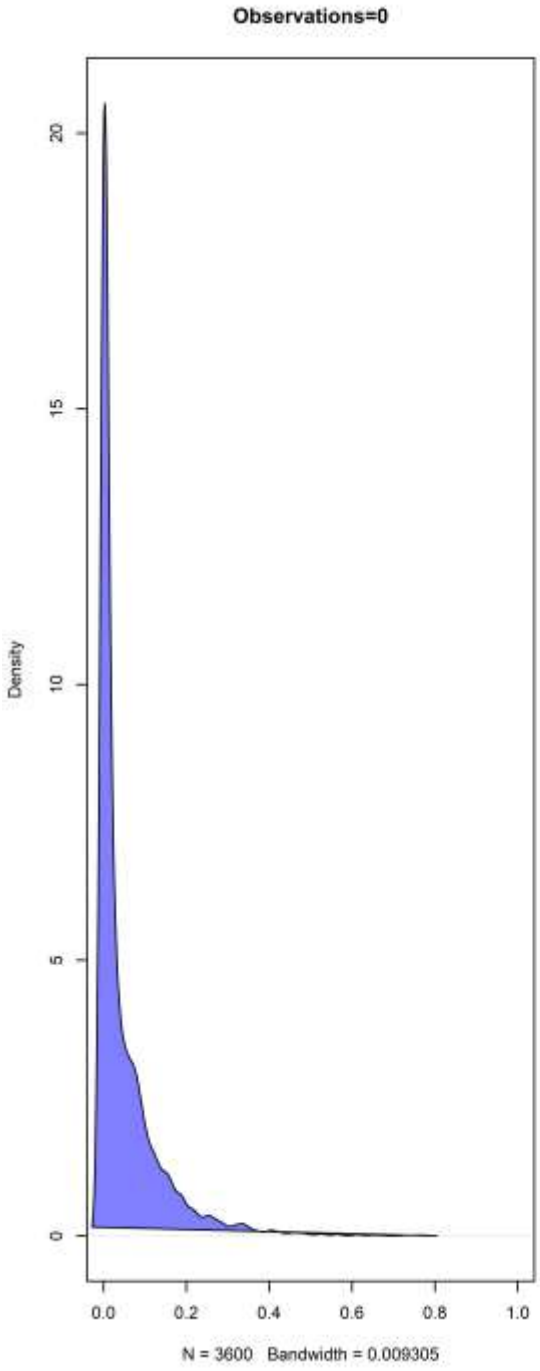

Figure 90. Logistic regression observed versus posterior plots for Red Drum Gear 3 Only Model 2I for 0 observations. There were no observations of 1 or more than 20.

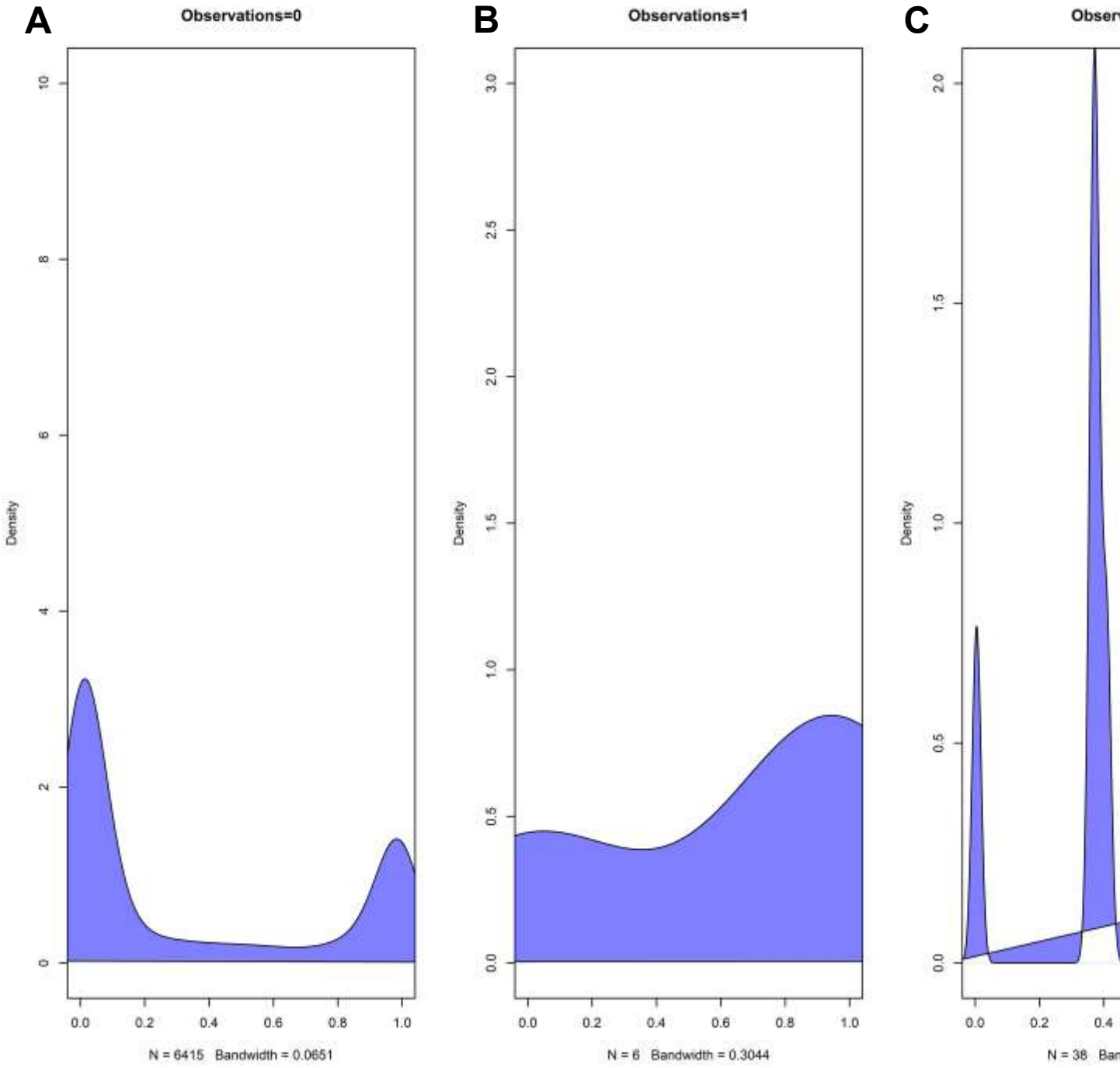

Figure 91. Logistic regression observed versus posterior plots for Snappers Gear 1 Only Model 2I for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

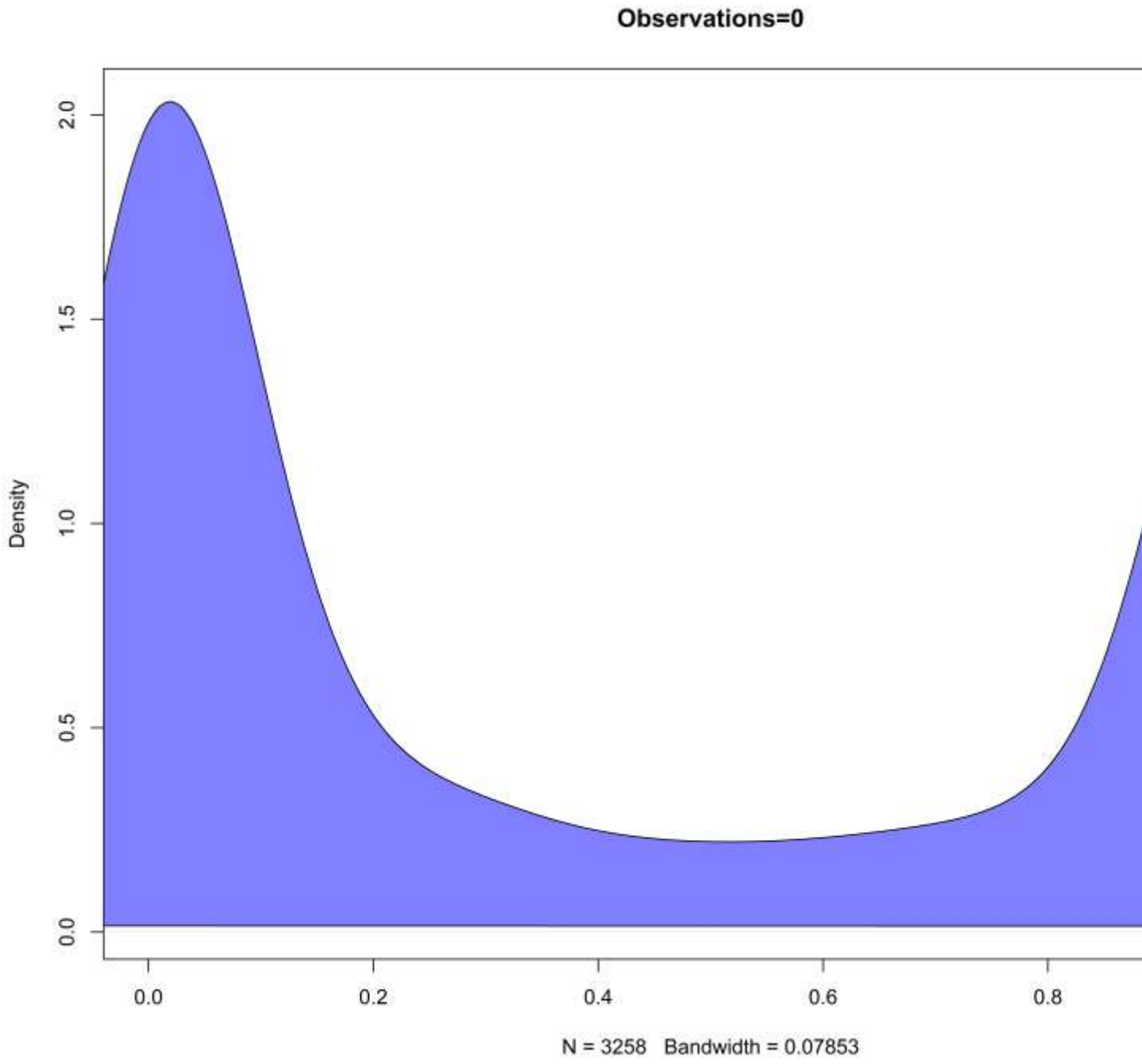

Figure 92. Logistic regression observed versus posterior plots for Snappers Gear 2 Only Model 2I for 0 observations. There were no observations of 1 or more than 20.

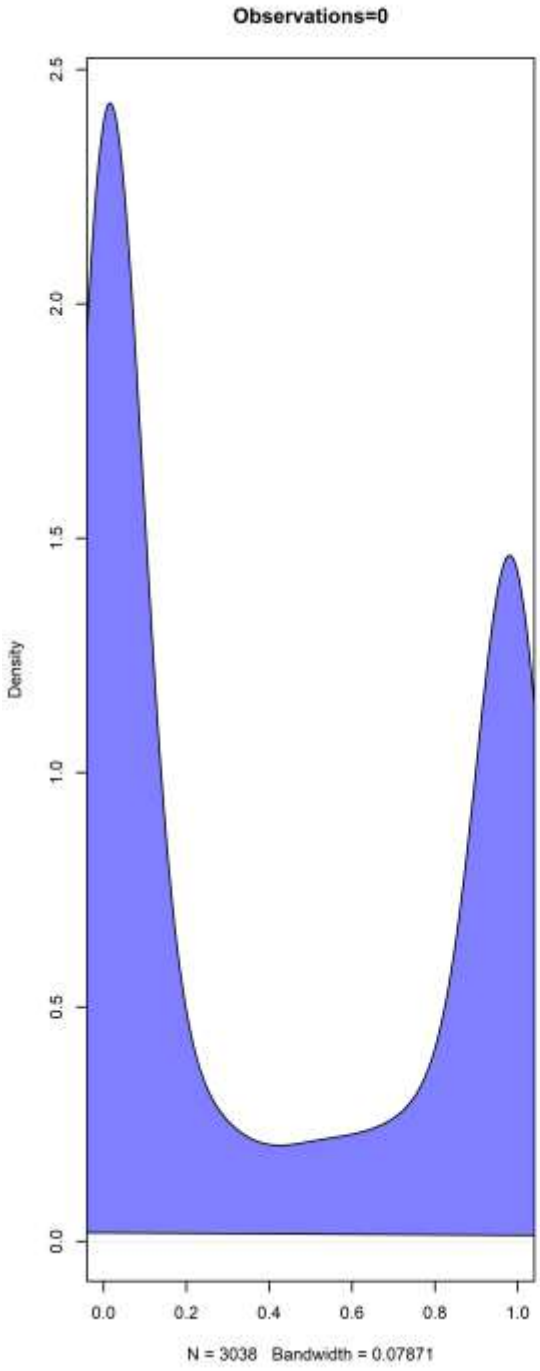

Figure 93. Logistic regression observed versus posterior plots for Snappers Gear 3 Only Model 2I for 0 observations. There were no observations of 1 or more than 20.

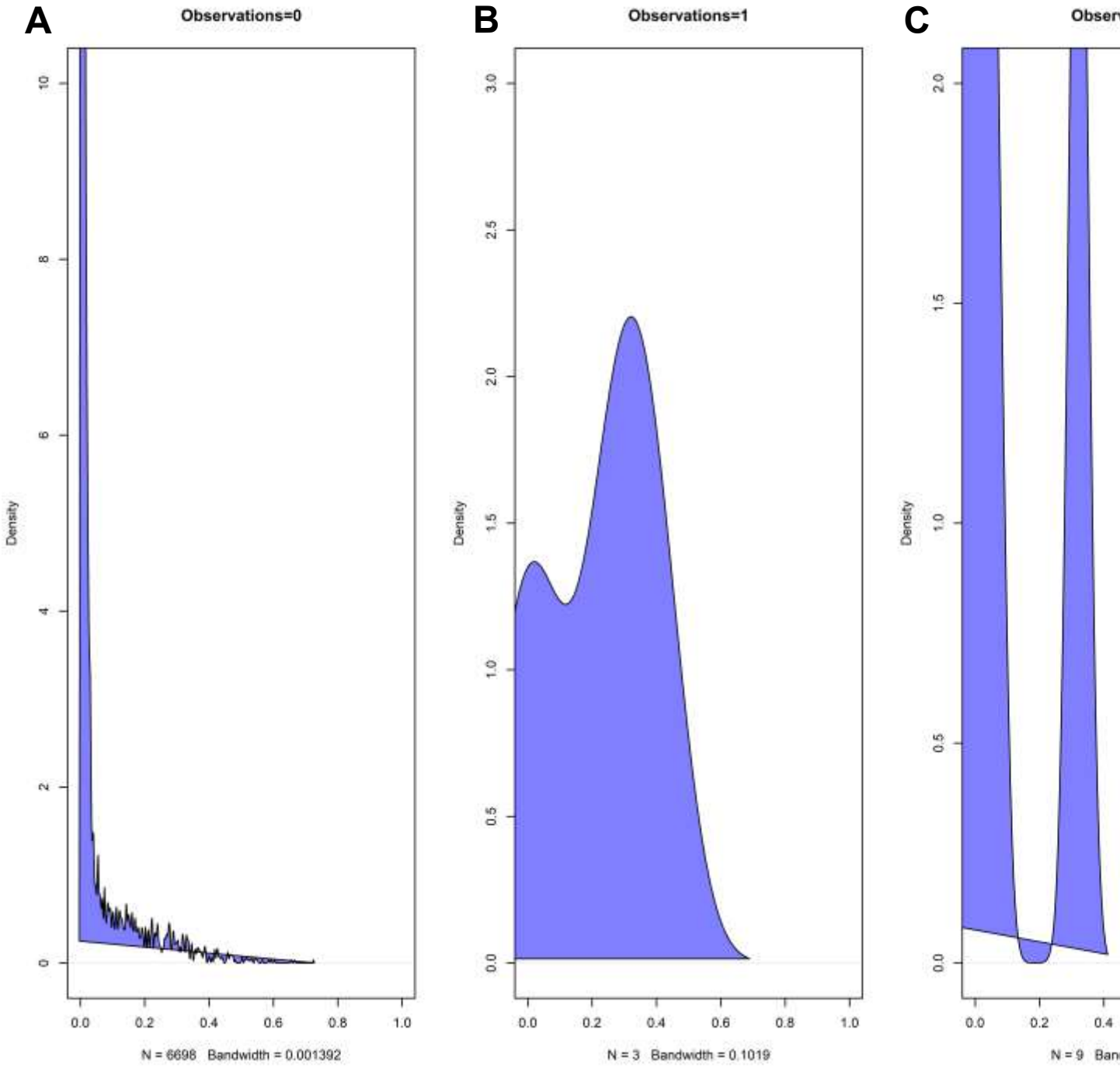

Figure 94. Logistic regression observed versus posterior plots for Crevalle Jack Gear 1 Only Model 2I for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

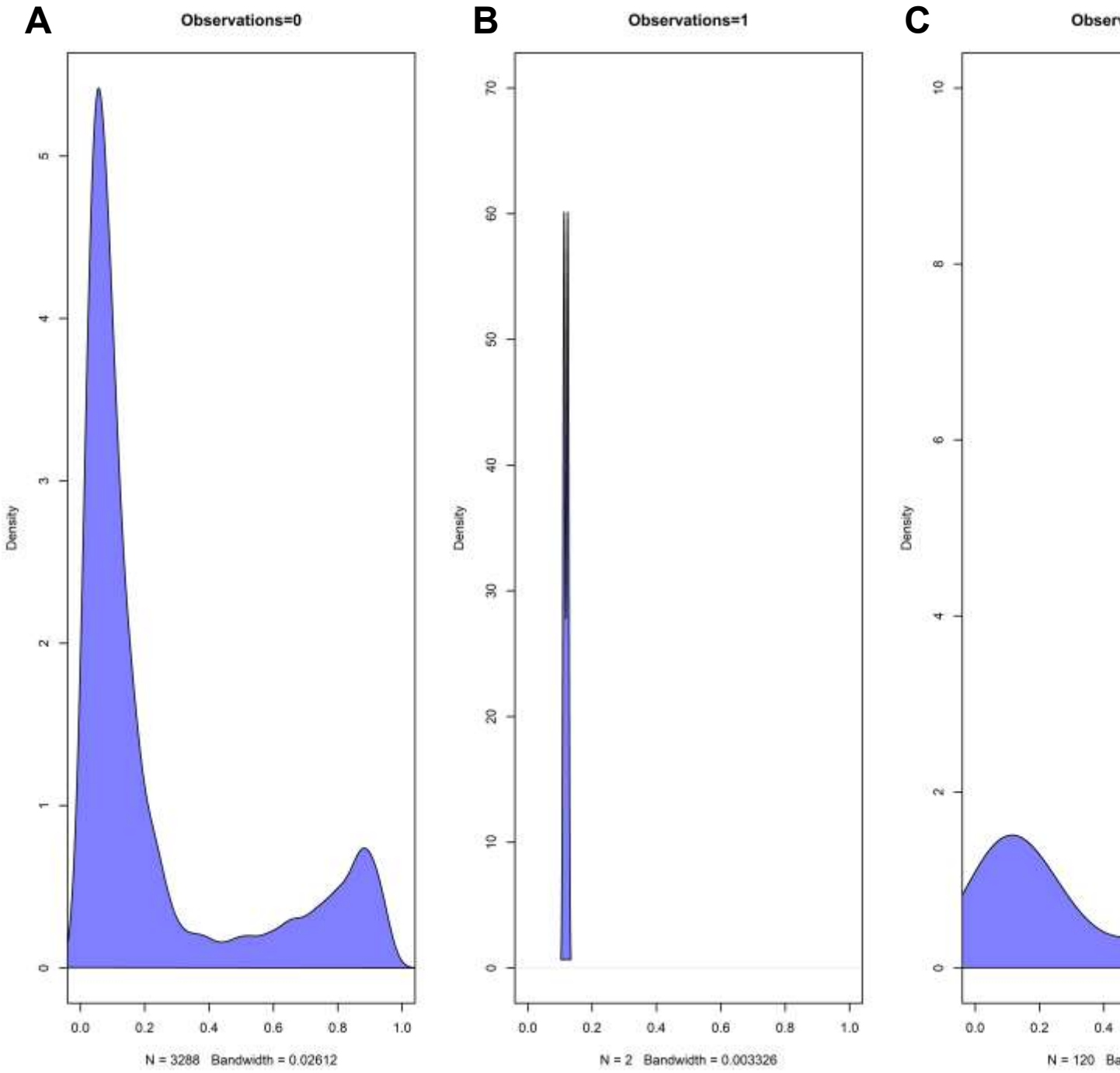

Figure 95. Logistic regression observed versus posterior plots for Crevalle Jack Gear 2 Only Model 2I for (A) 0 observations, (B) 1 observation, and (C) more than 20 observations.

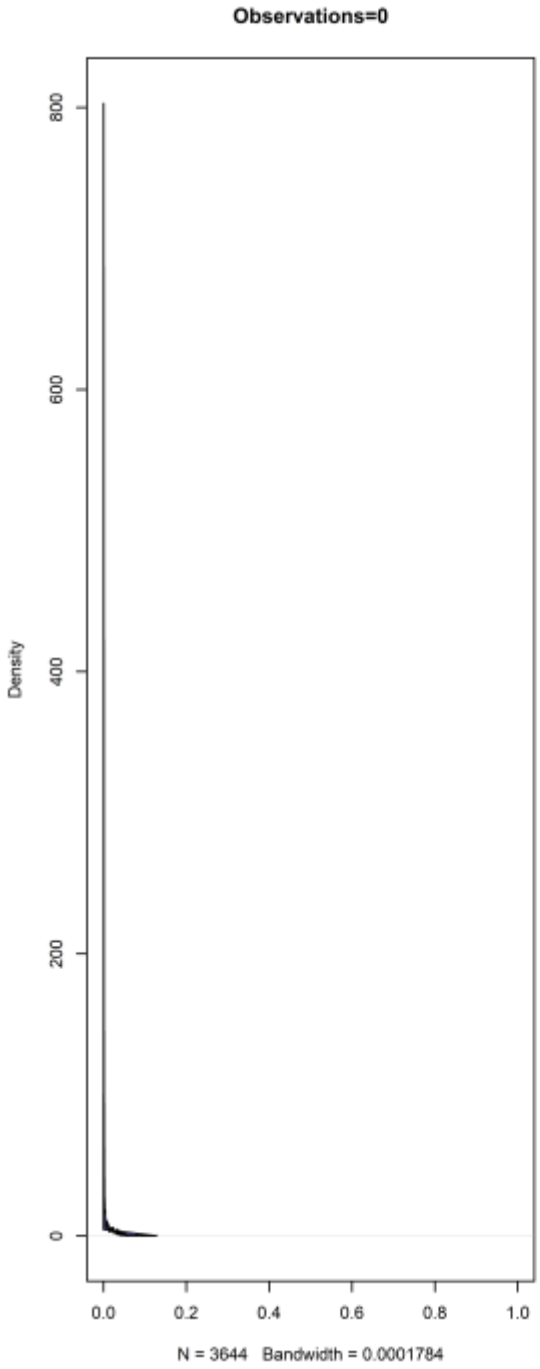

Figure 96. Logistic regression observed versus posterior plots for Crevalle Jack Gear 3 Only Model 2I for 0 observations. There were no observations of 1 or more than 20.